

Exploiting Structural Synergy in post-SIRT imaging: Anatomically Guided Joint Variational Reconstruction of Yttrium-90 PET/SPECT/CT

Sam Porter

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University College London

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I, Sam Porter, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the work.

Abstract

Selective Internal Radiotherapy (SIRT) uses Yttrium-90 (^{90}Y) microspheres to treat unresectable liver tumours. Accurate post-treatment imaging is required to verify microsphere distribution and perform dosimetric evaluations. However, imaging with ^{90}Y presents unique challenges. Bremsstrahlung SPECT offers high count statistics but suffers from poor spatial resolution and difficulties in correcting for scatter and other additive terms. ^{90}Y PET offers superior resolution, but suffers from high noise due to the extremely low internal pair branching ratio ($\sim 0.003\%$). Traditional reconstruction fails to exploit complementary information in these paired datasets.

This thesis proposes a synergistic framework that uses anatomically guided joint variational regularisation to exploit shared underlying structure. We introduce the Directional Total Nuclear Variation Prior, which promotes edge alignment between PET and SPECT without enforcing intensity correlation, while incorporating CT edge-guidance and suppressing unwanted noise.

We also develop a Monte Carlo-informed Bremsstrahlung SPECT model comprising a damped, pre-reconstruction estimate of non-primary contamination and an effective image-space response kernel.

To ensure computational efficiency, we propose a synergy-aware preconditioner which, in conjunction with a variance reduced stochastic gradient algorithms, results in fast convergence.

We validate our method against the state-of-the-art sequential hybrid kernel expectation maximisation scheme (HKEM). We demonstrate that dTNV shows improved quantitative accuracy and increased reproducibility over noise realisations

compared to existing methods.

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Acknowledgements

So, it turns out that a PhD is quite an undertaking. It's been a long old journey, but the end is in sight!

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Contents

List of Abbreviations	24
Notation	28
1 Introduction	32
Introduction	32
2 Physics and Measurement Models in Emission Tomography	35
2.1 Principles of Emission Tomography	36
2.2 Positron Emission Tomography (positron emission tomography (PET))	36
2.2.1 Coincidence Detection and Data Acquisition	38
2.2.2 The PET model	39
2.3 Single Photon Emission Computed Tomography (single photon emis- sion computed tomography (SPECT))	40
2.3.1 The SPECT model	41
2.4 Yttrium-90 Radioembolisation (selective internal radiotherapy (SIRT))	41
2.4.1 Decay Characteristics	42
2.4.2 Bremsstrahlung SPECT	42
2.4.3 Internal Pair Production PET	42
2.5 X-ray computed tomography (CT)	43
3 Penalised-Likelihood Reconstruction: Priors and Optimisation Algo- rithms	46
3.1 The Inverse Problem in Emission Tomography	47

3.2	Regularisation Strategies	49
3.2.1	Image-based Regularisation	50
3.2.2	Anatomically Guided Regularisation	51
3.2.3	Synergistic Regularisation	55
3.3	Optimisation Algorithms	58
3.4	Summary and Research Gap	65
4	Monte Carlo-Informed Modelling for ^{90}Y Bremsstrahlung SPECT	67
4.1	Pre-Reconstruction Estimation of Non-Primary Counts	68
4.1.1	Learned initialisation	68
4.1.2	Reconstruct–simulate refinement	68
4.1.3	Under-relaxed updates	70
4.2	Image-Space Bremsstrahlung Range Blur	72
4.2.1	Primary collimator–detector response	72
4.2.2	Effective Gaussian range-kernel approximation	73
4.2.3	Projection-response calibration and cross-code comparison	74
4.3	Scope, Limitations and Deployment	75
4.4	Summary	76
5	Anatomy-informed deconvolution for PET partial volume correction	77
5.1	Introduction and relationship to the thesis	77
5.2	Image-domain inverse problem and objective	78
5.3	Baseline: Richardson–Lucy deconvolution	79
5.3.1	Scaled-gradient interpretation and EM-type preconditioning	80
5.4	Kernelised Richardson–Lucy deconvolution	80
5.4.1	Kernel image model	80
5.4.2	KRL update	81
5.4.3	Hybrid KRL (HKRL)	81
5.5	Directional-TV regularised RL deconvolution	81
5.5.1	Penalised objective and update	82
5.6	Data and experimental setup	83

5.6.1	Hyperparameter tuning	85
5.7	Results	86
5.7.1	Spheres	86
5.7.2	τ -PET brain	87
5.8	Discussion	88
5.9	Summary	90
6	Introduction to Weighted Directional Total Nuclear Variation	91
6.1	Total Nuclear Variation Revisited	91
6.1.1	Mathematical properties and differentiability	92
6.2	Weighted Directional Total Nuclear Variation (w -dT _{NV})	95
6.2.1	Directional operators from guidance	96
6.2.2	Modality-wise weighting and the full w -dT _{NV} prior	97
6.3	Conclusion	98
7	Preconditioning and Subset Selection for Synergistic Reconstruction	99
7.1	Multi-Bed PET Acquisition	99
7.1.1	Notation	100
7.1.2	Why Not Stitch Independent Reconstructions?	100
7.1.3	Joint Reconstruction	101
7.2	Stochastic MAP Optimisation	102
7.2.1	Common Evaluation Protocol	103
7.3	Subset Selection	104
7.3.1	Data-Subset Organisation	104
7.3.2	Prior-Application Schedules	104
7.3.3	Experiment Design and Work Accounting	105
7.3.4	Results	106
7.4	Prior-Aware Preconditioning	107
7.4.1	Data Curvature	107
7.4.2	Prior Curvature Models	108
7.4.3	Combining Data and Prior Curvature	110

7.4.4	Preconditioner Comparison Methods	110
7.4.5	Preconditioner Comparison	111
7.4.6	Sensitivity to the Initial Step Size	112
7.4.7	Lehmer Combination	113
7.5	Conclusion	114

8 Weighted Directional Total Nuclear Variation for Joint ^{90}Y PET/SPECT

	Reconstruction with CTAC-derived Guidance	116
8.1	Motivation and clinical context	117
8.2	Joint MAP formulation for PET/SPECT	117
8.3	w - d TNV instantiated for ^{90}Y PET/SPECT with CTAC guidance . .	118
8.3.1	Directional guidance from the CTAC-derived μ -map	118
8.3.2	Global modality normalisation as a constant diagonal weight- ing	119
8.3.3	Directional finite-difference operator for PET/SPECT coupling	120
8.3.4	The w - d TNV Jacobian and regulariser (two modalities) . . .	120
8.3.5	Ablation comparators	121
8.4	Optimisation and implementation	122
8.4.1	Block-diagonal preconditioning	123
8.4.2	Comparator: sequential hybrid kernel EM	124
8.4.3	Software implementation	124
8.5	Forward models, corrections, and co-registration	124
8.5.1	PET forward model	124
8.5.2	SPECT forward model	125
8.5.3	Co-registration and warp operators	126
8.6	Data and evaluation protocol	127
8.6.1	Phantom data	127
8.6.2	Clinical data	129
8.7	Results	131
8.7.1	Hyperparameter selection	131
8.7.2	Phantom results	132

8.7.3 Clinical results	135
8.8 Discussion	136
8.9 Conclusion	141
9 Discussion, Conclusion & Future Work	142
Appendices	143
A Software Contributions	143
A.1 simind-python-connector	143
A.1.1 Purpose and Scope	143
A.1.2 Workflow and Interfaces	144
A.1.3 Conventions and Validation	145
A.2 Deep-learned scatter initialisation	146
A.3 Kernelised Richardson–Lucy deconvolution	147
B Convergence Diagnostics for the Long-Run Reference Reconstruction	148
C Automatic Lesion VOI Delineation	151
C.1 Candidate selection	151
C.2 Radial-gradient growth	151
D Colophon	154
References	155

List of Figures

2.1	A positron traverses tissue, losing kinetic energy, before annihilating with an electron, resulting in two (nearly) anti-parallel photons. . . .	37
2.2	Simple diagram of a PET scanner. Left, axial view showing detectors arranged in a ring. Right, sagittal/coronal view showing multiple rings forming the axial FoV.	38
2.3	Possible coincidences under PET electronic collimation. Panel (a) shows a scattered coincidence (the green dashed line is the effective line of response (LoR)) and a true coincidence; panel (b) shows a random coincidence (green dashed lines are the effective LoRs). . .	39
2.4	Left, non-parallel photon rejected by physical collimation. Right, photon parallel to the collimator can be detected, giving a line of response containing the emission source.	40
2.5	Representative diagram of bremsstrahlung production in tissue for an emitted β^- particle.	43
2.6	Bremsstrahlung spectra: (a) emitted spectrum from Geant4 Application for Tomographic Emission (GATE) FastY90 model [19, 20] and (b) simulated detected spectra using AnyScan Trio GATE scanner model [21] with a visible k-edge at ~ 90 keV resulting from photoelectric absorption in the collimator	44
2.7	Tangential profile of measured prompt counts and estimated additive terms (randoms+scatter), with the corresponding corrected prompts for an Yttrium-90 (^{90}Y) PET scan. The data are summed over all views and segments, leaving counts per tangential bin.	45

- 3.1 Visualisation of the Frobenius and nuclear norms (vertical axis) of a matrix X as functions of its (two) singular values (horizontal axis). The Frobenius norm is $\|X\|_F = \sqrt{\sigma_1(X)^2 + \sigma_2(X)^2}$ (red mesh), whereas the nuclear norm is $\|X\|_* = |\sigma_1(X)| + |\sigma_2(X)|$ (black mesh). For visualisation purposes we extend the singular-value axes to negative values. The nuclear norm exhibits sharp grooves along the coordinate axes, encouraging one singular value to vanish (i.e. promoting low-rank structure), while the Frobenius norm is smooth and does not preferentially drive a singular value exactly to zero. This behaviour can also be understood via subgradients: the nuclear-norm subgradient (cyan) points towards rank-1 grooves, preferentially shrinking the smaller singular value, whereas the Frobenius-norm subgradient (red) points towards the origin, shrinking both singular values equally. 59
- 3.2 Demonstration of limit cycle behaviour with an ordered subset gradient descent algorithm. Top row: 3 different simple binomial subfunctions representing subset likelihoods. Bottom row: left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating limit cycle behaviour (green dots show approximate points of limit cycle); right, likelihood value per iteration, demonstrating the limit cycle. 61
- 3.3 Demonstration of reduction in limit cycle behaviour with step size relaxation $\varsigma_k = \varsigma_0 / (1 + 0.05k)$. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating relaxed limit cycle behaviour; right, likelihood value per iteration, demonstrating relaxed limit cycle precession. 62

- 3.4 Demonstration of removal of limit cycle behaviour with an stochastic variance–reduced gradient (SVRG)-type (Ordered VRG) algorithm. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating removal of limit cycle behaviour; right, likelihood value per iteration, demonstrating removal of limit cycle precession 63
- 4.1 Projection-domain behaviour of the preparatory additive-term refinement for a measured NEMA phantom. Coloured curves are profiles of the estimated non-primary term $b^{(k)}$; every displayed iteration is one outer step of the preparatory reconstruct–simulate loop. Profiles are summed over views at an axial position through the spheres; the horizontal coordinate is the tangential detector bin and the vertical coordinate is counts. Full replacement produces large alternating changes within four outer iterations, whereas the under-relaxed profiles remain bounded through the tenth outer iteration. 71
- 4.2 Probability of external Bremsstrahlung photon emission as a function of displacement from the ^{90}Y decay site for four photon-energy intervals. Curves were derived from the fast-simulation data reported by Rault et al. [19]. 72
- 4.3 Detector-plane response standard deviation σ (cm) against source–collimator distance for (a) [39.8, 126.6], (b) [50, 150] and (c) [75, 225] keV. Markers show GATE MC, SIMIND MC, STIR with the geometric response, and STIR with the geometric response plus the image-space effective kernel. The geometric-only STIR projector underestimates the response width; composing it with the image-space Bremsstrahlung kernel substantially reduces the net width discrepancy across the evaluated distances and windows. . . . 75

5.1	Left, ground truth phantom, centre blurred and noisy reconstruction and right MRI guidance image for the simulated sphere phantom. Units are arbitrary.	83
5.2	Left, reconstructed τ -PET brain, right registered MRI guidance image. Units are arbitrary.	84
5.3	Figures showing central slice of reconstructions with best NRMSE. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.	86
5.4	Figures showing iteration versus global NRMSE for the best performing reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.	88
5.5	Figures showing RC versus CoV for all reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image. KRL and HKRL are visually indistinguishable in these plots.	89
5.6	Figures showing visually best reconstructions for the different reconstructions.	89
7.1	Qualitative comparison of separately reconstructed and fused PET bed positions with a joint multi-bed reconstruction using CT-guided edge-preserving regularisation. The shaded region marks the bed overlap and the cyan lines mark the displayed profiles. This example illustrates overlap-region behaviour only; no quantitative comparison was performed.	101

- 7.2 Convergence for the six combinations of subset organisation (paired, separate) and prior-application schedule (folded, sampled, once per epoch) using the MM local-diagonal curvature preconditioner. Lines show the mean per-run NRMSE over five stochastic realisations and shading shows the pointwise minimum–maximum over the first 20 matched data passes. In the legend, prior folded, prior/half, and prior once/epoch denote the folded, sampled, and once-per-epoch schedules; every schedule applies on average one complete prior contribution per matched data pass. 106
- 7.3 Convergence of data-only BSREM and the four prior-aware preconditioners in Table 7.1. All methods use the same objective and long-run reference within each modality. Lines show the mean per-run NRMSE over five stochastic realisations, and shading shows the pointwise minimum–maximum over the first 25 matched data passes. The PET vertical axis spans the full range reached by the data-only trajectory. 112
- 7.4 Sensitivity to the common initial step size for the displayed preconditioners. Solid curves use $\zeta^{(0)} = 1$ and dashed curves use $\zeta^{(0)} = 2$. Lines and envelopes again show the five-run mean and pointwise minimum–maximum over 25 matched data passes. 113
- 7.5 Parallel-sum and Lehmer combinations for the Lewis–Sendov block-diagonal prior model. This experiment changes only the combination rule; lines and shading show the five-run mean and pointwise minimum–maximum over 20 matched data passes. 114
- 8.1 NEMA phantom analysis regions. Left: warm-background (BG), lung insert (LI), and high-count (HC) VOIs. Right: CTAC-derived attenuation map showing sphere edges, the lung insert, and the under-filled sphere excluded from the HC VOI. 128

- 8.2 Training-patient VOIs. The five identified lesions and the background reference region are shown on the vendor PET reconstruction with registered attenuation maps. Top row: lesions 1–3. Bottom row: lesions 4–5 and background. 131
- 8.3 Training-patient hyperparameter sweep. Impact of varying PET weighting while holding SPECT regularisation constant (top row) and varying SPECT weighting while holding PET regularisation constant (bottom row) on TBR and CoV_{bkg} for lesions 1 and 2. The first and third columns show w -dT_{NTV}; the second and fourth show w -T_{NTV}. 133
- 8.4 Visual comparison of NEMA phantom reconstructions for a randomly chosen bootstrap. From left to right: w -dT_{NTV}, dTV, w -T_{NTV}, SHKEM (subiteration 27). PET reconstructions are shown on the top row and SPECT on the bottom row (SPECT guidance for SHKEM). Images are displayed in reconstruction intensity units, proportional to activity concentration. 133
- 8.5 Recovery coefficient (RC) for NEMA phantom spheres for w -dT_{NTV}, dTV, w -T_{NTV} and SHKEM (subiteration 27). Also shown are the equivalent RCs for a high-count, long-OSEM reconstruction. The grey line indicates perfect recovery. Statistical significance is reported for paired comparisons versus w -dT_{NTV} (Benjamini–Hochberg adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ 134
- 8.6 Mean voxel-wise coefficient of variation (CoV) across 20 bootstrapped PET realisations within each VOI. Statistical significance is reported for paired comparisons versus w -dT_{NTV} (BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ 135

- 8.7 Visual comparison of reconstruction methods for training patient SIRT3. Columns show w -dTNV, dTV, w -TNV and SHKEM (subiteration 27). The upper two quadrants show PET reconstructions; the lower two show SPECT reconstructions or guidance images. Images are displayed in reconstruction intensity units, proportional to activity concentration. 136
- 8.8 Clinical cohort summary: lesion-wise mean TBR (left) and patient-wise background CoV (right). Statistical significance is reported for comparisons versus w -dTNV(BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ 137
- 8.9 Training-patient CTAC-derived guidance. The CTAC-derived attenuation map μ (top row; cm^{-1}) and magnitude of the w -dTNV directional weighting field $\|\xi\|$ (bottom row) are shown for lesions 1 and 2. CTAC streak artefacts are visible in μ and are reproduced in $\|\xi\|$, providing a pathway for artefact transfer into the emission reconstruction. 139
- 8.10 NEMA phantom, central axial slice. CTAC-derived attenuation map μ (left; cm^{-1}) and magnitude of the w -dTNV directional weighting field $\|\xi\|$ (right). With a high-quality CTAC free of prominent streak artefacts, $\|\xi\|$ concentrates around true object boundaries and insert edges, enabling stable μ -map-guided directional smoothing. 140
- B.1 Long-run objective diagnostics. The solid curve is $\mathcal{O}_k - \mathcal{O}^*$ and the dashed curve is the relaxed step size. The linear view shows that most of the observed objective decrease occurs early, while the logarithmic view resolves the residual decrease over the complete 1000-epoch run. 149
- B.2 Relative PET iterate velocity, $\|x^{(k)} - x^{(k-1)}\|/\|x^{(k)}\|$, and relaxed step size during the long-run reference reconstruction. The final velocity is 7.285×10^{-8} per epoch while the step size remains 0.2174. 149

- B.3 Final PET and SPECT iterates of the long-run reference reconstruction, shown on a matched slice with independent colour bars. This pair is the reference x^* against which the NRMSE curves of Chapter 7 are measured; it is the long-run baseline rather than one of the candidate preconditioner settings compared there. 150
- C.1 Two-dimensional illustration of radial-gradient lesion delineation. Left: radial searches extend from the seed voxel to candidate boundary locations. Right: the corresponding image-gradient magnitude used to identify the lesion edge. This schematic illustrates the search principle; patient masks were generated in the image volume. 152
- C.2 Illustrative safeguards used to bound the radial-gradient search. The curves are schematic one-dimensional profiles rather than patient measurements. 153

List of Tables

5.1	Hyperparameter sweeps for kernel-based deconvolution	86
5.2	Tables summarising the hyperparameters and metric values of the best (measured by NRMSE) reconstructions of the various methods for (top) a uniform initialisation of ones and (bottom) the blurred PET image.	87
7.1	Preconditioner models used in the principal comparison. The four prior-aware models use the inverse-sum combination in Equation (7.23).	111
8.1	Selected regularisation weights (tuned on a single training patient) used for all phantom and clinical experiments.	132
8.2	Effect ratios $\exp(\Delta_{c,v})$ (comparator vs. baseline w -dTNN) with 95% confidence intervals and BH-adjusted p -values, reported by sphere diameter v . Values below one indicate lower recovery coefficient than w -dTNN.	134
8.3	Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p -values for mean tumour-to-background ratio (TBR), comparing each method to w -dTNN. Values below one indicate lower TBR than w -dTNN.	136
8.4	Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p -values for background CoV, comparing each method to w -dTNN. Values above one indicate higher background CoV than w -dTNN.	137

A.1	Main <code>simind-python-connector</code> interfaces relevant to thesis workflows.	145
B.1	Stationarity diagnostics for the 1000-epoch reference reconstruction.	149

List of Abbreviations

w-dTNV weighted directional total nuclear variation.

⁹⁰Y Yttrium-90.

⁹⁰Zr Zirconium-90.

BH Benjamini–Hochberg.

BSREM block-sequential regularised expectation maximisation.

CDR collimator–detector response.

CoV coefficient of variation.

CT computed tomography.

CTAC computed tomography attenuation correction.

DTV directional total variation.

EM expectation–maximisation.

ET emission tomography.

FDG ¹⁸F-fluoro-2-deoxyglucose.

FoV field of view.

FWHM full width at half maximum.

GATE Geant4 Application for Tomographic Emission.

HCC hepatocellular carcinoma.

HKEM hybrid kernelised expectation maximisation.

HKRL Hybrid Kernelised Richardson–Lucy.

HU Hounsfield units.

IEC International Electrotechnical Commission.

IPP internal pair production.

IRLS iteratively reweighted least squares.

JTV joint total variation.

KEM kernelised expectation maximisation.

KRL Kernelised Richardson–Lucy.

LoR line of response.

MAA macroaggregated albumin.

MAP maximum a posteriori.

ML maximum-likelihood.

MLEM maximum-likelihood expectation–maximisation.

MM majorisation–minimisation.

MR magnetic resonance.

MRI magnetic resonance imaging.

NEMA National Electrical Manufacturers Association.

- NRMSE** normalised root-mean-square error.
- OSEM** ordered subsets expectation–maximisation.
- PET** positron emission tomography.
- PLS** parallel level sets.
- PSF** point spread function.
- PVC** partial volume correction.
- RC** recovery coefficient.
- RDP** relative difference prior.
- RL** Richardson–Lucy.
- RL–DTV** Richardson–Lucy with Directional Total Variation.
- SAGA** stochastic average gradient amélioré.
- SHKEM** sequential hybrid kernelised expectation maximisation.
- SIMIND** Simulation of Imaging Nuclear Detectors.
- SIRF** Synergistic Image Reconstruction Framework.
- SIRT** selective internal radiotherapy.
- SPD** symmetric positive-definite.
- SPECT** single photon emission computed tomography.
- STIR** Software for Tomographic Image Reconstruction.
- SVD** singular value decomposition.
- SVRG** stochastic variance–reduced gradient.
- TARE** transarterial radioembolisation.

TBR tumour-to-background ratio.

TNV total nuclear variation.

ToF time of flight.

TV total variation.

VOI volume of interest.

Notation

x Emission activity image (voxel intensities), $x \in \mathbb{R}_+^J$.

y Measured projection data (counts), $y \in \mathbb{N}^I$.

$\bar{y}(x)$ Expected measured data under the forward model, $\bar{y}(x) = Ax + b$.

A System (forward-projection) matrix, $A = [a_{ij}] \in \mathbb{R}_+^{I \times J}$.

b Additive term (scatter, randoms, and other background contributions).

C Image-domain point-spread-function convolution (blurring) operator used in deconvolution.

μ Linear attenuation coefficient / CTAC-derived attenuation (μ -)map.

h Source-to-collimator distance (SPECT depth-dependent resolution).

I Number of measurement bins (lines of response / projection bins).

J (J_m) Number of voxels in the native / modality- m image space.

N Number of voxels on the shared reconstruction grid.

M Number of coupled modalities.

d Number of directional finite-difference channels (gradient dimension).

i Measurement-bin index.

j Voxel index (native grid).

- k Iteration / epoch index (written as a superscript, $x^{(k)}$).
- m Modality index.
- n Voxel index (shared reconstruction grid).
- s Data-subset index.
- v Sphere / volume-of-interest index.
- c Reconstruction-method (comparison) index; c_0 denotes the reference method.
- bed PET bed-position index, $\text{bed} = 1, \dots, N_{\text{bed}}$.
- bs Bootstrap-realisation index, $\text{bs} = 1, \dots, N_{\text{bs}}$.
- D Discrete (directional) finite-difference operator, $(Dx)_n \in \mathbb{R}^d$.
- div Discrete divergence, the negative adjoint of D .
- ∇ Gradient of a functional (e.g. $\nabla_x \mathcal{L}$, $\nabla_{\mathcal{G}} \phi$).
- $\Pi_{m,n}$ Anisotropic directional projection operator built from the guidance field.
- W_m Registration / resampling operator from native grid to shared grid; W_m^* its adjoint.
- K Kernel matrix (kernelised image model $x = K\alpha$).
- α Kernel coefficient image.
- $\mathcal{P}$ Preconditioner (diagonal or voxel-wise block).
- $\mathbb{I}$ Identity matrix ($\mathbb{I}_d$ when the dimension is made explicit).
- $\mathbf{1}$ All-ones vector.
- R_{bed} Restriction from the combined PET image to a bed position's axial support.
- $\mathcal{D}$ Data-fidelity term (Poisson negative log-likelihood).
- $\mathcal{L}$ Negative Poisson log-likelihood (up to an additive constant).

- $\mathcal{R}$ Regularisation functional (prior).
- $\mathcal{O}$ Full optimisation objective, $\mathcal{O} = \mathcal{D} + \mathcal{R} + \iota_{\mathbb{R}_+}$.
- $\mathcal{C}$ Feasible set (typically the non-negative orthant).
- $\iota_{\mathbb{R}_+}$ Indicator function of the non-negative orthant.
- β Regularisation strength (prior weight).
- $\rho_{m,n}$ Modality weight; collected in the diagonal matrix $B_n = \text{diag}(\rho_{1,n}, \rho_{2,n})$.
- γ, γ_m RDP edge-preservation parameter γ ; modality-specific directional-guidance strength $\gamma_m \in [0, 1]$.
- η Direction-field stabilisation parameter (distinct from the relaxation parameter η_{relax}).
- ε Numerical stabiliser in the RDP and Charbonnier / smoothed-TV parameter (subscripted variants denote other local stabilisers).
- ξ_n Normalised guidance direction field.
- u Anatomical guidance image (for the ^{90}Y study, $u = \mu$).
- $z_m, \mathbf{z}$ Modality- m image on the shared grid; $\mathbf{z} = (z_1, \dots, z_M)$ the stacked images.
- $\mathcal{G}_n$ Voxel-wise multi-modality Jacobian (columns are modality gradients).
- $\tilde{\mathcal{G}}_n$ Weighted directional Jacobian, $\tilde{\mathcal{G}}_n = \mathcal{G}_n^{\text{dir}} B_n$.
- ϕ Smoothed spectral potential applied to the Jacobian, $\phi(Y) = \sum_{\ell} g(\sigma_{\ell}(Y))$.
- g Charbonnier smoothing function of a singular value.
- σ_{ℓ} ℓ -th singular value.
- $\|\cdot\|_*$ Nuclear norm (sum of singular values).
- V_{vox} Voxel volume.

- $\varsigma^{(k)}$ (ς_k) Step size at iteration k (relaxed / diminishing over iterations).
- $\tilde{x}, \tilde{z}$ Stored snapshot iterate used by the variance-reduced estimators (SVRG/SAGA).
- $g^{(k)}$ (Variance-reduced) stochastic gradient estimator at iteration k .
- $H_{\mathcal{R}}$ Hessian of the prior (used for prior-aware preconditioning).
- p Matrix-Lehmer combination order ($p = 0$ gives the parallel sum).
- S Number of data subsets.
- N_d Number of stochastic data functions in a reconstruction.
- s_m Modality- m sensitivity image on the shared reconstruction grid.
- η_{relax} Step-size relaxation / decay parameter.
- ε_x Small positive image-domain stabiliser used in data preconditioning.
- $\tau_{m,n}$ Finite-difference participation factor in the diagonal prior-curvature approximation.
- $\lambda_{\min}$ Eigenvalue floor applied to preconditioner operands before matrix inversion.
- $\hat{x}$ Reconstruction estimate.
- x^* Digital ground truth or long-run reference image.
- $\mathcal{S}_v$ Set of voxel indices belonging to sphere v .
- $\mathcal{B}$ Background region of interest.
- $\mathcal{V}$ Volume of interest (VOI).
- $\bar{x}$ Regional / inter-bootstrap sample mean.
- SD Sample standard deviation.
- $\Delta_x, \Delta_y, \Delta_z$ Voxel dimensions; Δ_ℓ is the length of finite-difference offset ℓ .

Chapter 1

Introduction

Yttrium-90 (^{90}Y) selective internal radiotherapy (SIRT), also referred to as transarterial radioembolisation (TARE), is an established treatment for primary and secondary liver malignancies [1]. In this procedure, ^{90}Y -labelled microspheres are administered into the hepatic artery to preferentially irradiate tumour tissue while sparing healthy liver parenchyma. The administered activities are high (typically of the order 2 GBq), and clinically relevant absorbed-dose heterogeneity can arise at multiple spatial scales due to vascular delivery, microsphere clustering, and patient-specific hepatic anatomy [2]. Tumour doses of clinical interest often exceed 200 Gy [3], and recent evidence has shown potential benefit from additional subsequent treatment. Additional treatments could lead to more dose to healthy liver tissue without proper treatment planning and verification [4]. As a consequence, post-therapy quantitative imaging and dosimetry are central to treatment verification, outcome modelling, and the optimisation of any subsequent therapy.

Despite its importance, routine post-SIRT dosimetry is not universally performed in clinical practice, primarily due to practical workflow constraints and limitations in quantitative accuracy [5, 6]. Post-therapy imaging of ^{90}Y is inherently challenging: image quality is often degraded by low effective count statistics and by physics that is less favourable than standard diagnostic radionuclide imaging, leading to increased noise and modelling error [5]. More broadly, emission tomography is vulnerable to substantial degradation from Poisson counting statistics as well as inaccuracies in system modelling and hardware limitations [7]. Improving

reconstruction quality in this low-count, model-mismatch regime is therefore a key enabling step for accurate absorbed-dose estimation and for robust patient-specific outcome prediction [8].

In post-SIRT imaging, the two clinically relevant emission modalities are complementary. ^{90}Y positron emission tomography (PET) offers comparatively high spatial resolution, but is severely count-limited because of the extremely low internal pair-production branch. Bremsstrahlung single photon emission computed tomography (SPECT), by contrast, provides higher count statistics but suffers from poor spatial resolution and more severe modelling challenges, particularly for scatter, septal penetration, and additive corrections [5]. In contemporary workflows, these emission images are routinely paired with computed tomography (CT), which provides attenuation correction and anatomical context. This naturally motivates reconstruction approaches in which information is shared across modalities, with CT contributing structural guidance, and PET and SPECT contributing complementary functional information. Synergistic methods, where such information is incorporated during reconstruction rather than only after image formation, have gained prominence in related hybrid imaging contexts [9]. Within post- ^{90}Y SIRT imaging, these approaches have the potential to improve lesion-level quantification and absorbed-dose estimation without increasing acquisition time or administered activity.

The central aim of this thesis is therefore to develop and evaluate a synergistic reconstruction framework tailored to post- ^{90}Y SIRT, with a particular focus on improving the reliability of quantitative emission imaging and therefore downstream dosimetry under clinically realistic count levels and modelling limitations. To this end, the thesis develops practical improvements to the bremsstrahlung SPECT system model, introduces a weighted directional total nuclear variation prior for joint PET/SPECT reconstruction with CT-derived directional guidance, and investigates optimisation and preconditioning strategies for this coupled penalised-likelihood problem.

The thesis is organised as follows:

- **Chapter 1** motivates synergistic post-SIRT imaging and outlines the thesis contributions.
- **Chapter 2** introduces the physics and measurement models of PET, SPECT, and CT, with emphasis on post- ^{90}Y imaging.
- **Chapter 3** formulates penalised-likelihood emission reconstruction and reviews reconstruction priors and optimisation algorithms.
- **Chapter 4** develops practical improvements to the ^{90}Y bremsstrahlung SPECT system model.
- **Chapter 5** studies anatomy-informed regularisation in a controlled post-reconstruction deconvolution setting.
- **Chapter 6** introduces the weighted directional total nuclear variation prior and its differentiable formulation.
- **Chapter 7** investigates the optimisation of w -dT_{NTV}, including multi-bed PET reconstruction, subset organisation, and prior-aware preconditioning.
- **Chapter 8** evaluates w -dT_{NTV} for joint post- ^{90}Y PET/SPECT reconstruction with CTAC-derived guidance.
- **Chapter 9** discusses the findings, limitations, and directions for future work, and concludes the thesis.
- **The Appendix** documents additional software and reproducibility material that supports the thesis workflow.

Chapter 2

Physics and Measurement Models in Emission Tomography

This chapter introduces the physical principles and measurement models underpinning emission tomography, with emphasis on the effects that dominate reconstruction quality and quantitative reliability. In both positron emission tomography (PET) and single photon emission computed tomography (SPECT), the reconstruction task is to infer a non-negative radiotracer activity distribution from projection data generated by stochastic radioactive decay and corrupted by attenuation, scatter, and blurring effects. These are naturally captured by a forward model that maps the image to expected measurement counts with additive components representing background contributions.

We first summarise the detection physics of PET and SPECT and present convenient factorisations of the system matrix that separate geometry, attenuation, and detector effects. We then focus on the particularly challenging case of post-SIRT ^{90}Y imaging, where quantification is limited by extreme count starvation in internal pair production (IPP) PET and by spectral and resolution degradations in bremsstrahlung SPECT. Finally, we describe the role of computed tomography (CT) in this thesis as the source of attenuation information required for physically accurate emission modelling and a high-resolution anatomical reference for reconstruction regularisation in low-count regimes.

2.1 Principles of Emission Tomography

PET fundamentally differs from anatomical modalities such as CT. Rather than mapping the structural properties, emission tomography (ET) measures the spatio-temporal distribution of a radiolabelled substance introduced into the physiological system. A representative example is ^{18}F -fluoro-2-deoxyglucose (FDG)[10]. FDG is a glucose-analogue, meaning energy-hungry cells take up an increased amount of the molecules, and therefore radionuclide. Cells are unable to metabolise FDG, meaning it accumulates. The distribution of FDG can be imaged using a PET scanner, giving a functional view of glucose metabolism and highlighting any unusual uptake caused by overactive cells such as cancer.

While the instrumentation differs between SPECT and PET, both involve a similar inverse problem: estimating the activity distribution from projection measurements governed by Poisson counting statistics, attenuation, scatter, and detector resolution. The continuous activity distribution is represented on a finite voxel grid as a non-negative image vector $x \in \mathbb{R}_+^J$. The primary contribution to the expected measurements is modelled using the non-negative system matrix $A \in \mathbb{R}_+^{I \times J}$. Including additive background contributions gives the affine forward model

$$\bar{y} = Ax + b, \quad (2.1)$$

where $\bar{y} = \mathbb{E}[y] \in \mathbb{R}_+^I$ denotes the total expected measurement counts and $b \in \mathbb{R}_+^I$ contains contributions not described by the primary emission model. The precise discretisation and statistical measurement model are introduced later.

2.2 Positron Emission Tomography (PET)

PET primarily uses neutron-deficient isotopes (see Section 2.4 for a counter-example) that stabilise via β^+ decay. Within the nucleus, a proton converts into a neutron, emitting a positron (e^+) and an electron neutrino (ν_e) to conserve lepton number:

$$p \rightarrow n + e^+ + \nu_e. \quad (2.2)$$

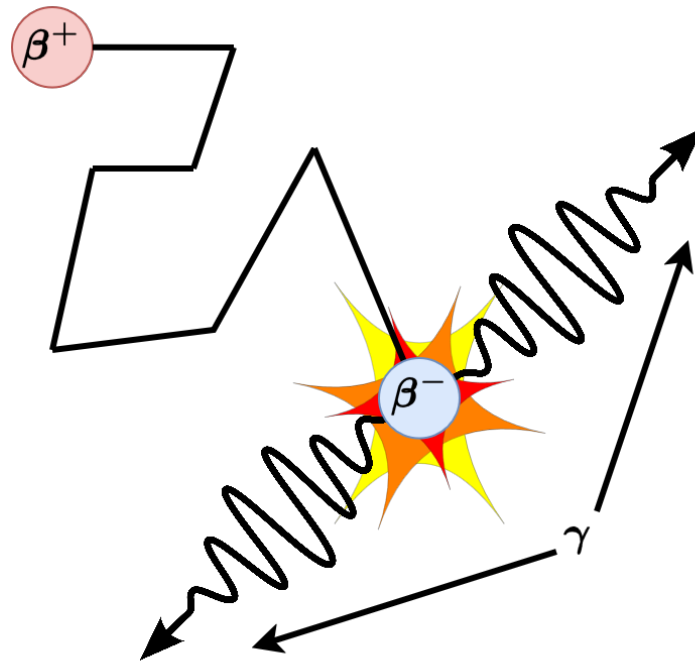


Figure 2.1: A positron traverses tissue, losing kinetic energy, before annihilating with an electron, resulting in two (nearly) anti-parallel photons.

The decay energy is shared between the positron and the neutrino, resulting in a continuous positron energy spectrum. This positron traverses the tissue losing kinetic energy via Coulomb interactions (a distance known as the positron range) before thermalising. This range imposes an effective blurring limit on spatial resolution, with an effective positron range of approximately 0.5–0.7 mm for ^{18}F and several millimetres for high-energy emitters such as ^{82}Rb [11].

Upon thermalisation, the positron interacts with an electron to form positronium, which rapidly decays via annihilation:



The rest mass energy of the pair ($2 \times 511 \text{ keV}$) is converted into two¹ (near) anti-parallel² photons (Figure 2.1).

¹Positron/electron pairs can also annihilate to produce more than two photons but this is much less likely ($p(3\gamma)/p(2\gamma) \approx 1/378$) [12]

²In order to conserve momentum, a certain a-collinearity will exist between the photons, introducing additional blurring [13].

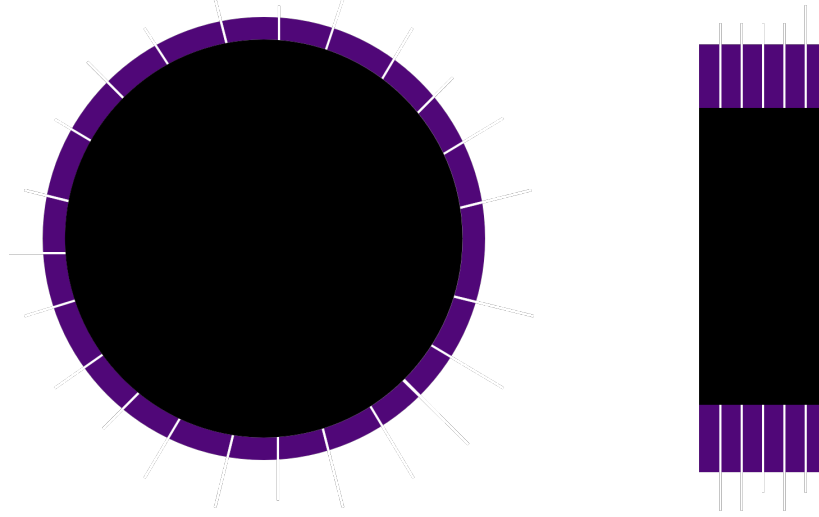


Figure 2.2: Simple diagram of a PET scanner. Left, axial view showing detectors arranged in a ring. Right, sagittal/coronal view showing multiple rings forming the axial FoV.

2.2.1 Coincidence Detection and Data Acquisition

PET detectors are typically assembled into modular detector blocks, which are then arranged circumferentially to form a ring around the patient. Modern clinical systems comprise multiple such rings stacked along the axial direction, increasing axial coverage and sensitivity by enlarging the axial field of view (FoV). Figure 2.2 illustrates this geometry showing a single ring in the axial view, and multiple rings in the sagittal view.

PET imaging relies on a technique known as electronic collimation. A valid event is recorded only if two detectors register photons within a coincidence timing window 2τ (typically 6–12 ns) [11]. The measured coincidences (or counts) along a LoR, connecting detectors 1 and 2, denoted as $y_{i,2}$, is a superposition of three components:

$$y_{i,2} = t_{i,2} + s_{i,2} + r_{i,2} \quad (2.4)$$

where t_{ij} represents true coincidences (Figure 2.3a, right), $s_{i,2}$ represents scatter (where one or both photons have undergone at least one Compton interaction; Figure 2.3a, left), and $r_{i,2}$ represents random coincidences (Figure 2.3b).

Random coincidences occur when two unrelated annihilation events are detected

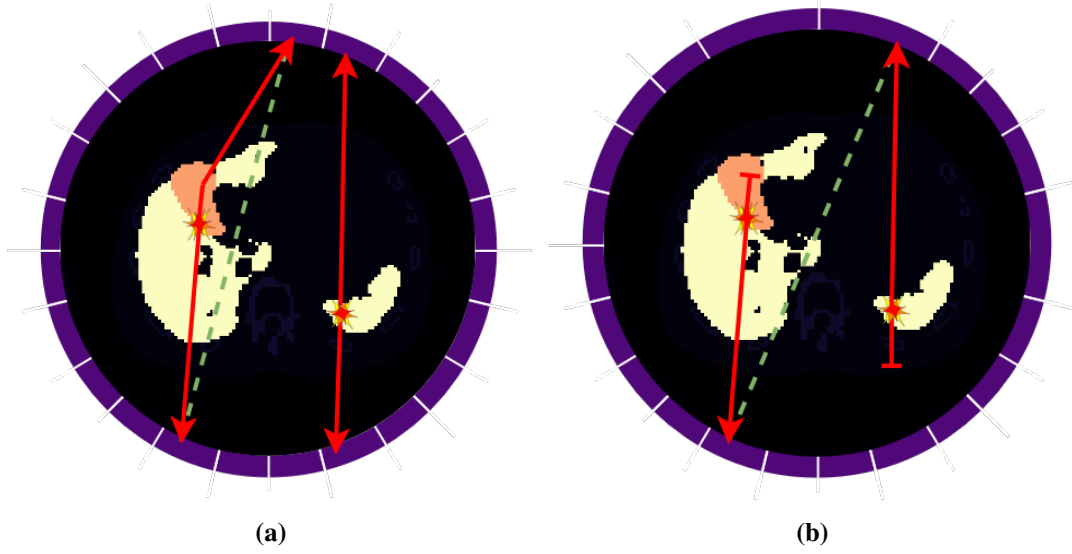


Figure 2.3: Possible coincidences under PET electronic collimation. Panel (a) shows a scattered coincidence (the green dashed line is the effective LoR) and a true coincidence; panel (b) shows a random coincidence (green dashed lines are the effective LoRs).

within the timing window. The rate of randoms on a specific LoR is governed by the singles rates of the individual detectors (R_1, R_2), often approximated as:

$$r_{i,2} = 2\tau R_1 R_2. \quad (2.5)$$

2.2.2 The PET model

In PET, the system matrix may be factorised as

$$A = A_{\text{det,sens}} A_{\text{det,blur}} A_{\text{att}} A_{\text{geom}} A_{\text{ToF}} A_{\text{positron}}. \quad (2.6)$$

Here, A_{positron} models positron-range blurring, A_{ToF} models the timing localisation accuracy for time of flight (ToF) PET (and is omitted for non-ToF PET), A_{geom} is the geometric projection operator mapping voxels to lines of response in the absence of attenuation, A_{att} is a diagonal operator containing attenuation factors along each line of response, $A_{\text{det,blur}}$ models detector-related resolution effects in reporting the true line-of-response positions, and $A_{\text{det,sens}}$ is a diagonal sensitivity factor incorporating multiplicative effects such as detector efficiencies and normalisation [14]. The

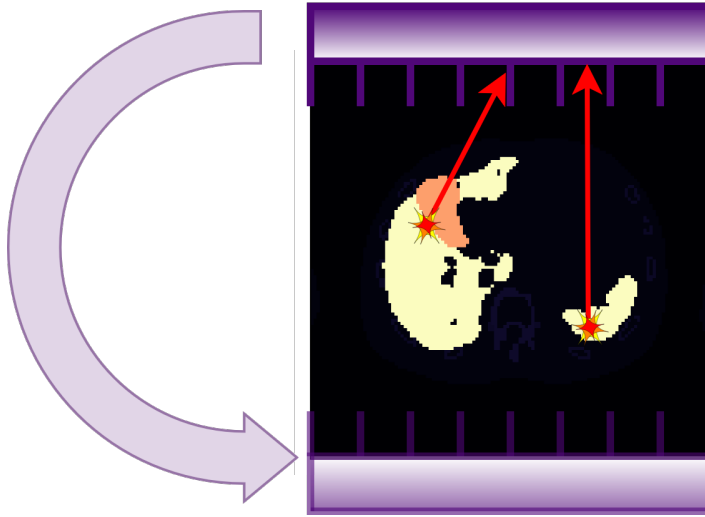


Figure 2.4: Left, non-parallel photon rejected by physical collimation. Right, photon parallel to the collimator can be detected, giving a line of response containing the emission source.

additive term is commonly written as $b = s + r$, corresponding to scatter and randoms, respectively.

2.3 Single Photon Emission Computed Tomography (SPECT)

SPECT imaging relies on single gamma emissions. As there is no coincident pair to define the LoR, physical collimation is required. A lead collimator, typically a parallel-hole geometry, allows only photons travelling through the collimator holes. A single (or multiple, most often a dual-head system) collimated gamma camera is rotated around the patient (Figure 2.4).

The sensitivity (g) of a parallel-hole collimator depends on the hole diameter (d), septal thickness (t), and the effective hole length (L_e). A commonly used geometric-efficiency scaling is

$$g \propto \frac{d^4}{L_e^2 (d + t)^2}, \quad (2.7)$$

which makes explicit the trade-off between resolution and sensitivity: reducing d improves geometric resolution but decreases sensitivity [11]. In practice, the

geometric efficiency of a (high-resolution) parallel-hole collimator can be very low (often $\lesssim 0.05\%$) [15]. In the context of bremsstrahlung imaging (Section 2.4.2), this low efficiency is compounded by the continuous energy spectrum of the emissions, which limits the effectiveness of standard photopeak-based energy windowing for scatter rejection.

2.3.1 The SPECT model

For SPECT, a convenient factorisation is

$$A = A_{\text{det,sens}} A_{\text{det,blur}} A_{\text{geom,att}} A_{\text{bremss}}, \quad (2.8)$$

where $A_{\text{det,sens}}$ collects multiplicative sensitivity factors, $A_{\text{det,blur}}$ represents intrinsic resolution effects within the gamma camera, and $A_{\text{geom,att}}$ is the geometric projection operator that encapsulates collimator effects (including depth-dependent resolution) together with depth- and view-dependent attenuation. The operator A_{bremss} is specific to bremsstrahlung SPECT and models the additional blurring associated with bremsstrahlung range (discussed in Section 2.4.2).

For a parallel-hole collimator, septal penetration can be incorporated via an energy-dependent effective hole length,

$$L_e(E) = l - \frac{2}{\mu(E)}, \quad (2.9)$$

where l is the physical hole length and $\mu(E)$ is the linear attenuation coefficient of the collimator material at photon energy E [11]. The combined intrinsic detector blur and collimator blur is commonly approximated by a depth-dependent Gaussian point-spread function embedded within $A_{\text{geom,att}}$, with standard deviation $\sigma(h, E)$ increasing with source-to-collimator distance h (and depending on E through penetration).

2.4 Yttrium-90 Radioembolisation (SIRT)

SIRT uses ^{90}Y microspheres for the locoregional treatment of hepatic malignancies. It is an established treatment for unresectable hepatocellular carcinoma (HCC) and

liver metastases [1], recommended by NICE for unresectable HCC and used for neuroendocrine and colorectal liver metastases in the UK [16, 17]. The reconstruction of post-SIRT microsphere distribution presents a severe ill-posed inverse problem, as ^{90}Y is designed for therapy, not imaging.

2.4.1 Decay Characteristics

^{90}Y is an almost pure β^- emitter decaying to the stable ground state of Zirconium-90 (^{90}Zr) with a half-life of 64.1 hours:



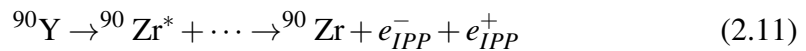
The electrons have a high endpoint energy of 2.28 MeV. The primary challenge for quantitative imaging is the absence of primary gamma photons. Imaging must therefore exploit two secondary, low-probability physical interactions.

2.4.2 Bremsstrahlung SPECT

As high-energy β^- particles decelerate in tissue through Coulomb interactions with charged particles, they emit Bremsstrahlung photons. These photons can be imaged with SPECT systems. However, the continuous spectrum of bremsstrahlung photons (Figure 2.6a) hugely reduces the suitability of traditional energy-window scatter-correction methods, while bremsstrahlung range blurring (Figure 2.5) reduces resolution. Consequently, bremsstrahlung SPECT is often characterised by poor spatial resolution and limited quantitative accuracy [18].

2.4.3 Internal Pair Production PET

A minor decay branch (31.86×10^{-6}) allows the nucleus to transition to an excited 0^+ state of ^{90}Zr , which de-excites via internal pair production (IPP):



The resulting positron annihilates to produce standard 511 keV photon pairs [22]. While this enables quantitative PET imaging, the acquisition is “count-starved” due to the extremely low branching ratio and noise-dominated due to the Bremsstrahlung

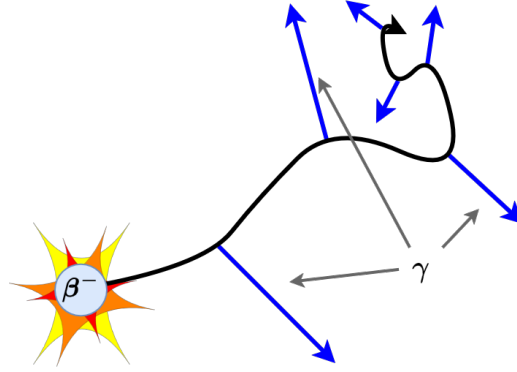


Figure 2.5: Representative diagram of bremsstrahlung production in tissue for an emitted β^- particle.

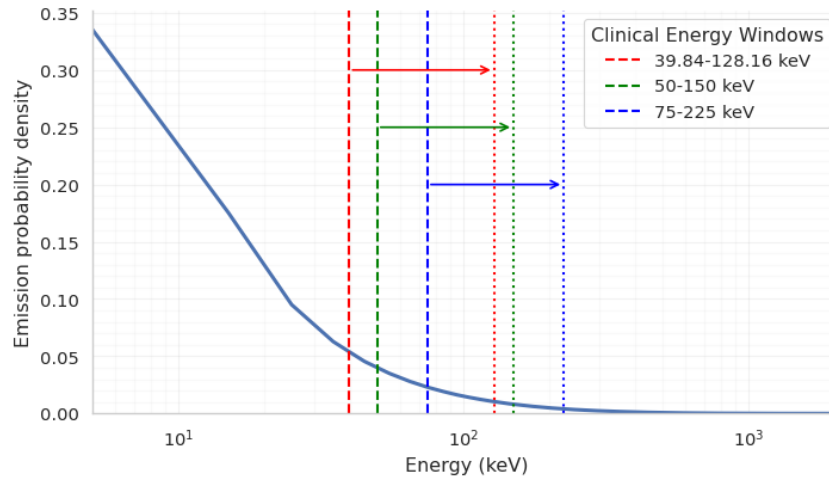
background (Figure 2.7).

2.5 X-ray CT

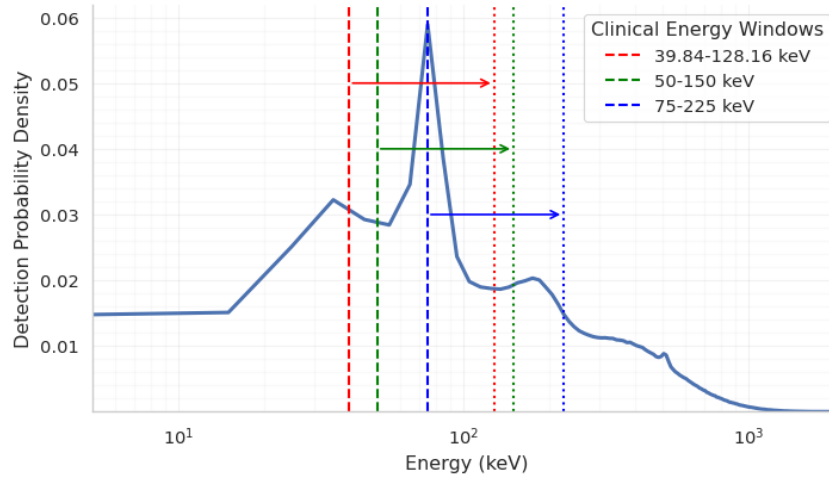
In this work, CT is used primarily to provide anatomical context and attenuation information for emission tomography reconstruction. X-ray CT measures the transmitted intensity of a photon beam after passage through an object or patient. In an ideal monoenergetic setting, transmission follows the Beer–Lambert law, so that the log-normalised measurements are proportional to line integrals of the linear attenuation coefficient. In practice, polychromaticity introduces effects such as beam hardening, which are mitigated by scanner calibration and correction steps to yield images that approximate an attenuation map.

Reconstructed CT images are typically reported in Hounsfield units (HU), calibrated such that water is 0 HU and air is -1000 HU. The high spatial resolution of CT provides structurally reliable edge information (e.g. liver boundaries and lesion margins) that can be used to guide regularisation in the ill-posed reconstruction of post-SIRT ^{90}Y PET and SPECT.

In addition, CT is routinely used for computed tomography attenuation correction (CTAC) in both PET and SPECT. The reconstructed HU image is first converted to an estimate of the linear attenuation coefficient map $\mu(E) \in \mathbb{R}_+^J$ at the relevant photon energy E (e.g. $E = 511$ keV for PET, or the photopeak energy for SPECT) [23, 24].



(a) Bremsstrahlung emission spectrum for ^{90}Y .



(b) Simulated detected spectrum (10 cm, AnyScan).

Figure 2.6: Bremsstrahlung spectra: (a) emitted spectrum from GATE FastY90 model [19, 20] and (b) simulated detected spectra using AnyScan Trio GATE scanner model [21] with a visible k-edge at ~ 90 keV resulting from photoelectric absorption in the collimator

Misregistration (e.g. respiratory motion), metal artefacts, and beam-hardening residuals can all propagate into CTAC and therefore bias quantitative PET/SPECT activity estimates [25, 26].

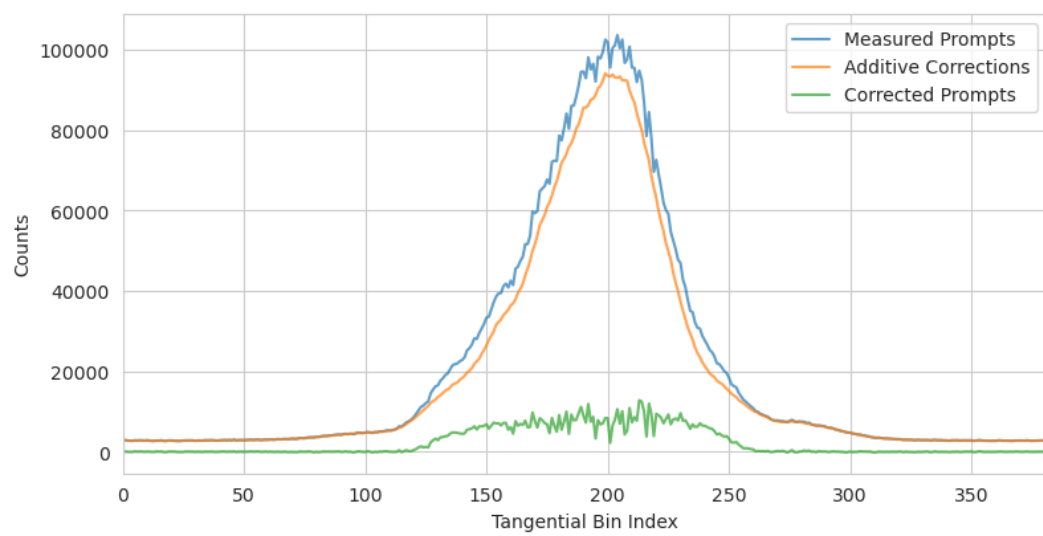


Figure 2.7: Tangential profile of measured prompt counts and estimated additive terms (randoms+scatter), with the corresponding corrected prompts for an ^{90}Y PET scan. The data are summed over all views and segments, leaving counts per tangential bin.

Chapter 3

Penalised-Likelihood Reconstruction: Priors and Optimisation Algorithms

Emission tomography reconstruction is an ill-posed inverse problem: the unknown activity must be inferred from noisy Poisson count data through a highly smoothing and poorly conditioned forward operator. While maximum-likelihood (ML) approaches remain the clinical workhorse [27, 28], their practical success relies heavily on early stopping, which acts as an implicit regulariser but can introduce iteration-dependent bias and compromise quantification [29, 30]. For quantitative, low-count, and multi-modality settings, precisely those that arise in modern PET/SPECT practice, explicit regularisation and fast, robust optimisation become essential.

This chapter reviews the modelling and algorithmic foundations that underpin the remainder of the thesis. We first formalise the Poisson statistical model and the penalised-likelihood reconstruction objective (often termed maximum a posteriori (MAP)), highlighting where ill-conditioning and noise amplification enter the problem. We then present some regularisation strategies used in emission tomography: from classical quadratic smoothness priors and total variation (TV) [31], through clinically deployed edge-preserving models such as the relative difference prior (RDP) [32], to anatomically guided methods that use aligned structural images (e.g. Bowsher [33], parallel level sets (PLS) [34], and kernelised representations [35, 36]. We then focus on synergistic regularisation, where multiple co-registered modalities

are coupled during reconstruction, and motivate Jacobian-based vectorial priors such as joint total variation (JTV) and total nuclear variation (TNV) [37, 38, 39], which will form the basis of the synergistic models developed later in the thesis.

Finally, we review optimisation algorithms suitable for large-scale maximum a posteriori (MAP) reconstruction. After discussing EM-type updates and their subset-accelerated variants, we highlight why subset methods can exhibit limit-cycle behaviour and why convergent penalised algorithms typically require relaxation (e.g. block-sequential regularised expectation maximisation (BSREM) [40]). We then introduce variance-reduced subset methods (e.g. stochastic average gradient amélior   (SAGA) and stochastic variance-reduced gradient (SVRG)) as a mechanism to retain low per-iteration cost while suppressing subset-induced stochasticity, and discuss the role of preconditioning, including “prior-aware” diagonal curvature approximations, in accelerating and stabilising convergence [41, 42].

3.1 The Inverse Problem in Emission Tomography

To discretely represent the spatial distribution of tracer concentration within a patient, the continuous activity distribution $f(r)$ is approximated using a basis expansion $f(r) \approx \sum_{j=1}^J x_j h_j(r)$. Typically, $\{h_j\}$ are chosen to be indicator functions on uniform cuboid voxels. In that case, it is customary to use a non-negative activity image vector $x = (x_1, \dots, x_J)^\top \in \mathbb{R}_+^J$.

The measured data consist of coincidence counts along lines of response (PET) or projection bins (SPECT). We denote the observed count in bin i as y_i , and model it as a realisation of a Poisson counting process,

$$y_i \sim \text{Poisson}(Y_i), \quad i = 1, \dots, I, \quad (3.1)$$

where $Y_i = \mathbb{E}[y_i]$ is the (unknown) mean count, which we model as $Y_i \approx \bar{y}_i(x)$ with

$$\bar{y}_i(x) = \sum_{j=1}^J a_{ij} x_j + b_i, \quad i = 1, \dots, I. \quad (3.2)$$

For notational simplicity, we will use the vector form

$$\bar{y}(x) = Ax + b, \quad \bar{y}_i(x) = (Ax + b)_i, \quad (3.3)$$

where $A = [a_{ij}] \in \mathbb{R}_+^{I \times J}$ and $b \in \mathbb{R}_+^I$. Assuming independence of measurement bins, we obtain

$$p(y | x) = \prod_{i=1}^I \frac{\bar{y}_i(x)^{y_i} \exp(-\bar{y}_i(x))}{y_i!}. \quad (3.4)$$

Taking logarithms yields the Poisson log-likelihood

$$\log p(y | x) = \sum_{i=1}^I (y_i \log \bar{y}_i(x) - \bar{y}_i(x) - \log(y_i!)). \quad (3.5)$$

Since the term $\sum_{i=1}^I \log(y_i!)$ does not depend on x , it can be dropped for optimisation purposes. Thus, maximising the likelihood w.r.t. x is equivalent to minimising

$$\mathcal{L}(x) := -\log p(y | x) + \text{const} = \sum_{i=1}^I (\bar{y}_i(x) - y_i \log \bar{y}_i(x)). \quad (3.6)$$

Unconstrained maximum-likelihood (ML) solutions are dominated by noise due to the poor conditioning of A and the Poisson noise inherent in y . While iterative expectation–maximisation (EM) methods such as maximum-likelihood expectation–maximisation (MLEM) and ordered subsets expectation–maximisation (OSEM) [27, 28] are the clinical standard, they rely on early stopping to act as an implicit regulariser. This yields clinically acceptable images but introduces solution-dependent bias that may not give reliable quantitative metrics [29]. In addition, finding the ML solution for quantification can be prohibitively slow [30].

A robust alternative to early stopping is the Bayesian formulation. By Bayes' rule, we have

$$p(x | y) = \frac{p(y | x) p(x)}{p(y)}, \quad (3.7)$$

where $p(y | x)$ is the Poisson likelihood induced by the forward model, $p(x)$ encodes

prior information about the unknown image, and the evidence

$$p(y) = \int p(y | x) p(x) dx \quad (3.8)$$

acts as a normalising constant independent of x .

A maximum a posteriori (MAP) estimate is then defined as

$$\hat{x}_{\text{MAP}} \in \arg \max_{x \in \mathcal{C}} p(x | y) = \arg \max_{x \in \mathcal{C}} p(y | x) p(x), \quad (3.9)$$

where $\mathcal{C}$ denotes a feasible set (typically enforcing non-negativity). Taking negative logarithms, as above, gives an equivalent minimisation problem:

$$\hat{x}_{\text{MAP}} \in \arg \min_{x \in \mathcal{C}} \{-\log p(y | x) - \log p(x)\}, \quad (3.10)$$

since $-\log p(y)$ is constant with respect to x .

It is common to express the prior in Gibbs form,

$$p(x) \propto \exp(-\beta \mathcal{R}(x)), \quad (3.11)$$

where $\mathcal{R}(x)$ is a regularisation functional and $\beta > 0$ controls the strength of the prior. Substituting this into the MAP objective yields the penalised-likelihood form

$$\hat{x}_{\text{MAP}} \in \arg \min_{x \in \mathcal{C}} \{\mathcal{L}(x) + \beta \mathcal{R}(x)\}. \quad (3.12)$$

3.2 Regularisation Strategies

The choice of $\mathcal{R}(x)$ determines the texture and quantitative characteristics of the reconstructed image. Historically, this has evolved from simple smoothing constraints to sophisticated, anatomically guided models.

3.2.1 Image-based Regularisation

One of the simplest forms of regularisation is Tikhonov regularisation, which imposes an ℓ_2 -norm penalty on local voxel-wise differences. This yields the quadratic prior

$$\mathcal{R}_{\text{Quad}}(x) = \sum_{j=1}^J \sum_{k \in \mathcal{N}_j} w_{jk} (x_j - x_k)^2, \quad (3.13)$$

where $\mathcal{N}_j$ denotes a local neighbourhood of voxel j (e.g. 6, 18, or 26-connected in 3D) and $w_{jk} \geq 0$ are weights (often distance-based and symmetric). While strongly convex and computationally efficient, the quadratic penalty strongly discourages large local differences, which tends to oversmooth edges and suppress high-contrast structures [43].

To better preserve edges, total variation (TV) [31] replaces the squared penalty with an ℓ_1 -type penalty on local differences. A neighbourhood-difference form that makes the connection to $\mathcal{R}_{\text{Quad}}$ explicit is

$$\mathcal{R}_{\text{TV}}(x) = \sum_{j=1}^J \sum_{k \in \mathcal{N}_j} w_{jk} |x_j - x_k|. \quad (3.14)$$

Relative to $\mathcal{R}_{\text{Quad}}$, the linear growth of $|x_j - x_k|$ reduces the marginal cost of large voxel-wise jumps in intensity, thereby promoting piecewise-constant reconstructions with sharper edges.

In much of the imaging literature, TV is instead written in terms of a discrete gradient operator. Defining $(Dx)_j \in \mathbb{R}^d$ to collect finite differences at voxel j along the d coordinate directions, a common choice on a regular grid is

$$((Dx)_j)^{(\ell)} = \frac{x_{j+\delta_\ell} - x_j}{\Delta_\ell}, \quad \ell = 1, \dots, d, \quad (3.15)$$

where Δ_ℓ denotes the Euclidean distance between voxels $j + \delta_\ell$ and j . The discrete TV functional is then

$$\mathcal{R}_{\text{TV}}(x) = \sum_{j=1}^J |(Dx)_j|, \quad (3.16)$$

with an isotropic alternative, more common in emission tomography [44, 45],

$$\mathcal{R}_{\text{TV}}(x) = \sum_{j=1}^J \|(Dx)_j\|_2. \quad (3.17)$$

For optimisation, it is common to use a smooth approximation to avoid nondifferentiability at zero, for example

$$\tilde{\mathcal{R}}_{\text{TV}}(x) = \sum_{j=1}^J \sqrt{\|(Dx)_j\|_2^2 + \varepsilon^2}, \quad \varepsilon > 0. \quad (3.18)$$

In clinical practice, adaptable edge-preserving priors are often preferred. The relative difference prior (RDP) [32], the core component of the GE “Q.Clear” algorithm, modifies the penalty to depend on the absolute intensity sums, penalising relative differences, as opposed to absolute ones.

$$\mathcal{R}_{\text{RDP}}(x) = \frac{1}{2} \sum_{j=1}^J \sum_{k \in \mathcal{N}_j} w_{jk} \frac{(x_j - x_k)^2}{x_j + x_k + \gamma|x_j - x_k| + \varepsilon}, \quad (3.19)$$

with γ controlling the amount of edge preservation and $\varepsilon \geq 0$ the STIR RDP stabiliser. Positive ε regularises zero-valued voxel pairs; when $\varepsilon = 0$, STIR assigns the zero-zero contribution its limiting value of zero. Note that if $(x_j + x_k + \varepsilon) \gg \gamma|x_j - x_k|$ then $\mathcal{R}_{\text{RDP}}$ behaves like a relative quadratic prior, whereas if $\gamma|x_j - x_k| \gg (x_j + x_k + \varepsilon)$ it behaves like an anisotropic TV penalty (up to a constant factor).

3.2.2 Anatomically Guided Regularisation

In low-count regimes, single-modality priors struggle to distinguish between noise and legitimate small structures. Anatomically guided reconstruction uses a high-resolution structural image u (e.g., CT) to modulate the regularisation of the emission image x .

Geometric Alignment methods assume that if an edge exists in the emission image, it should be supported by corresponding boundaries in a registered anatomical guidance image. The seminal Bowsher prior [33] implements this idea by defining, for each voxel j , a data-adaptive and dynamic neighbourhood $\mathcal{B}_j(u)$ using the

guidance image u . Concretely, from a fixed candidate neighbourhood $\mathcal{N}_j$, the Bowsher construction selects the B voxels whose intensities in u are most similar to u_j , i.e. those minimising a similarity metric such as

$$d_{jk}(u) := |u_j - u_k| \quad (k \in \mathcal{N}_j), \quad (3.20)$$

and sets $\mathcal{B}_j(u)$ to be the indices of the B smallest values of $d_{jk}(u)$. Regularisation is then applied only over this anatomically consistent subset, for example via a quadratic penalty

$$\mathcal{R}_{\text{Bow}}(x; u) = \sum_{j=1}^J \sum_{k \in \mathcal{B}_j(u)} w_{jk} (x_j - x_k)^2, \quad (3.21)$$

(or an alternative edge-preserving potential). By restricting the penalty to neighbours that are locally homogeneous in u , the prior encourages smoothing within anatomical regions while reducing smoothing across anatomical edges, thereby allowing sharp boundaries to form in x when they are supported by the guidance image.

An even more geometric framework is parallel level sets (PLS) [34]. These methods encourage structural consistency by encouraging the gradient vectors of the emission image and the guidance image to be aligned, i.e. $(Dx)_j \parallel (Du)_j$. A common formulation (often termed PLS1) penalises the magnitude of the component of $(Dx)_j$ orthogonal to $(Du)_j$, which is proportional to the sine of the enclosed angle θ_j :

$$R_{\text{PLS1}}(x, u) = \sum_{j=1}^J \|(Dx)_j\|_2 \|(Du)_j\|_2 |\sin \theta_j|. \quad (3.22)$$

This term vanishes when $(Dx)_j$ and $(Du)_j$ are parallel (or when either gradient is zero), and increases with the component of $(Dx)_j$ orthogonal to $(Du)_j$. While numerically stable, PLS1 scales with $\|(Du)_j\|_2$ and therefore inherits the spatially varying gradient magnitude of the anatomical image, which can lead to over-weighting at strong anatomical edges and under-weighting in low-contrast regions.

To decouple the directional information from the anatomical gradient magnitude,

directional total variation (DTV) [46] introduces a normalised direction field

$$\xi_j = \frac{(Du)_j}{\sqrt{\|(Du)_j\|_2^2 + \eta^2}}, \quad (3.23)$$

where $\eta > 0$ is a small stabilisation parameter that suppresses unreliable directions induced by noise or very fine-scale anatomical texture. Using ξ_j , one defines a local anisotropic projection operator

$$\Pi_j := \mathbb{I} - \gamma_j \xi_j \xi_j^\top, \quad (3.24)$$

where $\mathbb{I}$ is the $d \times d$ (number of dimensions, d , squared) identity matrix and $\gamma_j \in [0, 1]$ controls the strength of directional guidance. The resulting DTV prior is

$$R_{\text{DTV}}(x; u) = \sum_{j=1}^J \|\Pi_j(Dx)_j\|_2, \quad (3.25)$$

which penalises predominantly the gradient component orthogonal to the anatomical level-set direction, thereby encouraging edges in x to align with edges in u . In regions where u is locally flat (so $\xi_j \approx 0$), $\Pi_j \approx \mathbb{I}$ and DTV reduces to standard isotropic TV.

Regularisation strategies also exist outside the standard Bayesian (penalised-likelihood) framework. A prominent example is the kernelised expectation maximisation (KEM), in which prior information is encoded directly in the image representation rather than through an explicit penalty term. Following Wang and Qi [35] and its anatomically-aided extension [36], the emission image is modelled as a linear combination of kernel-derived basis functions:

$$x = K\alpha, \quad (3.26)$$

where $\alpha \in \mathbb{R}^J$ is a coefficient image and $K \in \mathbb{R}^{J \times J}$ is a (typically sparse) kernel matrix whose entries encode similarity between voxels based on features extracted from a guidance image [36]. Each voxel j is assigned a feature vector f_j (commonly a patch of intensities in the high-resolution guidance image), and a radial Gaussian

kernel is often used,

$$\kappa(f_j, f_k) = \left(-\frac{\|f_j - f_k\|_2^2}{2\sigma^2} \right), \quad (3.27)$$

with σ controlling edge sensitivity. The kernel matrix is then defined entrywise by $K_{jk} = \kappa(f_j, f_k)$, with additional sparsification for tractability: for each voxel j , only a small set of the most similar neighbours (e.g. via a local k -nearest-neighbour search within a window) are allowed to contribute, yielding a sparse K . To preserve overall intensity scaling (counts), K is typically row-normalised.

Substituting $x = K\alpha$ into the Poisson forward model gives a kernelised expectation for the data,

$$\bar{y}(\alpha) = A(K\alpha) + b, \quad (3.28)$$

so that reconstruction reduces to maximum-likelihood estimation of α under the same Poisson model, followed by $\hat{x} = K\hat{\alpha}$. Importantly, because the model remains linear in the unknown coefficients, an EM-type update can be derived with essentially the same structure as conventional MLEM/OSEM, but with K embedded within the projector/backprojector operations. This yields a procedure that is straightforward to implement in existing EM pipelines, while often delivering improved bias–variance trade-offs at low counts by enforcing anatomically-informed, spatially adaptive smoothing in the image representation [35, 36].

A known limitation of purely anatomy-derived kernels is PET-unique feature suppression: structures present in the anatomical image but absent (or weak) in the emission image can be biased towards anatomically plausible but functionally incorrect solutions [47]. To mitigate this, hybrid kernelised expectation maximisation (HKEM) augments the kernel with synergistic information from the evolving PET estimate itself, yielding an iteration-dependent kernel that combines anatomical and emission-driven similarity. In the HKEM formulation of Deidda *et al.*, the hybrid kernel entry factorises as

$$k_{fj}^{(n)} = k_m(v_f, v_j) k_p(z_f^{(n)}, z_j^{(n)}), \quad (3.29)$$

where v_j denotes the (fixed) anatomical feature vector at voxel j and $z_j^{(n)}$ is a feature

derived from the current PET update (updated each iteration) [47]. Both factors are commonly chosen as Gaussian similarities (Equation 3.27 extended) modulated by spatial proximity,

$$k_m(v_j, v_k) = \exp\left(-\frac{\|v_j - v_k\|_2^2}{2\sigma_a^2}\right) \exp\left(-\frac{\|x_j - x_k\|_2^2}{2\sigma_{da}^2}\right), \quad (3.30)$$

$$k_p(z_j^{(n)}, z_k^{(n)}) = \exp\left(-\frac{\|z_j^{(n)} - z_k^{(n)}\|_2^2}{2\sigma_p^2}\right) \exp\left(-\frac{\|x_j - x_k\|_2^2}{2\sigma_{df}^2}\right), \quad (3.31)$$

where x_j denotes the voxel coordinates and the parameters $\sigma_a, \sigma_f, \sigma_{da}, \sigma_{df}$ control the relative influence of anatomical similarity, functional-driven (emission) similarity, and spatial locality. The functional term helps address emission–guidance mismatch and, crucially, reduces the tendency of anatomy-only kernels to suppress PET-unique lesions by allowing the neighbourhood coupling to adapt to features emerging in the emission estimate. Empirically, Deidda *et al.* report that this hybridisation can reduce negative bias under low-count conditions and improve lesion contrast while retaining the denoising benefits of kernelised reconstruction [47].

Kernel approaches extend naturally to multiple guidance sources by enlarging the feature vectors (e.g. concatenating features from CT and/or additional functional channels [35]) and defining the kernel on the resulting joint feature space. This provides a flexible mechanism for sequential or multi-modality guidance without introducing an explicit Bayesian penalty term.

3.2.3 Synergistic Regularisation

If multiple modalities are available that depict the same underlying anatomy or functional distribution, it can be advantageous to reconstruct them synergistically, allowing information to be shared across modalities throughout the reconstruction process [9]. A prerequisite for such coupling is that the images are defined in a common spatial coordinate system. In practice, this requires geometric registration between modalities and resampling onto a common reconstruction grid, so that voxel index n refers to the same physical location across all channels.

When the native modalities, indexed using m , are acquired on different grids or

coordinate frames, we introduce spatial transforms W_m (registration + resampling operators) and define the co-registered images on a shared grid, with $W_m : \mathbb{R}_+^{J_m} \rightarrow \mathbb{R}_+^N$:

$$z_m := W_m x_m \in \mathbb{R}_+^N, \quad m = 1, \dots, M, \quad (3.32)$$

and collect them as the modality tuple

$$\mathbf{z} := (z_1, \dots, z_M). \quad (3.33)$$

Coupling priors are then naturally expressed in terms of $\mathbf{z}$, since neighbourhoods, gradients, and voxel-wise comparisons are only meaningful when images are spatially aligned.

In the Bayesian setting, a generic multi-modality reconstruction is given by

$$\hat{\mathbf{x}} \in \arg \min_{\mathbf{x} \in \mathcal{C}} \left\{ \sum_{m=1}^M \mathcal{L}_m(x_m) + \beta \mathcal{R}(\mathbf{W}\mathbf{x}) \right\}, \quad (3.34)$$

where $\mathbf{x} := (x_1, \dots, x_M)$ and $\mathbf{W} := \text{diag}(W_1, \dots, W_M)$, and $\mathcal{R}(\mathbf{W}\mathbf{x})$ promotes cross-modality consistency (e.g. co-located edges or shared support) in the co-registered domain.

A practical way to exploit multiple modalities without solving a fully joint inverse problem is sequential reconstruction, in which one modality is first reconstructed, and the resulting image is then used as side-information to guide the reconstruction of another modality. In the Bayesian regime, this is

$$\hat{x}_m \in \arg \min_{x_m \in \mathcal{C}} \left\{ \mathcal{L}_m(x_m) + \beta \mathcal{R}(x_m; W_{m \leftarrow (m-1)} x_{m-1}) \right\}, \quad (3.35)$$

where $W_{m \leftarrow (m-1)}$ maps the image reconstructed in the previous stage, x_{m-1} , into the spatial grid of modality m .

Deidda et al. have previously investigated a related method for ^{90}Y bremsstrahlung SPECT-informed PET reconstructions [48]. Bremsstrahlung SPECT images are first reconstructed using HKEM with CT and SPECT image-update

information to obtain an enhanced SPECT estimate. This SPECT image is then resampled to the PET grid and used as functional guidance for PET reconstruction.

$$\hat{\alpha}_m \in \arg \min_{\alpha_m \in \mathcal{C}} \left\{ \mathcal{L}_m(W_{m \leftarrow (m-1)} K_m \alpha_m) \right\}. \quad (3.36)$$

The paper reports that this sequential design is particularly beneficial for low-count ^{90}Y PET: SPECT provides complementary denoising/uptake support within PET–SPECT-consistent regions, while CT contributes primarily through resolution/edge information injected at the SPECT stage, and the PET image-update features reduce suppression of PET-unique structures and mitigate inter-modality mismatch.

Sequential guidance is limited by one-way dependency: errors in u may propagate directly to x and the shared information may not be fully utilised. Truly synergistic reconstruction estimates both images simultaneously, allowing information to flow bi-directionally. Perhaps the most obvious way to extend single modality regularisation to achieve a synergistic regulariser is joint total variation (JTV) [49], a simple extension of TV which sums gradient magnitudes:

$$\mathcal{R}_{\text{JTV}}(\mathbf{z}) = \sum_{n=1}^N \sqrt{\|Dz_{1,n}\|^2 + \|Dz_{2,n}\|^2 + \dots + \|Dz_{M,n}\|^2}. \quad (3.37)$$

This encourages joint sparsity, encouraging co-located edges. A convenient way to formalise multi-modality edge co-localisation is to view the set of images as a vector-valued field. For each voxel n , the voxel-wise Jacobian is defined by stacking image gradients (Equation 3.15) in columns:

$$\mathcal{G}_n(\mathbf{z}) := \begin{bmatrix} Dz_{1,n} & Dz_{2,n} & \dots & Dz_{M,n} \end{bmatrix} \in \mathbb{R}^{d \times M}. \quad (3.38)$$

When viewed in this way, JTV can be redefined as a Schatten 2-norm (also known as a Frobenius norm) applied to this jacobian [37],

$$\mathcal{R}_{\text{JTV}}(\mathbf{z}) = \sum_{n=1}^N \|\mathcal{G}_n(\mathbf{z})\|_F \quad (3.39)$$

where the Schatten p -norm is defined as

$$\|Y\|_{S_p} := \left(\sum_{r=1}^R \sigma_r(Y)^p \right)^{1/p} \quad (3.40)$$

for singular values $\sigma_\ell(Y)$ of a rank- R matrix Y . Note that $R = \min\{d, M\}$. The Schatten 1-norm, also known as the nuclear norm is the sum of singular values

$$\|Y\|_* = \sum_{r=1}^R \sigma_r(Y). \quad (3.41)$$

The nuclear norm is the convex envelope of the Rank function. Applied to the voxel-wise jacobians, it promotes singular value sparsity and has the effect of either driving functionally misaligned gradients to zero or parallel/anti-parallel, promoting edge-alignment. Applying this nuclear norm to the voxel-wise jacobians is known as total nuclear variation (TNV) [37, 38].

$$\mathcal{R}_{\text{TNV}}(\mathbf{z}) = \sum_{n=1}^N \|\mathcal{G}_n(\mathbf{z})\|_* \quad (3.42)$$

It has been applied to synergistic PET/MR reconstruction through its extended variant, total generalised nuclear variation [39].

The benefits of using the nuclear norm over the Frobenius norm can be visualised in Figure 3.1, where the nuclear norm exhibits sharp grooves along the coordinate axes that encourage one singular value to vanish (promoting low-rank structure), whereas the Frobenius norm is smooth and tends to shrink singular values jointly without explicitly favouring rank reduction.

Total Nuclear variation will form the basis for the synergistic prior demonstrated later on in this thesis.

3.3 Optimisation Algorithms

MLEM has the multiplicative update equation for iteration k :

$$x^{(k+1)} = \left(x^{(k)} \oslash A^\top \mathbf{1}_I \right) \odot A^\top (y \oslash \bar{y}), \quad (3.43)$$

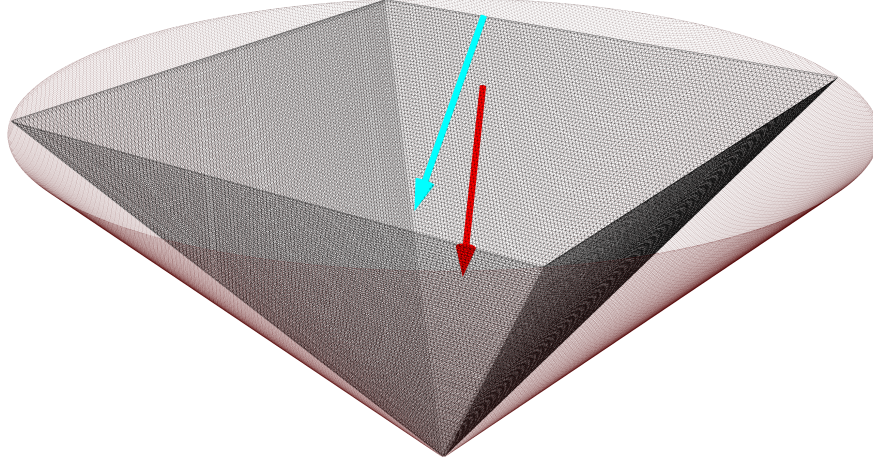


Figure 3.1: Visualisation of the Frobenius and nuclear norms (vertical axis) of a matrix X as functions of its (two) singular values (horizontal axis). The Frobenius norm is $\|X\|_F = \sqrt{\sigma_1(X)^2 + \sigma_2(X)^2}$ (red mesh), whereas the nuclear norm is $\|X\|_* = |\sigma_1(X)| + |\sigma_2(X)|$ (black mesh). For visualisation purposes we extend the singular-value axes to negative values. The nuclear norm exhibits sharp grooves along the coordinate axes, encouraging one singular value to vanish (i.e. promoting low-rank structure), while the Frobenius norm is smooth and does not preferentially drive a singular value exactly to zero. This behaviour can also be understood via subgradients: the nuclear-norm subgradient (cyan) points towards rank-1 grooves, preferentially shrinking the smaller singular value, whereas the Frobenius-norm subgradient (red) points towards the origin, shrinking both singular values equally.

where $\odot$ and $\oslash$ denote elementwise multiplication and division, respectively, $\mathbf{1}_I \in \mathbb{R}^I$ is an all-ones vector and k denotes the iteration number. For notational simplicity, we drop the explicit elementwise operators so that all divisions are understood componentwise. Then

$$x^{(k+1)} = \text{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right) A^\top \left(\frac{y}{\bar{y}}\right). \quad (3.44)$$

This can be rearranged to form the additive update equation

$$\begin{aligned} x^{(k+1)} &= x^{(k)} + \text{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right) A^\top \left(\frac{y}{\bar{y}(x^{(k)})} - \mathbf{1}_I\right) \\ &= x^{(k)} - \text{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right) \nabla_x \mathcal{L}(x^{(k)}). \end{aligned} \quad (3.45)$$

where $\mathbf{1}_I \in \mathbb{R}^J$ is an all-ones vector. This form highlights that the MLEM update can be interpreted as a preconditioned gradient descent algorithm with preconditioner $\mathcal{P}_{\text{EM}}^{(k)} = \text{diag}\left(\frac{x^{(k)}}{A^\top \mathbf{1}_I}\right)$, i.e.

$$x^{(k+1)} = x^{(k)} - \varsigma \mathcal{P}_{\text{EM}}^{(k)} \nabla_x \mathcal{L}(x^{(k)}), \quad (3.46)$$

with a step size $\varsigma = 1$.

A preconditioner aims to recondition the reconstruction space so that gradient steps are appropriately scaled across voxels, improving numerical conditioning and convergence. Intuitively, we seek larger steps in poorly sensitive directions and smaller steps in highly sensitive directions, thereby approximating curvature information.

Subsets are commonly used to accelerate early convergence by partitioning the data into S subsets and updating the image using only the contribution from a single subset (or a small batch of subsets) at each iteration. Writing the negative log-likelihood as a sum of subset terms,

$$\mathcal{L}(x) = \sum_{s=1}^S \mathcal{L}_s(x), \quad (3.47)$$

a generic subset scaled-gradient update takes the form

$$x^{(k+1)} = x^{(k)} - \varsigma \mathcal{P}^{(k)} \nabla_x \mathcal{L}_s(x^{(k)}). \quad (3.48)$$

If multiple subsets are used per iteration, $\nabla_x \mathcal{L}_s$ can be replaced by an average over the chosen subset indices. The most common algorithm found in clinical systems is

the OSEM algorithm. For subset s at iteration k , a standard OSEM update can be written as

$$x^{(k+1)} = \text{diag} \left(\frac{x^{(k)}}{A_s^T \mathbf{1}_I} \right) A_s^T \left(\frac{y_s}{\bar{y}_s(x^{(k)})} \right), \quad (3.49)$$

where A_s denotes the system matrix and data restricted to subset s . OSEM cycles through subsets incrementally.

OSEM typically shows a marked improvement in early-stage convergence by providing a cheaper approximation of the full-data EM update. However, because the subset gradient is only an approximation to the full gradient, OSEM is not, in general, a convergent algorithm for the maximum-likelihood problem: as the iterates approach a solution, incremental sub-gradient updates can lead to cyclical behaviour. An example illustrating this is shown in Figure 3.2.

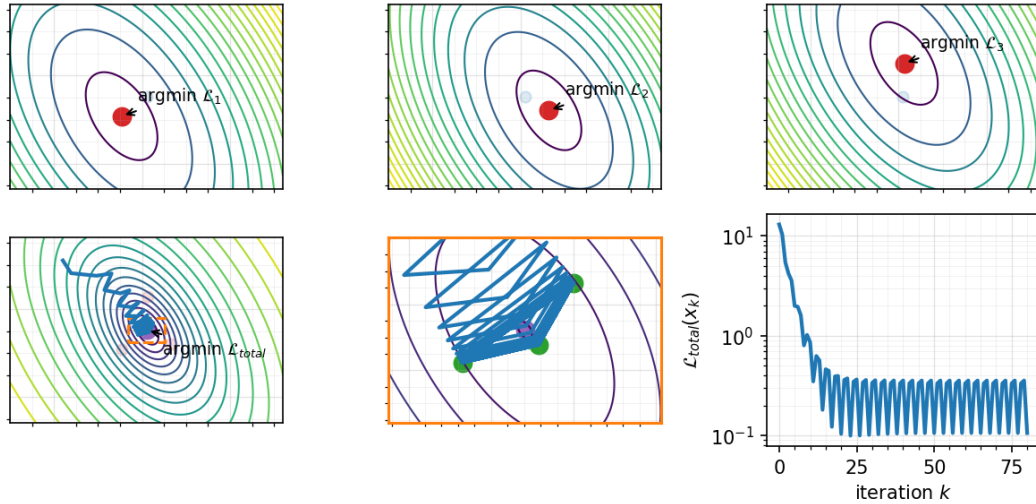


Figure 3.2: Demonstration of limit cycle behaviour with an ordered subset gradient descent algorithm. Top row: 3 different simple binomial sub-functions representing subset likelihoods. Bottom row: left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating limit cycle behaviour (green dots show approximate points of limit cycle); right, likelihood value per iteration, demonstrating the limit cycle.

We seek a fast and convergent algorithm for MAP reconstruction. A widely used approach is the (modified) block-sequential regularised expectation maximisation (BSREM) framework [40], which combines EM-type preconditioning and subset acceleration with a relaxation (diminishing step-size) to recover convergence while incorporating an explicit penalty. In a scaled-gradient form, a generic BSREM-type

update can be written as

$$x^{(k+1)} = x^{(k)} - \varsigma_k \mathcal{P}_{\text{EM}}^{(k)} \left(\nabla_x \mathcal{L}_s(x^{(k)}) + \frac{\beta}{S} \nabla_x \mathcal{R}(x^{(k)}) \right), \quad (3.50)$$

where $\varsigma_k > 0$ is an iteration dependent step size. Convergence is promoted by choosing $\{\varsigma_k\}$ to diminish appropriately (e.g. $\varsigma_k \rightarrow 0$ with $\sum_k \varsigma_k = \infty$), which suppresses the subset-induced limit-cycle behaviour while retaining rapid early progress (see Figure 3.3).

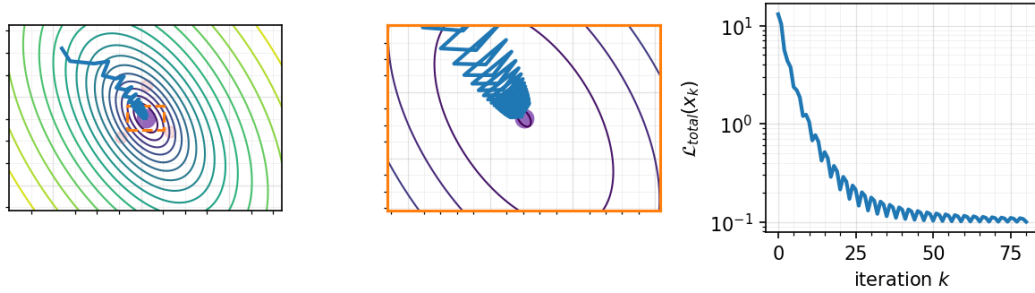


Figure 3.3: Demonstration of reduction in limit cycle behaviour with step size relaxation $\varsigma_k = \varsigma_0 / (1 + 0.05k)$. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating relaxed limit cycle behaviour; right, likelihood value per iteration, demonstrating relaxed limit cycle precession.

Variance-reduced subset methods have become increasingly popular in the PET reconstruction literature because they retain the low per-iteration cost of subset updates while substantially reducing the limit cycling observed in OSEM. The key idea is to replace the raw subset gradient, which is a high-variance estimator of the full gradient, with a variance-reduced estimator that is (approximately) unbiased and whose variance decreases as the iterates approach a solution [41, 42].

A stochastic subset gradient step uses $g_{s_k}(x^{(k)})$ in place of $\nabla \mathcal{L}(x^{(k)}) = \sum_s g_s(x^{(k)})$, i.e.

$$x^{(k+1)} = x^{(k)} - \varsigma_k \mathcal{P}^{(k)} g_{s_k}(x^{(k)}), \quad (3.51)$$

which is cheap but noisy. Variance reduction augments this step by incorporating additional information (stored gradients or periodic full-gradient computations) to form a corrected estimator $\tilde{g}_k$ with reduced variance. Two widely used examples are stochastic average gradient amélioré (SAGA) and SVRG:

SAGA [50] maintains a table of the most recently computed subset gradients $\{g_s(\cdot)\}_{s=1}^S$ and updates using

$$\tilde{g}_k = g_{s_k}(x^{(k)}) - g_{s_k}(\tilde{x}_{s_k}) + \frac{1}{S} \sum_{s=1}^S g_s(\tilde{x}_s), \quad (3.52)$$

where $\tilde{x}_s$ denotes the iterate at which the stored gradient for subset s was last evaluated.

SVRG [51] periodically computes a reference full gradient at a snapshot $\tilde{x}$ and then performs inner iterations using

$$\tilde{g}_k = g_{s_k}(x^{(k)}) - g_{s_k}(\tilde{x}) + \nabla_x \mathcal{L}(\tilde{x}). \quad (3.53)$$

In both cases, $\tilde{g}_k$ is designed so that $\mathbb{E}[\tilde{g}_k] \approx 1/S \nabla_x \mathcal{L}(x^{(k)})$ while its variance decreases as $x^{(k)}$ approaches a stationary point, thereby mitigating the oscillatory behaviour seen in OSEM and improving stability at high iteration counts (see Figure 3.4).

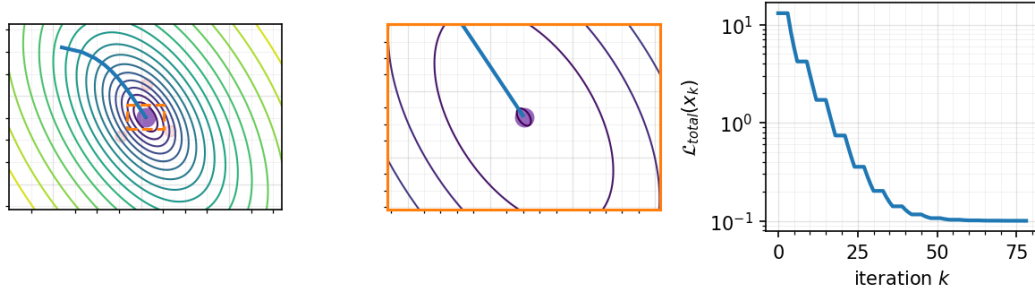


Figure 3.4: Demonstration of removal of limit cycle behaviour with an SVRG-type (Ordered VRG) algorithm. Left, full function with sub-gradient steps; centre, zoomed version of the full function demonstrating removal of limit cycle behaviour; right, likelihood value per iteration, demonstrating removal of limit cycle precession

A variance-reduced, prior-regularised update can then be written compactly as

$$x^{(k+1)} = x^{(k)} - \varsigma \mathcal{P}^{(k)} \left(\tilde{g}_k + \beta \nabla_x \mathcal{R}(x^{(k)}) \right). \quad (3.54)$$

Recent work indicates that incorporating the effect of the prior into $\mathcal{P}^{(k)}$ (“prior-aware preconditioning”) can be crucial for fast and robust convergence. Ehrhardt

et al. [41] showed substantially improved convergence rates with RDP-regularised PET reconstruction when using harmonic averaging of the EM-type preconditioner and a scaled diagonal estimate of the inverse Hessian of the prior. Using only the diagonal elements of the RDP Hessian

$$[\mathcal{H}_{\text{RDP}}(x)]_{jj} = \sum_{k \in \mathcal{N}_j} \frac{2w_{jk}(2x_k + \varepsilon)^2}{(x_j + x_k + \gamma|x_j - x_k| + \varepsilon)^3}, \quad (3.55)$$

they constructed a preconditioner as

$$\mathcal{P}^{(k)} = \text{diag} \left(\frac{x^{(k)} + \delta}{A^\top \mathbf{1} + \alpha \text{diag}(\beta \mathcal{H}_{\text{RDP}}(x^{(k)})) (x^{(k)} + \delta)} \right) \quad (3.56)$$

with scalar α tuned to correct for the loss of off-diagonal information.

Preconditioning and stochastic variance reduction for synergistic MAP reconstruction has not been well investigated in the literature so far and will form a part of this thesis.

A widely used alternative to EM-type and subset-gradient schemes is the limited-memory BFGS algorithm with bound constraints (L-BFGS-B) [52, 53, 54, 55, 56, 57], a quasi-Newton method that iteratively builds a low-rank approximation to the inverse Hessian from successive iterate and gradient differences. The “limited-memory” construction stores only a small number of correction pairs, making it viable for high-dimensional images, while the “B” variant enforces simple bounds (e.g. non-negativity) via an active-set strategy and projected search directions.

Tsai *et al.* propose a diagonally preconditioned variant (L-BFGS-B-PC) that embeds analytic curvature information into the optimisation via a change of variables [58]. We can choose a diagonal preconditioner $\mathcal{P}$ from a cheap curvature surrogate and run L-BFGS-B on the transformed problem $\mathcal{P}x$. This rescales the objective so that gradients are more isotropic in the transformed coordinates, while bound constraints are preserved. The resulting method combines low-cost, problem-specific diagonal curvature (from $\mathcal{P}$) with the low-rank curvature learned by L-BFGS-B, typically yielding substantially faster and more robust convergence than unpreconditioned L-BFGS-B in penalised reconstructions.

In practice, L-BFGS-B-PC can exhibit rapid local convergence on smooth penalised objectives, but each iteration requires one or more full objective and gradient evaluations (coupled with a line search), which is computationally burdensome in large-scale emission tomography when the likelihood is evaluated over all subsets. For this reason it is not used in the main reconstruction experiments in this thesis. Nevertheless, it provides a useful reference optimiser and is employed later for a small demonstrative deconvolution problem in Chapter 5.

3.4 Summary and Research Gap

This chapter reviewed the statistical and algorithmic foundations of emission tomography reconstruction under a Poisson measurement model. Starting from maximum-likelihood estimation, we motivated penalised-likelihood (MAP) formulations as a principled alternative to early stopping, particularly in low-count and quantitatively demanding settings. We then surveyed regularisation strategies ranging from quadratic smoothness and total variation to clinically deployed edge-preserving models such as the relative difference prior, before focusing on anatomically guided approaches (e.g. Bowsher, PLS/DTV) and kernelised representations that incorporate structural side-information through the image model.

A central theme in modern PET/SPECT practice is the availability of multiple co-registered modalities. While sequential (one-way) guidance can be effective, it is inherently limited by asymmetric information flow and by the risk of propagating errors or mismatches from the guidance image into the emission reconstruction. This motivates synergistic reconstruction, in which modalities are coupled through joint priors. In this setting, jacobian-based penalties such as JTV and TNV provide a natural mechanism to promote co-located, directionally consistent edges across modalities while allowing modality-specific contrast.

From an optimisation perspective, clinical subset-accelerated EM methods achieve fast early progress but can exhibit limit-cycle behaviour and lack convergence guarantees in penalised problems without relaxation. Convergent frameworks such as BSREM recover stability via diminishing step sizes, but may converge

slowly when the curvature induced by strong, spatially varying priors dominates the data term. Variance-reduced subset methods (e.g. SAGA/SVRG) offer a promising compromise, retaining low per-iteration cost while suppressing subset-induced stochasticity. However, their practical performance depends critically on effective preconditioning.

Research gap. Despite substantial progress in both synergistic regularisation and variance-reduced optimisation, two gaps remain directly relevant to quantitative ^{90}Y imaging:

1. **Synergistic, anatomy-informed coupling:** existing Jacobian-based priors (e.g. TNV) are typically formulated to promote cross-modality edge consistency, but are not routinely combined with explicit directional guidance derived from high-resolution CT edges in a way that remains robust in low-count regimes.
2. **Synergy-aware optimisation:** variance-reduced subset algorithms and prior-aware diagonal preconditioners have mainly been studied in single-modality PET with clinically popular priors (e.g. RDP). Their extension to synergistic MAP reconstruction, where cross-modality coupling induces strong and spatially heterogeneous curvature, is comparatively underexplored.

Thesis direction. Motivated by these gaps, the remainder of this thesis develops a directionally guided, differentiable total nuclear variation prior that uses CT-derived edge information while coupling functional modalities through voxel-wise Jacobians. We then design and evaluate a variance-reduced subset optimisation scheme (SVRG-type) equipped with a synergy-aware preconditioner that accounts for both data fidelity and coupled prior curvature, targeting fast, stable convergence in low-count ^{90}Y reconstruction.

Chapter 4

Monte Carlo-Informed Modelling for ^{90}Y Bremsstrahlung SPECT

Chapter 2 described the principal limitations of ^{90}Y Bremsstrahlung SPECT: the continuous photon spectrum makes conventional energy-window scatter correction unsuitable, high-energy photons undergo septal penetration and collimator scatter, and external Bremsstrahlung photons may be emitted away from the original decay site as the β^- particle traverses tissue. This chapter develops the practical SPECT model used later for joint PET/SPECT reconstruction. It addresses two specific components: a SIMIND-derived estimate of non-primary counts and an image-space approximation to Bremsstrahlung range blur.

The resulting model is

$$\bar{y}(x) = A_{\text{prim}}K_{\text{brem}}x + b_{\text{MC}} \quad (4.1)$$

where A_{prim} contains attenuation, geometric projection, the intrinsic detector response and the geometric collimator response; K_{brem} is the image-space range-blur operator and b_{MC} represents detected counts not described by this primary-event model. The latter includes patient scatter, collimator scatter and septal penetration, so these effects are not also included in A_{prim} . It is therefore referred to throughout as a non-primary additive term, rather than as patient scatter alone.

The two components in Equation 4.1 are estimated in a preparatory stage.

Once estimated, both the acquisition model and b_{MC} are fixed during the target reconstruction. The purpose of the reconstruct–simulate loop below is to prepare an object- and acquisition-specific additive estimate, which avoids altering the objective function, and therefore the solution, during reconstruction.

4.1 Pre-Reconstruction Estimation of Non-Primary Counts

Monte Carlo estimates of scatter and penetration can be combined with a faster analytic projector by alternating reconstruction and simulation [59]. The approach used here follows that established principle. The practical questions addressed in this chapter are how to initialise the simulation for a different acquisition protocol and how to prevent an alternating update from becoming unstable before the final additive term is frozen.

4.1.1 Learned initialisation

The scatter-derived initial estimate $b^{(0)}$ is produced by the convolutional neural network of Xiang et al., which predicts scatter projections from measured emission data and CT-derived attenuation information [60]. The available network was trained for a different scanner, collimator and energy-window configuration from the acquisitions used in this thesis. Its output is consequently treated only as a computationally inexpensive warm start for the broader non-primary term, not as a final quantitative correction or as an independently validated contribution. The adapted SIRF inference workflow is documented in Appendix A.2.

4.1.2 Reconstruct–simulate refinement

Starting from $b^{(0)}$, an outer iteration consists of the following operations:

1. reconstruct an activity estimate $x^{(k)}$ using the current additive term $b^{(k)}$;
2. smooth the intermediate image with an isotropic 5 mm-full width at half maximum (FWHM) Gaussian and restrict it to the body mask, giving $\tilde{x}^{(k)}$;

3. use the smoothed image as the source distribution for a SIMIND simulation; and
4. extract a newly simulated non-primary estimate $\widehat{b}^{(k+1)}$ and update the running estimate.

Each intermediate reconstruction used 12 staggered subsets and 10 epochs (120 subset updates), warm-started from the preceding outer iterate. Ten outer iterations were used as a pre-specified computational budget. This endpoint is not presented as evidence of asymptotic convergence. The simulations were driven from Python using the interface described in Appendix A.1.

At outer iteration k , one simulation produces the total simulated detected projection counts $p_{\text{tot}}^{(k)}$ and the attenuated, geometrically collimated primary component $p_{\text{geo}}^{(k)}$. Inspired by the work by Dewaraja et. al in [59], the Monte Carlo projections are first placed on the scale of the analytic acquisition model $A_{\star} = A_{\text{prim}}K_{\text{brem}}$. Let $\mathcal{I}_k$ contain bins with non-zero $p_{\text{geo},i}^{(k)}$ after the lowest and highest 5% of the per-bin ratios $(A_{\star}\widetilde{x}^{(k)})_i/p_{\text{geo},i}^{(k)}$ have been excluded. The scalar calibration is

$$s^{(k)} = \frac{\sum_{i \in \mathcal{I}_k} (A_{\star}\widetilde{x}^{(k)})_i}{\sum_{i \in \mathcal{I}_k} p_{\text{geo},i}^{(k)}}, \quad (4.2)$$

and the newly simulated additive term is

$$\widehat{b}^{(k+1)} = s^{(k)} \left(p_{\text{tot}}^{(k)} - p_{\text{geo}}^{(k)} \right). \quad (4.3)$$

Thus, Equation 4.3 includes all simulated counts outside the geometrically collimated primary component; it is not an estimate of patient Compton scatter alone. Before simulation, the body mask was formed by thresholding the attenuation map at 0.05 cm^{-1} , applying a morphological opening of radius two voxels and retaining the largest connected component in order to remove CT acquisition noise and the patient bed.

4.1.3 Under-relaxed updates

The most direct update replaces the running estimate with the new simulation,

$$b^{(k+1)} = \widehat{b}^{(k+1)}. \quad (4.4)$$

Using this methods, we observed some divergence in non-primary additive term where subsequent updates would result in increasing oscillating underestimation followed by overestimation. To reduce the resulting iteration-to-iteration variation, an under-relaxed update was used,

$$b^{(k+1)} = (1 - \alpha)b^{(k)} + \alpha\widehat{b}^{(k+1)}, \quad \alpha = 0.2. \quad (4.5)$$

The value $\alpha = 0.2$ was selected heuristically rather than through an optimisation study.

Figure 4.1 shows the diagnostic that motivated this choice. With full replacement, the scale of successive projection profiles oscillates and grows rapidly. By the fourth outer iteration, the estimated additive profile well exceeds the measured total-emission profile. Under-relaxation keeps the profiles bounded over the ten-step refinement budget and suppresses the large iteration-to-iteration swings.

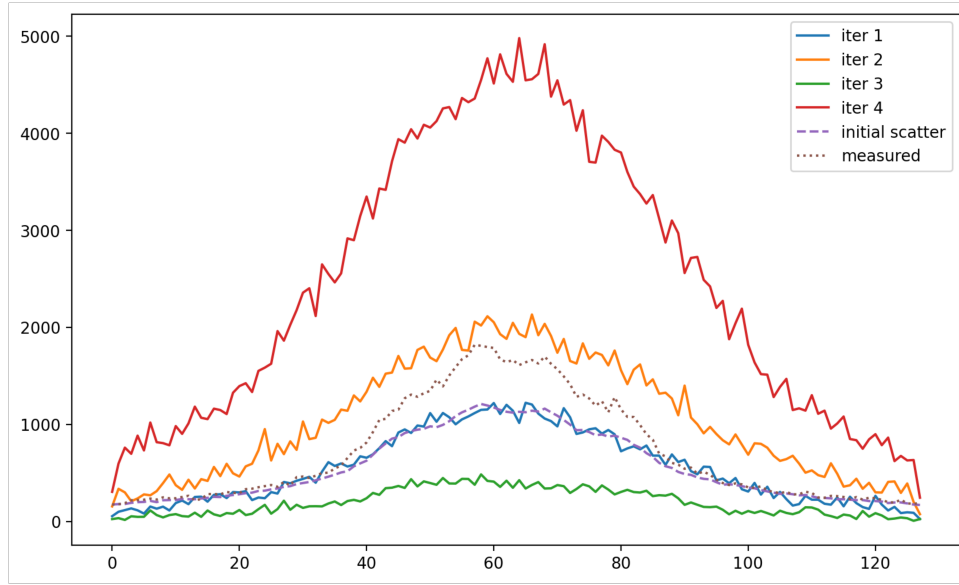
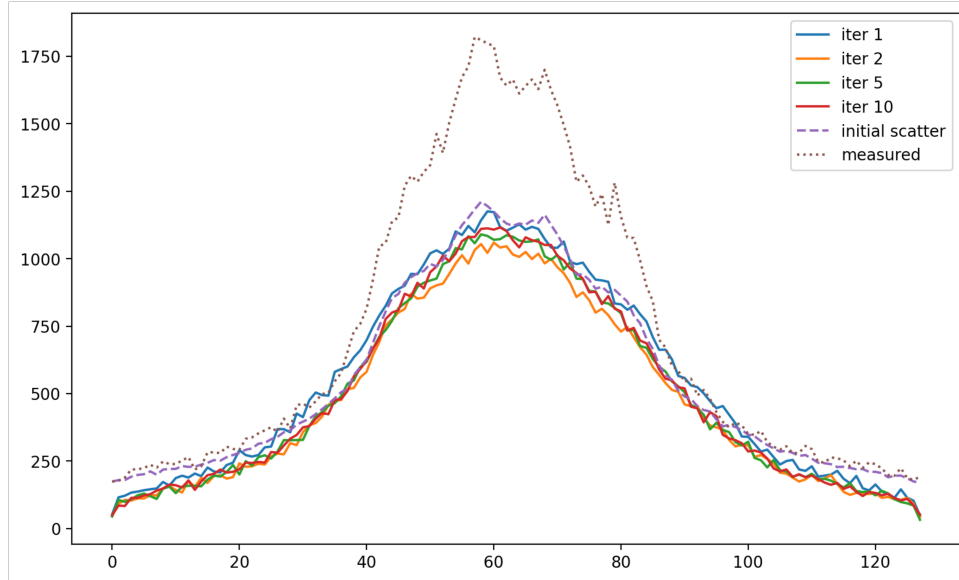
(a) Full replacement, $\alpha = 1$.(b) Under-relaxed update, $\alpha = 0.2$.

Figure 4.1: Projection-domain behaviour of the preparatory additive-term refinement for a measured NEMA phantom. Coloured curves are profiles of the estimated non-primary term $b^{(k)}$; every displayed iteration is one outer step of the preparatory reconstruct–simulate loop. Profiles are summed over views at an axial position through the spheres; the horizontal coordinate is the tangential detector bin and the vertical coordinate is counts. Full replacement produces large alternating changes within four outer iterations, whereas the under-relaxed profiles remain bounded through the tenth outer iteration.

After the tenth outer iteration, the final estimate is denoted $b_{\text{MC}} = b^{(10)}$ and is frozen. No additional SIMIND simulation or additive-term update is performed

during the target reconstruction. This makes b_{MC} a data- and activity-dependent plug-in estimate and the target reconstruction has a fixed objective.

4.2 Image-Space Bremsstrahlung Range Blur

External Bremsstrahlung photons may be emitted after the β^- particle has travelled away from the ^{90}Y decay site, whereas internal Bremsstrahlung is emitted at the decay site. The displacement distribution of the external component is energy dependent, as illustrated in Figure 4.2, and its net contribution adds a resolution loss distinct from the collimator–detector response [19, 61, 62]. Detailed Monte Carlo projectors can model this transport directly, but repeatedly evaluating such a model is impractical for the iterative and joint reconstructions developed later in this thesis.

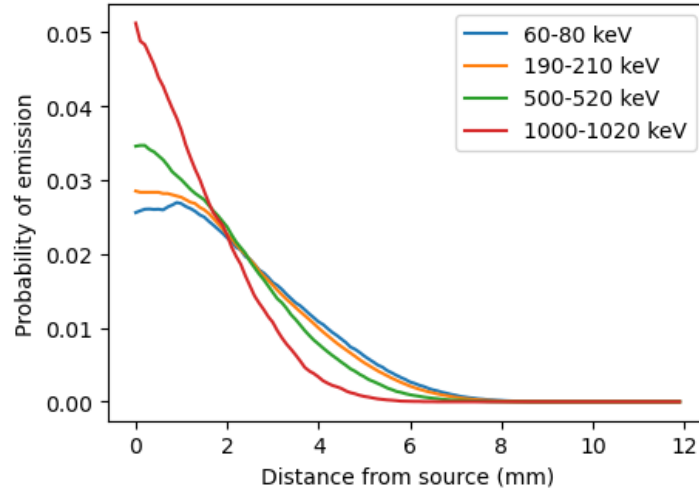


Figure 4.2: Probability of external Bremsstrahlung photon emission as a function of displacement from the ^{90}Y decay site for four photon-energy intervals. Curves were derived from the fast-simulation data reported by Rault et al. [19].

4.2.1 Primary collimator–detector response

The analytic primary projector blurs each source plane with a depth-dependent Gaussian point-spread function that represents the intrinsic and geometric components of the collimator–detector response (CDR); scatter and penetration remain in b_{MC} . For photon energy E and source–collimator distance h , this point spread function (PSF) is

$$K_{\text{cdr}}(\mathbf{r}; h, E) = \mathcal{N}(\mathbf{r}; \mathbf{0}, \sigma_{\text{cdr}}(h; E)^2 \mathbb{I}), \quad (4.6)$$

an isotropic Gaussian in the detector plane whose standard deviation σ_{cdr} sets the blur width. Software for Tomographic Image Reconstruction (STIR) models this width as affine in depth,

$$\sigma_{\text{cdr}}(h; E) = c_0(E) + c_1(E)h, \quad (4.7)$$

so the two coefficients $c_0(E), c_1(E)$ fully specify the depth-dependent CDR for a given energy. Physically, however, the CDR standard deviation is the quadrature sum of a depth-independent intrinsic term and a depth-dependent parallel-hole geometric term, $\sqrt{\sigma_{\text{int}}(E)^2 + \sigma_{\text{geo}}(h; E)^2}$, which is not affine in h . We therefore fit STIR's affine model (4.7) by least squares over $h \in [2, 20]$ cm to this quadrature sum, evaluated from the measured intrinsic and analytic geometric standard deviations. Over this range the quadrature sum is close to linear, so the affine surrogate reproduces the conventional distance dependence of the collimator response to within the accuracy required inside the projector [63].

4.2.2 Effective Gaussian range-kernel approximation

An exact reduced transport model would distinguish internal Bremsstrahlung, represented by an identity component at the decay site, from the displaced external component. The reconstruction implementation instead uses one zero-mean isotropic Gaussian as an empirical surrogate for their net central-response broadening,

$$G_{\text{brem}}(\mathbf{r}) = \mathcal{N}(\mathbf{r}; \mathbf{0}, \sigma_{\text{eff}}^2 \mathbb{I}). \quad (4.8)$$

The Gaussian form is motivated by the external-photon displacement distributions of Rault et al., while its effective width was estimated from the water-equivalent Monte Carlo response study described below [19]. A single 6.9 mm-FWHM kernel ($\sigma_{\text{eff}} \approx 2.93$ mm) was then used throughout the reconstructions. This is not a microscopic displacement PDF: by applying one Gaussian to all source activity, it approximates the combined internal/external response with one fitted width.

The resulting operator K_{brem} is a three-dimensional Gaussian convolution applied in image space. It is composed with the analytic projector in the forward

operation and with the analytic backprojector in reverse order in the adjoint,

$$A_{\star} = A_{\text{prim}} K_{\text{brem}}, \quad A_{\star}^{\text{T}} = K_{\text{brem}}^{\text{T}} A_{\text{prim}}^{\text{T}}. \quad (4.9)$$

For the symmetric Gaussian kernel, $K_{\text{brem}}^{\text{T}} = K_{\text{brem}}$. This construction therefore preserves a matched forward–adjoint pair while adding only an image-space convolution to each operation.

4.2.3 Projection-response calibration and cross-code comparison

The approximation was assessed using compact ^{90}Y source distributions in water and four source–collimator distances (2, 5, 10 and 20 cm). Detector-plane response widths were compared for three representative energy windows: [39.8, 126.6], [50, 150] and [75, 225] keV. The four models were GATE Monte Carlo, SIMIND Monte Carlo, STIR with the geometric CDR alone, and STIR with the geometric response composed with K_{brem} . GATE provides the response used to calibrate and check the effective width, while SIMIND provides a cross-code comparison.

The first interval is the window used for this response simulation and is not identical to the 39.84–128.16 keV acquisition window used for the joint phantom study in Chapter 8. The single 6.9 mm effective width was retained across these closely related but non-identical protocols rather than re-estimated for each acquisition.

Figure 4.3 reports the fitted detector-plane standard deviation. The geometric-only STIR model is systematically narrower than both Monte Carlo responses. Adding the image-space kernel moves the STIR prediction towards the GATE and SIMIND values at every evaluated distance and for all three energy windows. The agreement is not exact, particularly for the widest window at large distance, but the comparison supports the Gaussian kernel as a practical reduced-order representation of the missing net response broadening. The study did not include an event-class ablation separating external range, internal emission, scatter and penetration in the fitted response widths. It therefore does not independently validate the decomposition between A_{prim} and b_{MC} .

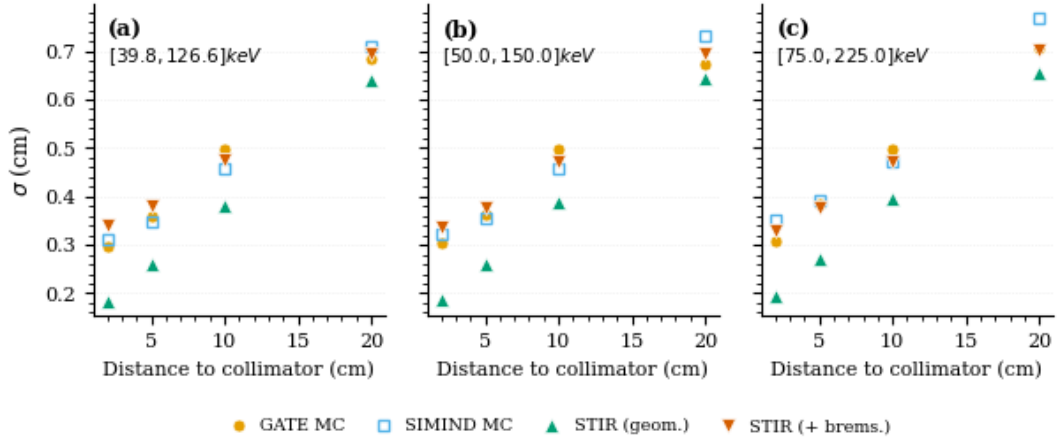


Figure 4.3: Detector-plane response standard deviation σ (cm) against source–collimator distance for (a) [39.8, 126.6], (b) [50, 150] and (c) [75, 225] keV. Markers show GATE MC, SIMIND MC, STIR with the geometric response, and STIR with the geometric response plus the image-space effective kernel. The geometric-only STIR projector underestimates the response width; composing it with the image-space Bremsstrahlung kernel substantially reduces the net width discrepancy across the evaluated distances and windows.

4.3 Scope, Limitations and Deployment

The work in this chapter is deliberately a component-level commissioning study. Iterative Monte Carlo estimation of scatter is established methodology, and under-relaxation is a standard numerical device. The contribution here is the integration of a domain-shifted learned initial estimate with a calibrated SIMIND refinement, the empirical identification and suppression of unstable full replacement, and the definition of a frozen additive term compatible with the subsequent reconstruction. Likewise, the physical existence of Bremsstrahlung range broadening is established; the contribution is an inexpensive fixed-width effective Gaussian surrogate that can be composed with a standard matched projector, calibrated against GATE and cross-checked against SIMIND.

Several limitations follow from these choices. The network output is used outside its training protocol and is only a scatter-derived initialisation; because the update is damped, it continues to influence the finite-budget estimate. The scalar calibration in Equation 4.2 cannot correct spatially varying differences between SIMIND and the analytic projector. The relaxation factor and ten-iteration budget

were fixed heuristically, and Figure 4.1 is a stability diagnostic rather than a convergence proof. The range kernel is isotropic and water-equivalent, so it does not model material-dependent transport in lung or bone, nor the detailed angular and spectral structure represented by a full Monte Carlo projector. A single effective width is also used across protocols and does not explicitly model the internal/external Bremsstrahlung mixture. The response study contains no event-class ablation and therefore cannot isolate range blur from every broadening mechanism in the Monte Carlo curves. The GATE result is a calibration-consistency comparison rather than an independent validation; SIMIND supplies the cross-code check. Finally, the analysis isolates selected model components and does not by itself establish improved clinical quantification.

The configuration carried forward is consequently explicit. A learned additive estimate is refined for ten outer iterations using SIMIND and Equation 4.5; the final non-primary term is frozen; and the 6.9 mm effective range kernel is retained within the matched SPECT acquisition model. Chapter 8 uses this fixed model for the joint ^{90}Y PET/SPECT experiments.

4.4 Summary

This chapter established the two Bremsstrahlung-specific components required by the later reconstruction experiments. First, a learned projection estimate was used to initialise object- and acquisition-specific SIMIND simulations of non-primary contamination. Full replacement was unstable in the measured NEMA example, whereas an under-relaxed update with $\alpha = 0.2$ remained bounded over the prescribed refinement budget. The resulting additive estimate was then frozen. Second, Monte Carlo response broadening was represented by a fixed 6.9 mm-FWHM image-space Gaussian. Composing this kernel with the analytic projector brought projected response widths substantially closer to GATE and SIMIND across three energy windows. Together, these components define the fixed ^{90}Y Bremsstrahlung SPECT model used in the final joint-reconstruction study.

Chapter 5

Anatomy-informed deconvolution for PET partial volume correction

5.1 Introduction and relationship to the thesis

Chapter 3 established the penalised-likelihood framework for emission tomography, reviewed representative methods for injecting prior knowledge into the optimisation problem (including anatomically guided and kernelised models), and motivated optimisation strategies that remain stable under regularisation. The core of this thesis concerns regularised and synergistic reconstruction under the full Poisson forward model $\bar{y}(x) = Ax + b$, where the ill-conditioning of A and the low-count regime make prior design, preconditioning, and choice of optimisation algorithm critical.

In this chapter we consider a deliberately simplified, image-domain inverse problem: post-reconstruction partial volume correction (PVC) via deconvolution with an estimated point spread function (PSF). Although this is not a substitute for incorporating resolution modelling within reconstruction, it provides a controlled sandbox in which the effects of anatomical guidance and regularisation can be studied without confounding factors such as subset noise, projector modelling error, and complex system geometry. In particular, the chapter serves two main purposes within the broader narrative:

1. it connects the priors reviewed in Chapter 3 (kernel methods, directional regularisation) to a tractable inverse problem where their qualitative behaviour

is easily interpreted;

2. it provides a practical PVC baseline that is straightforward to deploy on retrospectively reconstructed images.

Richardson–Lucy (RL) deconvolution is a purely image-domain analogue of MLEM under the Poisson model $\bar{y}(x) = Cx + b$: it is the same EM update, with the system matrix A replaced by the PSF convolution operator C [28, 64]. In this setting the “sensitivity image” simplifies substantially, since for a unit-sum PSF one has $C^\top \mathbf{1} \approx \mathbf{1}$, so the usual MLEM normalisation by $A^\top \mathbf{1}$ becomes approximately trivial. We extend Richardson–Lucy (RL) to a penalised deconvolution regime by incorporating a directional total variation prior (Equation 3.25), and we construct kernelised deconvolution updates by substituting the kernel image model $x = K\alpha$, exactly as in KEM/HKEM. The resulting anatomy-aware deconvolution variants are referred to throughout as Kernelised Richardson–Lucy (KRL), Hybrid Kernelised Richardson–Lucy (HKRL), and Richardson–Lucy with Directional Total Variation (RL–DTV). Their basic update equations are derived in the following sections.

5.2 Image-domain inverse problem and objective

Let $x \in \mathbb{R}_+^J$ denote the unknown (sharper) PET activity image on a voxel grid, and let $y \in \mathbb{R}_+^J$ denote an observed blurred image. Although reconstructed PET images are not independent raw count data, their local noise is often treated as approximately Poisson in post-reconstruction restoration models [64]. We therefore use the following image-domain approximation:

$$y_j \sim \text{Poisson}(\bar{y}_j(x)), \quad \bar{y}(x) = Cx + b, \quad j = 1, \dots, J, \quad (5.1)$$

where $C \in \mathbb{R}_+^{J \times J}$ is a discretised convolution operator representing a space-invariant PSF, and $b \in \mathbb{R}_+^J$ is an optional additive background term (set to 0 in the experiments unless stated otherwise). The operator C plays the same role as the system matrix A in (3.3), but in the image domain.

Within this approximation, assuming independence across voxels in the image-

domain observation model, the Poisson negative log-likelihood is

$$\mathcal{L}(x) = \sum_{j=1}^J (\bar{y}_j(x) - y_j \log \bar{y}_j(x)), \quad \bar{y}(x) = Cx + b, \quad (5.2)$$

dropping constants independent of x as in Chapter 3.

A MAP (penalised-likelihood) deconvolution estimate is then

$$\hat{x} \in \arg \min_{x \in \mathcal{C}} \{ \mathcal{L}(x) + \beta \mathcal{R}(x) \}, \quad \mathcal{C} = \mathbb{R}_+^J, \quad (5.3)$$

with $\beta > 0$ and an appropriate regulariser $\mathcal{R}$.

Throughout, C is non-negative and (approximately) intensity preserving: for a unit-sum PSF kernel and negligible boundary effects, $C\mathbf{1} \approx \mathbf{1}$ and $C^\top \mathbf{1} \approx \mathbf{1}$. The adjoint $C^\top$ corresponds to convolution with the flipped PSF. For a symmetric PSFs common in PET, $C^\top \approx C$.

5.3 Baseline: Richardson–Lucy deconvolution

The RL algorithm is the EM algorithm for the Poisson model (5.1) with $b = 0$ [28]. Introducing elementwise multiplication and division, denoted $\odot$ and $\oslash$, the unit-sensitivity RL update is

$$x^{(k+1)} = x^{(k)} \odot C^\top \left(y \oslash (Cx^{(k)}) \right). \quad (5.4)$$

More generally, the right-hand side of (5.4) contains an elementwise division by $C^\top \mathbf{1}$; in this chapter this denominator is neglected because the PSF kernels are normalised and boundary effects are small. As with MLEM/OSEM under the full forward model, RL iterations tend to sharpen edges rapidly but amplify high-frequency noise as iterations increase. Consequently, unregularised RL is typically used with early stopping, which acts as an implicit regulariser but yields iteration-dependent bias.

5.3.1 Scaled-gradient interpretation and EM-type preconditioning

To connect with Chapter 3, we rewrite (5.4) as an additive scaled-gradient step. The gradient of the Poisson negative log-likelihood (5.2) (with $b = 0$) is

$$\nabla_x \mathcal{L}(x) = C^\top (\mathbf{1} - y \odot (Cx)). \quad (5.5)$$

Using the same rearrangement as for MLEM in Chapter 3, and again using $C^\top \mathbf{1} \approx \mathbf{1}$, we obtain

$$x^{(k+1)} = x^{(k)} \odot \left(\mathbf{1} - \nabla_x \mathcal{L}(x^{(k)}) \right). \quad (5.6)$$

This corresponds to an EM-type diagonal preconditioner analogous to $\mathcal{P}_{\text{EM}}^{(k)} = \text{diag}(x^{(k)} / (A^\top \mathbf{1}))$ in the full reconstruction problem.

5.4 Kernelised Richardson–Lucy deconvolution

Kernel methods were reviewed in Chapter 3 as an implicit regularisation strategy in which prior information enters through the image model $x = K\alpha$ rather than through an explicit penalty [35, 36]. The same idea can be applied to image-domain deconvolution.

5.4.1 Kernel image model

Let u be a co-registered anatomical guidance image (magnetic resonance (MR) or CT), resampled to the PET grid. We represent

$$x = K\alpha, \quad \alpha \in \mathbb{R}_+^J, \quad K \in \mathbb{R}_+^{J \times J}, \quad (5.7)$$

where K is a sparse, row-normalised kernel matrix constructed from features extracted from u (and optionally spatial coordinates), as in Chapter 3. Row normalisation ($K\mathbf{1} = \mathbf{1}$) ensures that constant coefficient images are mapped to constant activity images, preserving the intended intensity scale of the kernel representation.

Substituting into (5.1) gives the kernelised forward model

$$\bar{y}(\alpha) = C(K\alpha) + b. \quad (5.8)$$

5.4.2 KRL update

Maximising the Poisson likelihood with respect to α under (5.8) yields an EM-type multiplicative update analogous to KEM:

$$\alpha^{(k+1)} = \alpha^{(k)} \odot \frac{K^T C^T (y \oslash (CK\alpha^{(k)} + b))}{K^T C^T \mathbf{1}}, \quad x^{(k+1)} = K\alpha^{(k+1)}. \quad (5.9)$$

As in KEM, the denominator can be precomputed once, and the update preserves non-negativity given $\alpha^{(0)} \geq 0$. Under the same unit-sensitivity approximation used for vanilla RL, $C^T \mathbf{1} \approx \mathbf{1}$, the denominator reduces to $K^T \mathbf{1}$; this term is generally not equal to $\mathbf{1}$ and cannot be ignored.

5.4.3 Hybrid KRL (HKRL)

A known limitation of purely anatomy-derived kernels is suppression of PET-unique structures when they are not supported by the guidance image [47]. Hybrid kernels mitigate this by incorporating features derived from the evolving PET estimate [47].

Because iteration-dependent kernels can propagate noise back into the coupling structure at late iterations, we adopt a frozen-kernel strategy consistent with the motivation in HKEM, freezing after 5 iterations (although this proved to be unnecessary for the sphere phantom as the best iteration was usually before this point).

5.5 Directional-TV regularised RL deconvolution

directional total variation (DTV) was introduced in Chapter 3 as an anatomically guided regulariser that promotes edge alignment between the emission image and a structural guidance image [46]. Here we use DTV as an explicit penalty for deconvolution.

5.5.1 Penalised objective and update

The penalised-likelihood deconvolution objective is

$$\hat{x} \in \arg \min_{x \in \mathbb{R}_+^J} \{ \mathcal{L}(x) + \beta R_{\text{sDTV}}(x; u) \}, \quad \mathcal{L}(x) = \sum_j ((Cx + b)_j - y_j \log(Cx + b)_j). \quad (5.10)$$

Here R_{sDTV} is a smoothed DTV penalty. A natural baseline solver is the scaled-gradient RL–DTV iteration

$$x^{(k+1)} = \left\{ x^{(k)} \odot \left(\mathbf{1} - \varsigma_k \left[\nabla_x \mathcal{L}(x^{(k)}) + \beta \nabla_x R_{\text{sDTV}}(x^{(k)}; u) \right] \right) \right\}_{\geq 0}, \quad (5.11)$$

where $\{\}_{\geq 0}$ denotes projection onto non-negativity. Step sizes $\{\varsigma_k\}$ may be taken constant (with conservative tuning) or diminishing to improve stability. This update is the image-domain analogue of the scaled-gradient MAP schemes discussed in Chapter 3 (compare with BSREM-type forms), but without subsets.

In contrast to kernelised RL variants, RL–DTV is typically numerically stable once β is chosen, and the objective in (5.10) is smooth with simple bound constraints. This makes it well suited to a quasi-Newton solver: in the experiments of this chapter we therefore accelerate convergence by optimising (5.10) with the L-BFGS-B algorithm reviewed in Section 3.3, using bound constraints to enforce non-negativity. In this setting, L-BFGS-B replaces the hand-tuned step-size sequence $\{\varsigma_k\}$ by an automatic line search and curvature-informed scaling, often reaching a stationary point of (5.10) in substantially fewer iterations than the scaled-gradient RL–DTV update (5.11).

For RL and KRL/HKRL, early stopping is essential and is treated as part of the regularisation. In experiments we report results across iteration counts and select a stopping point based on NRMSE (Equation 5.15) for the phantom and qualitative assessment for the brain example. For RL–DTV, iteration sensitivity is reduced, but a stopping point is still selected to avoid unnecessary computation once the relative

objective decrease satisfies

$$\frac{\mathcal{O}(x^{(k)}) - \mathcal{O}(x^{(k+1)})}{\max\{|\mathcal{O}(x^{(k)})|, |\mathcal{O}(x^{(k+1)})|, 1\}} \leq f_{\text{tol}}, \quad \mathcal{O}(x) = \mathcal{L}(x) + \beta R_{\text{sDTV}}(x; u). \quad (5.12)$$

Here f_{tol} is chosen to be 10^{-6} . In L-BFGS-B implementations that expose the historical `factr` parameter, the equivalent objective tolerance is $f_{\text{tol}} = \text{factr} \varepsilon_{\text{mp}}$, where ε_{mp} is machine precision.

5.6 Data and experimental setup

Two datasets are used to characterise behaviour under controlled and realistic conditions.

1. **Spherical phantom:** a hot-sphere phantom enabling assessment of edge recovery and spill-over (see Figure 5.1).
2. **τ -PET brain with magnetic resonance imaging (MRI) guidance:** a representative in vivo dataset (see Figure 5.2).

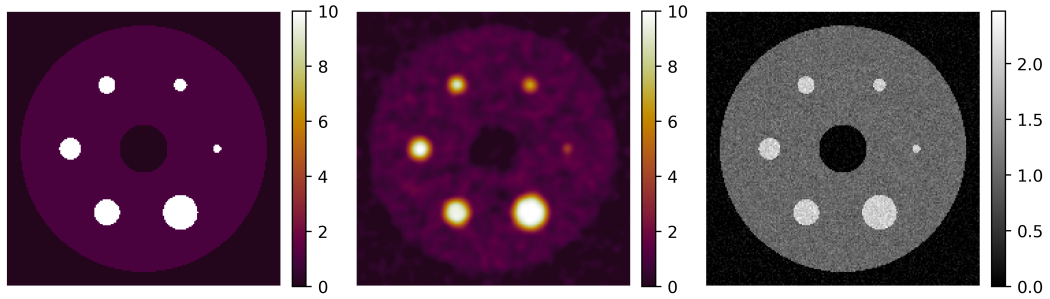


Figure 5.1: Left, ground truth phantom, centre blurred and noisy reconstruction and right MRI guidance image for the simulated sphere phantom. Units are arbitrary.

In both cases, the anatomical image u is registered to the PET grid and resampled to the reconstruction voxel size. The registered anatomy is used either to define the kernel K (KRL/HKRL) or the directional field $\{\xi_j\}$ (RL-DTV).

To obtain a controlled test case with known truth while retaining a realistic resolution/noise regime, the sphere dataset is simulated from a digital activity distribution by forward projecting with an isotropic Gaussian resolution model of

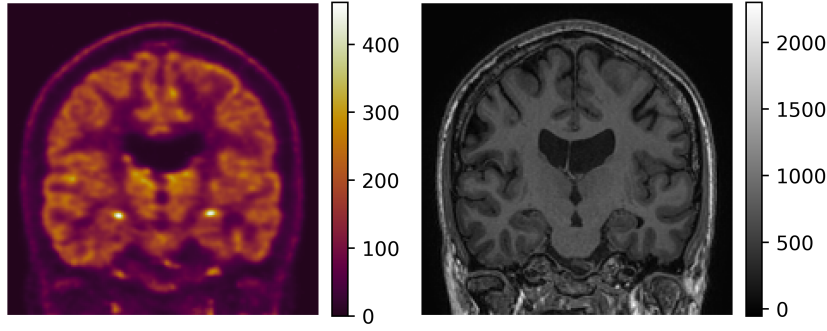


Figure 5.2: Left, reconstructed τ -PET brain, right registered MRI guidance image. Units are arbitrary.

FWHM = 6 mm embedded in the forward projector, adding Poisson noise to the resulting sinograms, and then reconstructing using an operator without additional blurring. This deliberately introduces a resolution-modelling mismatch between simulation and reconstruction. The deconvolution operator used in the subsequent image-domain correction was the same 6 mm Gaussian model, implemented with the same boundary handling as the simulation blur.

Let x^* denote the digital ground-truth activity image and $\hat{x}$ the reconstructed image. For each sphere v , let S_v be the set of voxel indices belonging to that sphere, and let $\mathcal{S} = \bigcup_v S_v$ denote the union over all spheres. We report an aggregate recovery coefficient computed by pooling all sphere voxels,

$$\text{RC} = \frac{\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \hat{x}_i}{\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} x_i^*}. \quad (5.13)$$

To quantify background noise, let $\mathcal{B}$ denote the set of voxel indices in a warm background region. We compute the coefficient of variation (CoV) within $\mathcal{B}$ as

$$\text{CoV}_{\mathcal{B}} = \frac{\sigma_{\mathcal{B}}}{\bar{x}_{\mathcal{B}}}, \quad \bar{x}_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \hat{x}_i, \quad \sigma_{\mathcal{B}} = \sqrt{\frac{1}{|\mathcal{B}| - 1} \sum_{i \in \mathcal{B}} (\hat{x}_i - \bar{x}_{\mathcal{B}})^2}. \quad (5.14)$$

Finally, overall fidelity to the truth is summarised by the normalised root mean

squared error (NRMSE) over the full image grid Ω ,

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{|\Omega|} \sum_{i \in \Omega} (\hat{x}_i - x_i^*)^2}}{\sqrt{\frac{1}{|\Omega|} \sum_{i \in \Omega} (x_i^*)^2}}. \quad (5.15)$$

The in vivo dataset is a τ -PET scan acquired on a healthy volunteer using the tracer ^{18}F -MK-6240, with a co-registered anatomical MRI serving as guidance. This dataset is used to qualitatively illustrate how the different priors behave in the presence of heterogeneous uptake patterns and fine anatomical structure, where over-regularisation can lead to anatomical “imprinting”. The FWHM of the deconvolution operator was estimated from a Hoffman brain phantom scan acquired on the same Siemens Vision scanner. The phantom reconstruction used PSF modelling; after removing extra-cerebral activity, the reconstructed image was co-registered to a Hoffman template, and the template was smoothed over a grid of Gaussian FWHM values, independently in the in-plane and axial directions. The smoothing that minimised the sum-of-squared differences to the reconstructed phantom image was selected, giving $\text{FWHM} = (4.6 \text{ mm}, 4.6 \text{ mm}, 6.4 \text{ mm})$.

5.6.1 Hyperparameter tuning

Hyperparameters were tuned by grid search.

For DTV reconstructions, we swept the regularisation weight

$$\beta \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0, 5.0\}.$$

Kernel methods used a fixed number of neighbours (48 nearest neighbours) within a cubic $9 \times 9 \times 9$ neighbourhood [36]. Feature channels were normalised before computing Gaussian similarities, as in the kernel construction described in Chapter 3; the reported σ_a and σ_f values are therefore interpreted in these normalised feature units. For KRL we swept the anatomical weighting σ_a . For HKRL we additionally swept the functional weighting σ_f . Because we used a k -nearest-neighbours strategy, distance weights (σ_{df}, σ_{da}) were not used. The implementation of the deconvolution methods used in these experiments is described in Appendix A.3.

Table 5.1: Hyperparameter sweeps for kernel-based deconvolution

Method	Hyperparameter	Values swept
KRL/HKRL	σ_a	{0.1, 0.2, 0.5, 1.0, 2.0, 5.0}
HKRL	σ_f	{0.1, 0.2, 0.5, 1.0, 2.0, 5.0}

The requirement for early stopping in RL, KRL and HKRL means that the initialising image plays a pivotal role in regularisation. We investigate the effect that initialisation could have on image quality by running identical sweeps over the sphere dataset initialising with a uniform image of ones and the blurred PET image.

τ -PET brain data reconstructions were initialised using a uniform image of ones for RL/KRL/HKRL and the blurred PET image for RL-DTV.

5.7 Results

5.7.1 Spheres

The results for the best reconstructions are summarised in Table 5.2 and visualised in Figure 5.3.

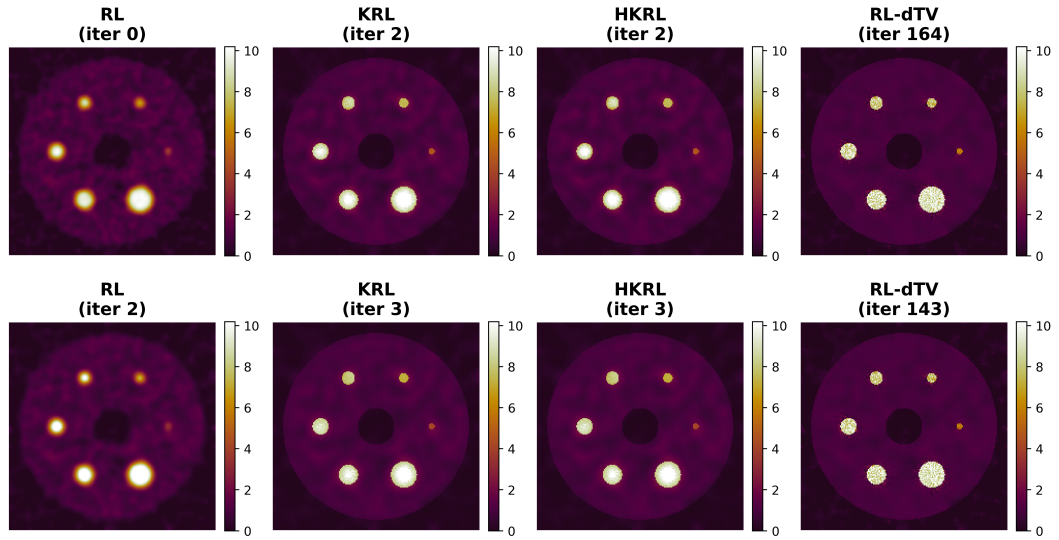


Figure 5.3: Figures showing central slice of reconstructions with best NRMSE. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.

Across both initialisation strategies, RL-DTV achieved the lowest NRMSE (0.0177), the highest aggregate recovery coefficient (approximately 0.889), and

Uniform initialisation							
Method	Min NRMSE	Iter.	RC	Bkg CoV	σ_a	σ_f	β
RL	0.0327	2	0.7645	0.2106	–	–	–
KRL	0.0190	3	0.8781	0.1546	0.5	–	–
HKRL	0.0191	3	0.8776	0.1546	0.5	5	–
RL-DTV	0.0177	143	0.8890	0.1233	–	–	2

Blurred initialisation							
Method	Min NRMSE	Iter.	RC	Bkg CoV	σ_a	σ_f	β
RL	0.0338	0	0.7307	0.2278	–	–	–
KRL	0.0208	2	0.8841	0.1805	5	–	–
HKRL	0.0212	2	0.8805	0.1805	5	5	–
RL-DTV	0.0177	164	0.8893	0.1233	–	–	2

Table 5.2: Tables summarising the hyperparameters and metric values of the best (measured by NRMSE) reconstructions of the various methods for (top) a uniform initialisation of ones and (bottom) the blurred PET image.

the lowest background CoV (approximately 0.123). KRL and HKRL substantially improved on unregularised RL, but were almost indistinguishable from one another in this simple sphere phantom. With blurred initialisation, the best RL result occurred at iteration 0, indicating that unconstrained RL deconvolution immediately increased the NRMSE once applied to this noisy reconstruction.

The effect of initialisation on the quality of reconstructions can be seen in Figures 5.4 and 5.5.

5.7.2 τ -PET brain

τ -PET brain data were only qualitatively analysed. Optimal images were chosen by visual quality, which is an obvious limitation of the study design. The chosen images are shown in Figure 5.6.

In the brain example, RL produced sharp but visibly noisy cortical structure, while KRL and HKRL suppressed noise but showed stronger dependence on the anatomical guidance image. RL-DTV gave the best qualitative compromise in this dataset: cortical definition was improved relative to the blurred image and the kernel methods, while noise amplification was lower than for unregularised RL. This

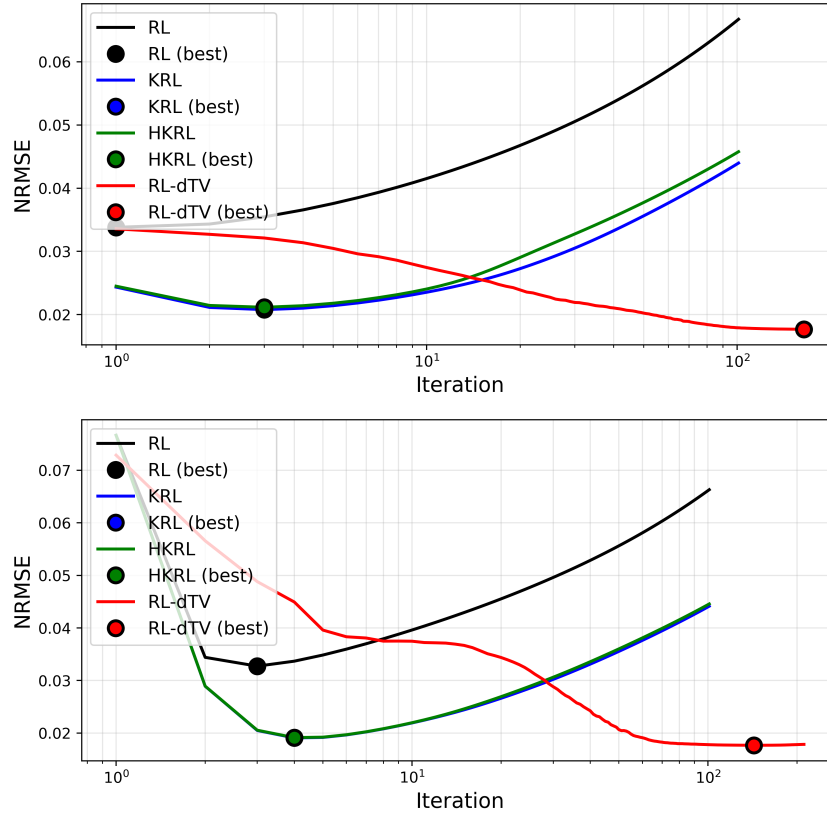


Figure 5.4: Figures showing iteration versus global NRMSE for the best performing reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image.

visual assessment supports the quantitative sphere result, but should be interpreted cautiously because it is based on a single healthy volunteer scan and does not include a reference truth.

5.8 Discussion

Deblurring is inherently ill-posed: the PSF attenuates high frequencies, so naive inversion magnifies noise. Anatomical guidance provides a structured way to preferentially preserve edges that are plausible given u while damping noise-like oscillations, either by restricting the solution space (kernel methods) or by penalising anatomically inconsistent gradients (DTV).

All guidance-based methods risk bias when the guidance image does not reflect tracer uptake boundaries. KRL is the most restrictive because it enforces $x \in \text{range}(K)$; RL-DTV is less restrictive but can still encourage alignment where

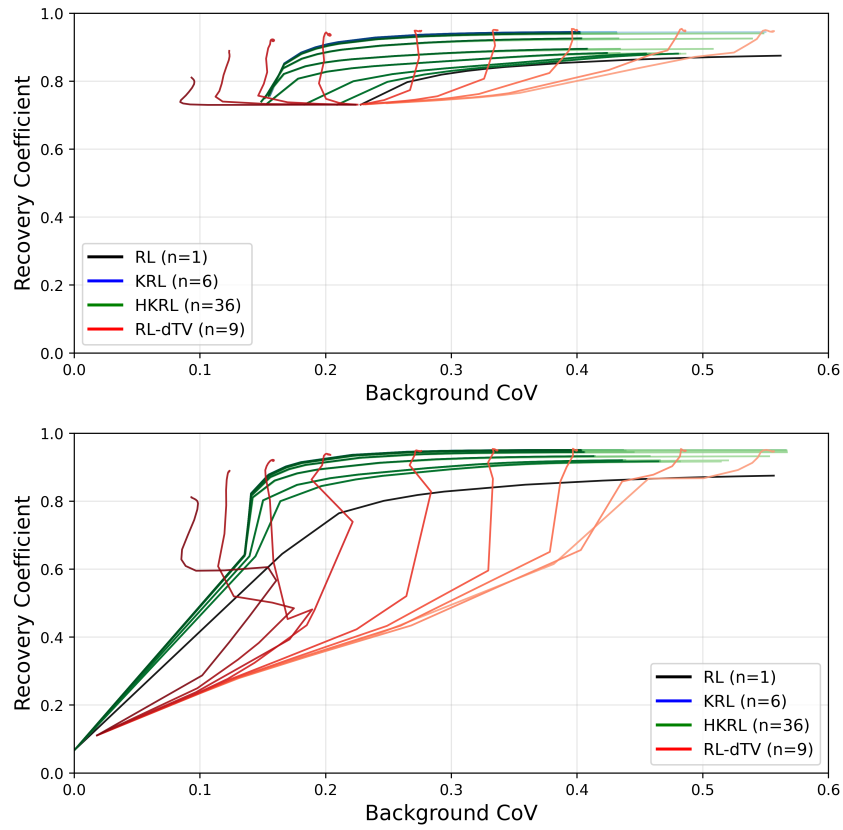


Figure 5.5: Figures showing RC versus CoV for all reconstructions. Top, initialised by the blurred PET image and bottom, initialised by a uniform image. KRL and HKRL are visually indistinguishable in these plots.

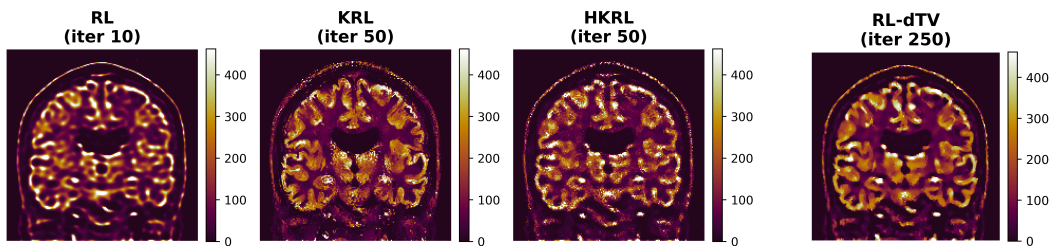


Figure 5.6: Figures showing visually best reconstructions for the different reconstructions.

the data term is weak (low counts, low contrast). HKRL partially mitigates mismatch by allowing the coupling structure to adapt to PET features, but must be controlled (e.g. freezing) to avoid noise-driven feedback.

The behaviours observed here anticipate challenges in the full reconstruction setting: strong priors induce spatially varying curvature and require effective scaling (preconditioning), and guidance mechanisms must be designed to avoid suppressing modality-unique information. These considerations directly motivate the synergy-aware prior and optimisation strategy developed later in the thesis.

The present study should therefore be viewed as a proof-of-principle rather than a definitive validation. A stronger evaluation would require a larger set of patient scans and more demanding controlled simulations, for example BrainWeb-style digital brain phantoms with inserted PET-unique lesions or tumours. Such work is ongoing, supported by additional clinical datasets, and would allow the qualitative observations here to be tested with population-level variability and known ground truth.

5.9 Summary

This chapter formulated post-reconstruction PET PVC as an approximate image-domain Poisson inverse problem $\bar{y}(x) = Cx + b$ and studied RL deconvolution and three anatomy-aware regularised variants. KRL and HKRL implement implicit regularisation through kernelised image models, consistent with the kernel framework reviewed in Chapter 3 [35, 36, 47]. RL-DTV implements explicit anatomically guided regularisation via a DTV penalty [46] using the same directional notation as Chapter 3. Together, these methods provide a practical PVC baseline and a controlled setting in which to diagnose guidance-driven bias and noise amplification, informing the design choices for the synergistic priors and optimisation algorithms developed in subsequent chapters.

Chapter 6

Introduction to Weighted Directional Total Nuclear Variation

In this chapter, we introduce a smoothed weighted directional total nuclear variation (w -dTNV) prior for synergistic emission tomography. The chapter focuses on the regulariser itself: its relationship to total nuclear variation, its differentiability after singular-value smoothing, the use of anatomical information through directional operators, and the modality-wise weighting needed when coupling heterogeneous emission images.

The optimisation machinery used to apply this prior in large-scale PET/SPECT reconstruction, including prior-aware preconditioning, subset selection, and multi-bed PET handling, is developed separately in Chapter 7.

6.1 Total Nuclear Variation Revisited

This section builds on the definitions in Section 3.2.3. For convenience, we restate the key definitions here:

- Registered images: $\mathbf{z} := \{z_m\}_{m=1}^M$, where $z_m := W_m x_m$ and $W_m : \mathbb{R}_+^{J_m} \rightarrow \mathbb{R}_+^N$.
- Finite-difference operator: for $\ell = 1, \dots, d$,

$$\left((Dx)_n\right)^{(\ell)} := \frac{x_{n+\delta_\ell} - x_n}{\Delta_\ell}.$$

- Voxel-wise Jacobian:

$$\mathcal{G}_n(\mathbf{z}) := \begin{bmatrix} Dz_{1,n} & Dz_{2,n} & \cdots & Dz_{M,n} \end{bmatrix} \in \mathbb{R}^{d \times M}.$$

- Nuclear norm: for a matrix Y with singular values $\{\sigma_k(Y)\}_{k=1}^r$ and $r = \text{rank}(Y)$,

$$\|Y\|_* := \sum_{k=1}^r \sigma_k(Y).$$

- Total nuclear variation:

$$\mathcal{R}_{\text{TNV}}(\mathbf{z}) := \sum_{n=1}^N \|\mathcal{G}_n(\mathbf{z})\|_*.$$

6.1.1 Mathematical properties and differentiability

Briefly revisiting total variation, optimisation of $\mathcal{R}_{\text{TV}}$ (cf. (3.17)) is inconvenient in gradient-based methods because these functionals are not globally smooth: the TV gradient is undefined when $(Dx_m)_j = 0$. A standard remedy is to introduce a small smoothing parameter $\varepsilon > 0$.

If we define

$$\tilde{\mathcal{R}}_{\text{TV}}(x_m) := \sum_{j=1}^{J_m} \|(Dx_m)_j\|_{2,\varepsilon}, \quad \|v\|_{2,\varepsilon} := \sqrt{\|v\|_2^2 + \varepsilon^2}, \quad (6.1)$$

then the image-space gradient is

$$\nabla_{x_m} \tilde{\mathcal{R}}_{\text{TV}}(x_m) = -\text{div} \left(\frac{Dx_m}{\|Dx_m\|_{2,\varepsilon}} \right), \quad (6.2)$$

where the division is understood pointwise over voxels.

For nuclear-norm vectorial total variation, the unsmoothed singular-value map is not differentiable at zero singular values, and hence the corresponding nuclear norm is not C^1 at rank-deficient points. We therefore apply the Charbonnier function

to the singular values:

$$g(s) = \sqrt{s^2 + \varepsilon^2} - \varepsilon, \quad g'(s) = \frac{s}{\sqrt{s^2 + \varepsilon^2}}, \quad g''(s) = \frac{\varepsilon^2}{(s^2 + \varepsilon^2)^{3/2}}. \quad (6.3)$$

The constant $-\varepsilon$ matches the implemented objective value and does not affect gradients or optimisation.

The nuclear norm is orthogonally invariant: for any orthogonal $U \in \mathbb{R}^{d \times d}$ and $V \in \mathbb{R}^{M \times M}$,

$$\|Y\|_* = \|UYV^\top\|_*. \quad (6.4)$$

More generally, any function of the form

$$\phi(Y) := f(\sigma(Y)), \quad f(s) = \sum_{\ell=1}^r g(s_\ell), \quad r = \min\{d, M\}, \quad (6.5)$$

is orthogonally invariant and depends on Y only through its singular values $\sigma(Y) = (\sigma_1(Y), \dots, \sigma_r(Y))$. Since $g \in C^\infty([0, \infty))$ for $\varepsilon > 0$, the symmetric function f is smooth and hence ϕ is differentiable, with gradient given by the standard formula for orthogonally invariant matrix functions [65, 66].

Jacobian-space gradient. Let $\phi(\mathcal{G}) = \sum_{\ell=1}^{\min(d, M)} g(\sigma_\ell(\mathcal{G}))$, where $\sigma_\ell(\mathcal{G})$ are the singular values of the voxel-wise Jacobian matrix $\mathcal{G} \in \mathbb{R}^{d \times M}$. A thin singular value decomposition (SVD)

$$\mathcal{G} = U \Sigma V^\top, \quad U \in \mathbb{R}^{d \times r}, \Sigma = \text{diag}(\sigma) \in \mathbb{R}^{r \times r}, V \in \mathbb{R}^{M \times r}, \quad (6.6)$$

with $r = \min\{d, M\}$ gives

$$\nabla_{\mathcal{G}} \phi(\mathcal{G}) = U \text{diag}(g'(\sigma)) V^\top, \quad g'(\sigma_\ell) = \frac{\sigma_\ell}{\sqrt{\sigma_\ell^2 + \varepsilon^2}}. \quad (6.7)$$

In practice, it is unnecessary to form U if we exploit the small Gram matrix

$$G := \mathcal{G}^\top \mathcal{G} \in \mathbb{R}^{M \times M}, \quad G \succeq 0. \quad (6.8)$$

Let $G = V\Lambda V^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_M)$. Then $\lambda_\ell = \sigma_\ell^2$ for the nonzero spectrum and V coincides with the right singular vectors of $\mathcal{G}$. On the range of $\mathcal{G}$ we have $U = \mathcal{G}V\Sigma^{-1}$, so substituting into (6.7) yields

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) = \mathcal{G}V \text{diag}\left(\frac{g'(\sigma_\ell)}{\sigma_\ell}\right) V^\top, \quad (6.9)$$

with the continuous extension at $\sigma_\ell = 0$. For Charbonnier smoothing,

$$\frac{g'(\sigma_\ell)}{\sigma_\ell} = \frac{1}{\sqrt{\sigma_\ell^2 + \varepsilon^2}} = \frac{1}{\sqrt{\lambda_\ell + \varepsilon^2}}, \quad (6.10)$$

hence

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) = \mathcal{G}V \text{diag}\left(\frac{1}{\sqrt{\lambda_\ell + \varepsilon^2}}\right) V^\top = \mathcal{G}(G + \varepsilon^2\mathbb{I})^{-1/2}. \quad (6.11)$$

Equation (6.11) shows that the Jacobian-space gradient is obtained by applying the inverse square-root of the $M \times M$ symmetric positive-definite (SPD) matrix $G + \varepsilon^2\mathbb{I}$ to the columns of $\mathcal{G}$.

For $M = 2$ modalities, even an eigendecomposition of G can be avoided. Writing

$$G = \begin{pmatrix} a & c \\ c & b \end{pmatrix}, \quad a = \|\mathcal{G}_{:,1}\|_2^2, \quad b = \|\mathcal{G}_{:,2}\|_2^2, \quad c = \langle \mathcal{G}_{:,1}, \mathcal{G}_{:,2} \rangle, \quad (6.12)$$

define $S := G + \varepsilon^2\mathbb{I} = \begin{pmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{pmatrix}$ with $s_{11} = a + \varepsilon^2$, $s_{22} = b + \varepsilon^2$, $s_{12} = c$, and the invariants

$$t := \text{tr}(S) = s_{11} + s_{22}, \quad s := \sqrt{\det(S)} = \sqrt{s_{11}s_{22} - s_{12}^2}. \quad (6.13)$$

Since $\varepsilon > 0$ implies $S \succ 0$, the following identity holds for 2×2 SPD matrices:

$$S^{-1/2} = \sqrt{t + 2s}(S + s\mathbb{I})^{-1} = \frac{\sqrt{t + 2s}}{2\det(S) + st} \begin{pmatrix} s_{22} + s & -s_{12} \\ -s_{12} & s_{11} + s \end{pmatrix}. \quad (6.14)$$

Therefore, for two modalities the Jacobian-space gradient can be evaluated via

$$\nabla_{\mathcal{G}}\phi(\mathcal{G}) = \mathcal{G}S^{-1/2}, \quad S = G + \varepsilon^2\mathbb{I}, \quad (6.15)$$

using only dot products and scalar square-roots per voxel.

Image-space gradient. Define voxel-wise

$$Q_n := \nabla_{\mathcal{G}}\phi(\mathcal{G}_n(\mathbf{z})) \in \mathbb{R}^{d \times M}, \quad (6.16)$$

and let $Q = \{Q_n\}_{n=1}^N$. Applying the chain rule through the linear map $\mathbf{z} \mapsto D\mathbf{z}$ yields

$$\delta\tilde{\mathcal{R}}_{\text{TNV}}(\mathbf{z})[\delta\mathbf{z}] = \sum_{n=1}^N \langle Q_n, (D\delta\mathbf{z})_n \rangle_F = \langle Q, D\delta\mathbf{z} \rangle. \quad (6.17)$$

By definition of the adjoint D^* ,

$$\langle Q, D\delta\mathbf{z} \rangle = \langle D^*Q, \delta\mathbf{z} \rangle. \quad (6.18)$$

Defining the discrete divergence as $\text{div} := -D^*$, we obtain

$$\nabla_{\mathbf{z}}\tilde{\mathcal{R}}_{\text{TNV}}(\mathbf{z}) = -\text{div } Q, \quad (6.19)$$

where div acts column-wise on Q .

6.2 Weighted Directional Total Nuclear Variation (w-dTNV)

This section extends TNV [38, 37] in two orthogonal ways:

- high-resolution anatomical information is incorporated directionally rather than as an additional coupled intensity channel;
- voxel-wise, modality-wise diagonal weighting compensates for heterogeneous scaling and spatially varying reliability across modalities.

The starting point is the standard TNV construction: penalising the nuclear norm of a multi-modality Jacobian promotes joint sparsity and alignment of edges, since a low nuclear norm favours a locally low-rank Jacobian.

6.2.1 Directional operators from guidance

To incorporate guidance information, such as a high-resolution anatomical image, we borrow the operator from directional total variation (Equation 3.24). The key design choice is that the guidance image is used only to steer the penalty, not to add an extra column to the joint Jacobian. This avoids directly coupling an anatomical channel, with distinct units, resolution, and artefacts, to the emission channels through the singular spectrum.

If a guidance image were appended as an additional modality, then a single global scaling would have data-dependent and spatially non-uniform consequences. Where emission gradients are weak, even modest guidance gradients can dominate the local Jacobian and hence the singular values; where emission gradients are strong, the same scaling may become negligible. Since the nuclear norm couples modalities through the eigenstructure of the local Gram matrix, such reweighting can also transfer structured guidance artefacts into the coupled reconstruction. By contrast, a directional operator modifies which gradient components are penalised, without forcing the presence of edges where the emission data do not support them.

Applying $\Pi_{m,n}$ to each modality gradient defines the directional multi-modality Jacobian

$$\mathcal{G}_n^{\text{dir}}(\mathbf{z}) = [\Pi_{1,n}(Dz_1)_n \mid \cdots \mid \Pi_{M,n}(Dz_M)_n] \in \mathbb{R}^{d \times M}. \quad (6.20)$$

Here D denotes the chosen discrete directional-difference operator and each $\Pi_{m,n} \in \mathbb{R}^{d \times d}$ is constructed from the guidance field at voxel n . In the implementation used in this work, a guidance image u defines

$$\xi_n = \frac{(Du)_n}{\sqrt{\|(Du)_n\|_2^2 + \eta^2}}, \quad \Pi_{m,n} = \mathbb{I}_d - \gamma_m \xi_n \xi_n^\top, \quad \gamma_m \in [0, 1]. \quad (6.21)$$

When η is not specified explicitly it is set from a robust scale estimate of $\|(Du)_n\|_2$,

namely 0.01 times the difference between its 99th and 1st percentiles. In homogeneous regions of the guidance image, ξ_n is small and $\Pi_{m,n}$ is close to the identity, so the regulariser reduces to an approximately isotropic TNV-type coupling.

Using CT only to construct the directional operators decouples anatomical steering from emission-to-emission weighting. Modality weights determine how strongly emission gradients are coupled to one another through the nuclear norm, while $\Pi_{m,n}$ determines how strongly the penalty prefers guidance-consistent edge directions. Retaining a two-modality Jacobian also preserves the efficient 2×2 Gram-matrix structure derived in Section 6.1.

6.2.2 Modality-wise weighting and the full w -dTNV prior

PET and SPECT images of the same administered Y-90 activity differ substantially in dynamic range: the effective count level, sensitivity, and noise characteristics vary by orders of magnitude between the two modalities, and in clinical imaging the two reconstructions may also operate at different spatial resolutions. If the directional emission gradients $\Pi_{m,n}(Dz_m)_n$ are used directly without normalisation, the nuclear-norm coupling is dominated by the modality with the larger gradient magnitudes.

To compensate, we introduce a voxel-wise, modality-wise diagonal weight matrix

$$B_n := \text{diag}(\rho_{1,n}, \rho_{2,n}) \in \mathbb{R}^{2 \times 2}, \quad \rho_{m,n} > 0. \quad (6.22)$$

The weighted directional Jacobian is

$$\tilde{\mathcal{G}}_n(\mathbf{z}) := \mathcal{G}_n^{\text{dir}}(\mathbf{z}) B_n = [\Pi_{1,n}(Dz_1)_n \mid \Pi_{2,n}(Dz_2)_n] \text{diag}(\rho_{1,n}, \rho_{2,n}) \in \mathbb{R}^{d \times 2}. \quad (6.23)$$

The columns of $\tilde{\mathcal{G}}_n$ are $\tilde{g}_{m,n} := \rho_{m,n} \Pi_{m,n}(Dz_m)_n \in \mathbb{R}^d$.

The weighted directional Total Nuclear Variation (w -dTNV) prior is then

$$\mathcal{R}_{w\text{-dTNV}}(\mathbf{z}) := V_{\text{vox}} \sum_{n=1}^N \phi(\tilde{\mathcal{G}}_n(\mathbf{z})), \quad \phi(Y) := \sum_{\ell=1}^{\min(d,M)} g(\sigma_\ell(Y)), \quad (6.24)$$

where V_{vox} is the voxel volume and g is the Charbonnier smoothing from (6.3).

Choosing equal weights $\rho_{m,n} \equiv 1$ recovers unweighted d TNV. Choosing $\rho_{m,n}$ proportional to the inverse of a characteristic modality scale brings both channels to a comparable range before the nuclear-norm coupling is evaluated. The PET/SPECT normalisation used in the application study is specified in Chapter 8.

Effect on the singular spectrum. Right-multiplying by B_n scales the m -th column of $\tilde{\mathcal{G}}_n$ by $\rho_{m,n}$. This scales the singular values of $\tilde{\mathcal{G}}_n$ in a non-trivial way, since the singular values of YB are not simply $\sigma_\ell(Y)\rho_m$. The intended effect is instead on the Gram matrix: $\tilde{\mathcal{G}}_n^\top \tilde{\mathcal{G}}_n$ contains modality self-terms and cross-terms after scaling, so suitable $\rho_{m,n}$ values allow both modalities to contribute comparably to the coupled singular spectrum. The 2×2 Gram-matrix algebra in Section 6.1 applies directly to this weighted case.

6.3 Conclusion

This chapter introduced a smoothed, weighted, and directional Total Nuclear Variation (w -dTNV) prior for synergistic reconstruction. Charbonnier smoothing of the singular values yields a differentiable spectral functional whose Jacobian-space gradient can be written in Gram-matrix form. For two coupled emission modalities this reduces the key spectral operation to a 2×2 inverse square-root, allowing the prior and its gradient to be evaluated robustly without a general per-voxel SVD.

The chapter also separated two roles that are easily confounded in multimodality reconstruction. Anatomical information is used directionally through $\Pi_{m,n}$, steering the penalty without adding CT as another coupled intensity channel. Modality scaling is handled separately through the diagonal weight matrix B_n , allowing PET and SPECT gradients to enter the nuclear-norm coupling on comparable scales. Chapter 7 uses these definitions to construct practical preconditioners and subset-selection strategies for large-scale w -dTNV optimisation.

Chapter 7

Preconditioning and Subset Selection for Synergistic Reconstruction

Chapter 6 defined the smoothed weighted directional total nuclear variation prior. Applying this coupled, edge-preserving prior at scale raises three linked optimisation questions. First, overlapping PET bed positions must be reconstructed into a common image if the prior is to act consistently across their overlap. Second, PET and SPECT subsets can be paired or treated as separate stochastic functions, and the prior can be folded into every sampled function, sampled alongside the data, or applied once per data pass. Third, the very different data and prior curvature scales require an effective preconditioner. This chapter addresses these questions in turn and uses the resulting evidence to motivate the optimisation protocol carried forward into Chapter 8. That application protocol had already been frozen before the wider preconditioner comparison reported here.

7.1 Multi-Bed PET Acquisition

Whole-liver PET commonly requires multiple partially overlapping bed positions. Reconstructing each bed independently and fusing the resulting images is awkward for an edge-preserving prior because the regularisation decisions on either side of the overlap are then made independently. A joint formulation instead estimates one activity image over the combined field of view.

7.1.1 Notation

Let x be the PET activity image on the combined reconstruction grid. For bed position $\text{bed} = 1, \dots, N_{\text{bed}}$, let R_{bed} restrict the combined image to the axial support reconstructed by that bed. The expected PET data for bed position bed are

$$\bar{y}_{\text{bed}}(x) = A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}, \quad (7.1)$$

where A_{bed} is the bed-specific PET system matrix and r_{bed} collects additive corrections. The joint PET data term is

$$\mathcal{D}_{\text{PET}}^{\text{multi}}(x) = \sum_{\text{bed}=1}^{N_{\text{bed}}} \mathcal{D}_{\text{bed}}(y_{\text{bed}} \mid A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}), \quad (7.2)$$

with gradient

$$\nabla \mathcal{D}_{\text{PET}}^{\text{multi}}(x) = \sum_{\text{bed}=1}^{N_{\text{bed}}} R_{\text{bed}}^{\top} A_{\text{bed}}^{\top} \nabla_{\bar{y}_{\text{bed}}} \mathcal{D}_{\text{bed}}(y_{\text{bed}} \mid A_{\text{bed}} R_{\text{bed}} x + r_{\text{bed}}). \quad (7.3)$$

The corresponding PET sensitivity image on the combined grid is

$$s_{\text{PET}}^{\text{multi}} = \sum_{\text{bed}=1}^{N_{\text{bed}}} R_{\text{bed}}^{\top} A_{\text{bed}}^{\top} \mathbf{1}. \quad (7.4)$$

This sensitivity is used in the EM/BSREM data-curvature term in exactly the same way as the single-bed sensitivity in Equation (7.15).

7.1.2 Why Not Stitch Independent Reconstructions?

A common practical alternative is to reconstruct each PET bed independently and fuse the images using sensitivity-weighted stitching in the overlap. With an edge-preserving prior, however, each reconstruction makes its own noise-suppression and edge decisions before fusion. The qualitative example in Figure 7.1 shows overlap-region noise and discontinuities after separate reconstruction and fusion. It is included to motivate the joint formulation; no quantitative comparison was performed. **TODO:** Add the Zenodo-managed citation for the multi-bed PET work.

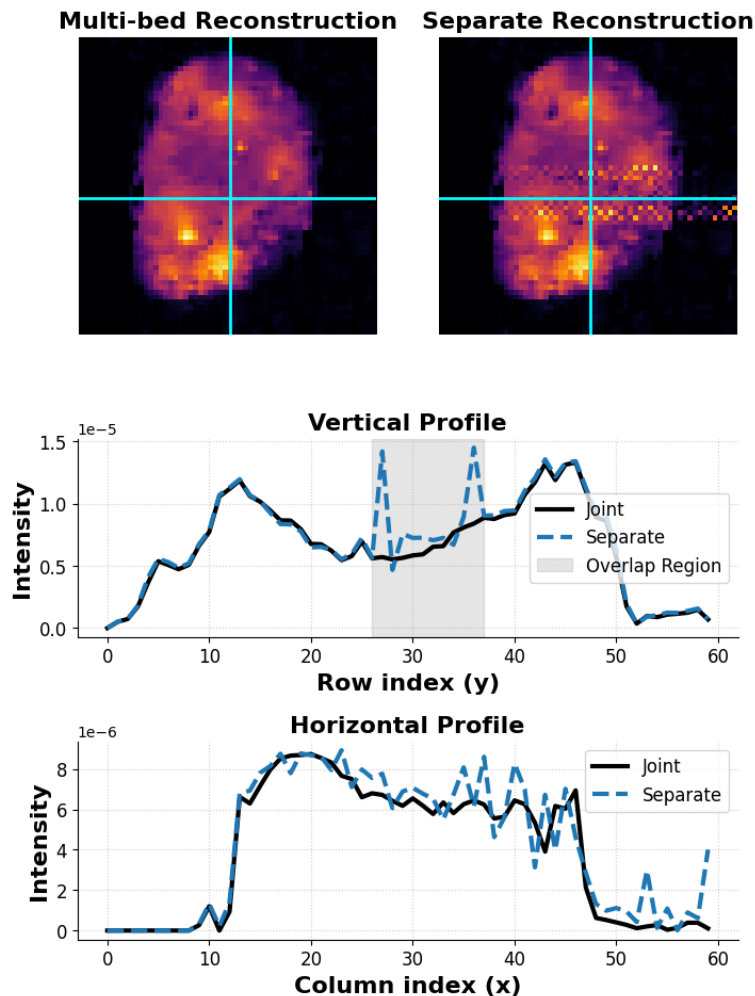


Figure 7.1: Qualitative comparison of separately reconstructed and fused PET bed positions with a joint multi-bed reconstruction using CT-guided edge-preserving regularisation. The shaded region marks the bed overlap and the cyan lines mark the displayed profiles. This example illustrates overlap-region behaviour only; no quantitative comparison was performed.

7.1.3 Joint Reconstruction

The joint formulation in Equation (7.2) estimates a single combined PET image. Each bed contributes its own data term and sensitivity, while the image-domain prior is evaluated over the complete combined field of view. The combined reconstruction therefore permits the edge-preserving prior to act consistently across the overlap, which is the behaviour required for the dual-bed application in Chapter 8. This construction does not itself guarantee an artefact-free reconstruction; the evidence here is qualitative.

7.2 Stochastic MAP Optimisation

Let $\mathbf{z} = (z_1, z_2)$ denote the PET and SPECT images represented on a common reconstruction grid. For N_d sampled data functions, define $\mathcal{D}(\mathbf{z}) = \sum_{i=1}^{N_d} \mathcal{D}_i(\mathbf{z})$. The objective is

$$\mathcal{O}(\mathbf{z}) = \mathcal{D}(\mathbf{z}) + \mathcal{R}_{\text{w-dTNV}}(\mathbf{z}) + \iota_{\mathbb{R}_+}(\mathbf{z}), \quad (7.5)$$

where $\mathcal{D}_i$ is a Poisson negative log-likelihood term composed with the relevant projector, image-space registration operator, and optional resolution model; $\mathcal{R}_{\text{w-dTNV}}$ is the prior from Chapter 6; and $\iota_{\mathbb{R}_+}$ is the indicator function of the non-negative orthant,

$$\iota_{\mathbb{R}_+}(\mathbf{z}) = \begin{cases} 0, & \mathbf{z} \geq 0, \\ +\infty, & \text{otherwise,} \end{cases} \quad (7.6)$$

with the inequality understood componentwise. As $\iota_{\mathbb{R}_+}$ is non-smooth, it is kept out of the gradient estimator $g^{(k)}$ below and instead handled by proximal (projected) gradient splitting: the smooth terms $\mathcal{D} + \mathcal{R}$ are advanced by a stochastic gradient step, and the result is passed through $\text{prox}_{\zeta^{(k)} \iota_{\mathbb{R}_+}}$, which for an indicator function reduces to Euclidean projection onto $\mathbb{R}_+$, i.e. clipping negative voxels to zero. This is exactly the $\{\cdot\}_{\geq 0}$ operation in the update Equation (7.8).

The stochastic experiments use SVRG [51, 42, 41]. Writing $\mathcal{R} = \mathcal{R}_{\text{w-dTNV}}$, the implemented estimator keeps the complete prior outside the sampler. At anchor point $\tilde{\mathbf{z}}$ it is

$$g^{(k)} = \nabla \mathcal{R}(\mathbf{z}^{(k)}) + \nabla \mathcal{D}(\tilde{\mathbf{z}}) + N_d \left[\nabla \mathcal{D}_{i_k}(\mathbf{z}^{(k)}) - \nabla \mathcal{D}_{i_k}(\tilde{\mathbf{z}}) \right], \quad i_k \sim \text{Unif}\{1, \dots, N_d\}. \quad (7.7)$$

The data functions are sampled uniformly with replacement. The complete prior gradient is evaluated at every stochastic update, and the full data-gradient snapshot is refreshed once per epoch. The subset-selection experiment in Section 7.3 also evaluates coarser prior schedules that keep the same average prior weight per data pass.

The image update is a projected, preconditioned gradient step,

$$\mathbf{z}^{(k+1)} = \left\{ \mathbf{z}^{(k)} - \varsigma^{(k)} \mathcal{P}^{(k)} g^{(k)} \right\}_{\geq 0}, \quad (7.8)$$

where $\mathcal{P}^{(k)}$ is either a diagonal preconditioner or a voxel-wise 2×2 block preconditioner coupling the PET and SPECT update components. The step size follows the linear decay rule

$$\varsigma^{(k)} = \frac{\varsigma^{(0)}}{1 + \eta_{\text{relax}} k}, \quad \varsigma^{(0)} = 1, \quad \eta_{\text{relax}} = 0.002, \quad (7.9)$$

for the 20-pass comparisons described below.

7.2.1 Common Evaluation Protocol

The convergence experiments used an anthropomorphic phantom scanned on the Mediso AnyScan at the National Physical Laboratory. PET and SPECT were each partitioned into 18 subsets. Each plotted comparison curve comprises five stochastic realisations and covers the first 20 complete data passes, extended to 25 for the preconditioner comparisons of Section 7.4. A single 1000-epoch reconstruction of the same objective, run with more conservative optimisation settings, provided the reference image x^* ; it is a final long-run iterate rather than a proven global minimiser. Appendix B reports the objective and iterate diagnostics supporting its effective stationarity relative to the best observed iterate.

For each stochastic run and each saved iteration, the normalised root-mean-square error (NRMSE) in region R is

$$\text{NRMSE}(x_k, x^*; R) = \frac{\sqrt{N^{-1} \sum_{j \in R} (x_{k,j} - x_j^*)^2}}{\sqrt{N^{-1} \sum_{j \in R} (x_j^*)^2}}, \quad N = |R|. \quad (7.10)$$

The regions are evaluated separately over the whole image, the hot-sphere mask, and the cold-sphere mask. The plotted centre line is the arithmetic mean of the five per-run NRMSE values at each iteration, not the NRMSE of an averaged reconstruction; the envelope is their pointwise minimum–maximum. The long-run reference itself

was generated once.

7.3 Subset Selection

Subset methods accelerate early iterations in emission tomography by replacing full data passes with stochastic or ordered blocks of projection data [67]. Joint PET/SPECT reconstruction introduces two further choices. Corresponding modality subsets can be added to form paired stochastic functions, or each modality subset can remain a separate function; and the prior can be applied on different schedules relative to the sampled data functions. The experiment below crosses these two subset organisations with three prior-application schedules. All three schedules deliver the same average prior weight, one complete prior contribution per data pass, and differ only in how finely that contribution is distributed across updates. **TODO:** Add the Zenodo-managed citation for IEEE Xplore document 10654962.

7.3.1 Data-Subset Organisation

The one-bed phantom data were divided into 18 PET and 18 SPECT subsets. Two organisations were tested.

Separate subsets. PET and SPECT subsets are treated as distinct stochastic functions. For a one-bed acquisition with $N_{\text{PET}} = 18$ and $N_{\text{SPECT}} = 18$, this gives

$$\{\mathcal{D}_i\}_{i=1}^{36} = \{\mathcal{D}_{\text{PET},1}, \dots, \mathcal{D}_{\text{PET},18}, \mathcal{D}_{\text{SPECT},1}, \dots, \mathcal{D}_{\text{SPECT},18}\}. \quad (7.11)$$

Paired subsets. PET and SPECT subsets with matching indices are summed into one stochastic function. For one bed this gives

$$\mathcal{D}_j^{\text{pair}} = \mathcal{D}_{\text{PET},j} + \mathcal{D}_{\text{SPECT},j}, \quad j = 1, \dots, 18. \quad (7.12)$$

7.3.2 Prior-Application Schedules

Writing N_d for the number of data functions in the chosen organisation ($N_d = 36$ separate, $N_d = 18$ paired), three schedules for applying the prior $\mathcal{R}$ were compared. All runs use the SVRG estimator.

Folded. The scaled prior $\mathcal{R}/N_d$ is added to every data function, so each sampled function is $\mathcal{D}_i + \mathcal{R}/N_d$ and every update evaluates the complete prior gradient. Because the prior enters every sampled function with the same weight, the estimator reduces algebraically to Equation (7.7): folding is mathematically identical to holding the complete prior outside the sampler and evaluating it at every update. This is the schedule used by the preconditioner experiments later in this chapter and by the application protocol of Chapter 8.

Sampled. The prior, scaled by $1/N_d$, is kept outside the data functions and instead enters the sampler as additional stochastic functions on the same footing as the data. Prior-only updates therefore occur on half of the draws in expectation, roughly every other update, and accumulate one complete prior contribution per data pass on average. This schedule is labelled prior/half in the figure legend.

Once per epoch. The sampled data updates exclude the prior entirely; the complete prior gradient is instead applied in a single dedicated update once per data pass.

7.3.3 Experiment Design and Work Accounting

All stochastic functions were sampled uniformly with replacement. To compare the configurations on a common horizontal axis, one matched data pass is defined by 36 modality-specific data-subset evaluations. Paired organisation reaches this count after 18 joint PET+SPECT data updates, each evaluating one PET and one SPECT subset; separate organisation reaches it after 36 modality-specific data updates. Prior work per matched pass then depends on the schedule. The folded schedule evaluates the complete prior gradient at every data update (36 evaluations per pass separate, 18 paired). The sampled schedule interleaves on average as many prior-only updates as data updates, roughly doubling the parameter-update count per pass for a comparable number of prior-gradient evaluations. The once-per-epoch schedule adds a single prior-gradient evaluation and update per pass.

This accounting matches data-projection work and average prior weight, but it does not match parameter-update count, prior-gradient work, or elapsed computation. Prior-only updates do not advance the horizontal axis, and a prior-gradient evaluation is much cheaper than a data-subset evaluation, so the additional prior work of the

folded and sampled schedules, up to N_d evaluations per pass, is not reflected in the plotted axis. All configurations used the same objective and the MM local-diagonal curvature preconditioner, and were assessed with the common 20-pass protocol in Section 7.2.1.

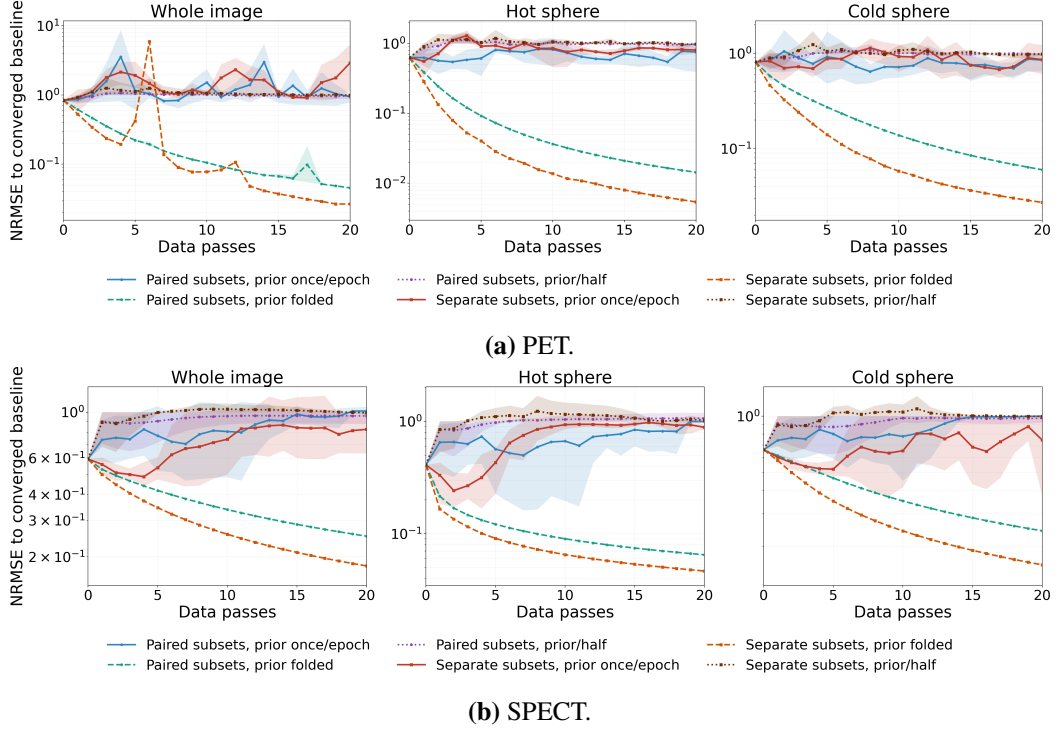


Figure 7.2: Convergence for the six combinations of subset organisation (paired, separate) and prior-application schedule (folded, sampled, once per epoch) using the MM local-diagonal curvature preconditioner. Lines show the mean per-run NRMSE over five stochastic realisations and shading shows the pointwise minimum–maximum over the first 20 matched data passes. In the legend, prior folded, prior/half, and prior once/epoch denote the folded, sampled, and once-per-epoch schedules; every schedule applies on average one complete prior contribution per matched data pass.

7.3.4 Results

Figure 7.2 separates the six configurations by prior schedule far more than by subset organisation. Only the folded schedule converged. Separate folded subsets reached the lowest NRMSE in every region for both modalities, with paired folded subsets close behind and the clearest gaps in the hot- and cold-sphere regions; the separate folded PET whole-image mean shows one large transient excursion near pass six before recovering. Under the sampled and once-per-epoch schedules,

neither organisation made net progress within the 20 passes: the curves stagnate near their starting values, with the once-per-epoch runs additionally showing large excursions and, for SPECT, early transient progress that subsequently regressed. Because all six configurations deliver the same average prior weight per matched data pass, the failures are attributable to the granularity of prior application rather than to its total contribution. A plausible explanation is that stable progress at these step sizes requires the data and prior gradients to be combined within each preconditioned update: the sampled and once-per-epoch schedules concentrate the prior into intermittent prior-only corrections, which repeatedly disturb iterates that the intervening data-only updates have advanced without regularisation.

Within the folded schedule, the organisation effect is plausibly an update-frequency effect: the separate organisation performs 36 parameter updates per matched data pass, compared with 18 paired updates, and therefore makes more frequent progress for the same number of data-subset projections. The experiment does not isolate modality grouping from update frequency, and should be interpreted as a data-projection-matched comparison of the implemented configurations, not as a compute-matched speed benchmark or a causal test of modality splitting alone. All runs used the MM local-diagonal curvature preconditioner with the common step-size schedule, so the schedule sensitivity reported here is an observation under that protocol rather than an established general property.

7.4 Prior-Aware Preconditioning

Preconditioning is critical for subset-based MAP reconstruction because the data term and the edge-preserving prior can have very different local curvature scales [68, 40, 69, 41]. The comparison below includes data-only BSREM scaling and four prior-aware curvature models for w -dTNN.

7.4.1 Data Curvature

For modality m , let s_m denote the sensitivity image on the shared grid,

$$s_m = W_m A_m^\top \mathbf{1}, \quad (7.13)$$

where A_m is the native-grid system matrix and W_m maps native images to the shared prior grid. The data-only EM/BSREM preconditioner is

$$\mathcal{P}_{\text{BSREM},m} = (z_m + \epsilon_x) \oslash s_m, \quad (7.14)$$

with element-wise division. This scaling models the data term only; it contains no prior curvature. Equivalently, the corresponding diagonal data Hessian surrogate is

$$\hat{H}_{\text{data},m} = s_m \oslash (z_m + \epsilon_x). \quad (7.15)$$

The implementation optionally smooths the numerator image used in the EM preconditioner with a Gaussian filter. In the experiments below the smoothing FWHM is 5 mm in each direction.

7.4.2 Prior Curvature Models

Four prior-curvature constructions were considered: three based on a local majorisation–minimisation (MM)/iteratively reweighted least squares (IRLS) surrogate and one based on the Lewis–Sendov Hessian of the smoothed spectral penalty. The term MM describes the derivation of the first family; none of the implemented preconditioners has been established as a majoriser of the complete objective, and no monotone-descent guarantee is claimed.

For the MM family, at voxel n let

$$\tilde{\mathcal{G}}_n = [\rho_{1,n}\Pi_{1,n}(Dz_1)_n \mid \rho_{2,n}\Pi_{2,n}(Dz_2)_n] \in \mathbb{R}^{d \times 2} \quad (7.16)$$

be the weighted directional Jacobian from Chapter 6 (Equation (6.23)). The active code forms the modality-space MM weight

$$C_n = \left(\tilde{\mathcal{G}}_n^\top \tilde{\mathcal{G}}_n + \epsilon^2 \mathbb{I}_2 \right)^{-1/2} = \begin{pmatrix} c_{11,n} & c_{12,n} \\ c_{12,n} & c_{22,n} \end{pmatrix}. \quad (7.17)$$

This matrix is the same 2×2 inverse square-root used in the gradient calculation in Chapter 6 (Equation (6.15)), now interpreted as the local quadratic-surrogate weight.

For diagonal preconditioning, the finite-difference stencil and directional projectors are collapsed to a per-voxel participation factor $\tau_{m,n}$. This factor includes the number of valid forward/backward stencil incidences at voxel n , the squared finite-difference scale factors, and the diagonal entries of $\Pi_{m,n}^\top \Pi_{m,n}$. The local diagonal w -dTNV curvature approximation is

$$\hat{H}_{\mathcal{R},m,n}^{\text{loc-diag}} = V_{\text{vox}} \rho_{m,n}^2 c_{mm,n} \tau_{m,n}. \quad (7.18)$$

This is the diagonal extraction of the local MM curvature surrogate. A more conservative Gershgorin-inflated diagonal surrogate adds the magnitude of the cross-modal term,

$$\hat{H}_{\mathcal{R},1,n}^{\text{diag-G}} = V_{\text{vox}} \rho_{1,n}^2 (c_{11,n} + |c_{12,n}|) \tau_{1,n}, \quad \hat{H}_{\mathcal{R},2,n}^{\text{diag-G}} = V_{\text{vox}} \rho_{2,n}^2 (c_{22,n} + |c_{12,n}|) \tau_{2,n}. \quad (7.19)$$

For block preconditioning, the PET and SPECT update components at each voxel are coupled by a 2×2 block. Let $\mathcal{N}(n)$ be the set of target voxels whose finite-difference stencil depends on source voxel n . For target voxel $r \in \mathcal{N}(n)$, let $c_{r \leftarrow n} \in \mathbb{R}^d$ be the finite-difference impulse vector produced in the directional channels when source voxel n is perturbed. The local voxel-block curvature approximation uses

$$\alpha_{r \leftarrow n}^{mm'} = c_{r \leftarrow n}^\top \Pi_{m,r}^\top \Pi_{m',r} c_{r \leftarrow n}, \quad (7.20)$$

and

$$\hat{H}_{\mathcal{R},n}^{\text{loc-block}} = V_{\text{vox}} \sum_{r \in \mathcal{N}(n)} [B_r C_r B_r] \circ A_{r \leftarrow n}, \quad A_{r \leftarrow n} = \begin{pmatrix} \alpha_{r \leftarrow n}^{11} & \alpha_{r \leftarrow n}^{12} \\ \alpha_{r \leftarrow n}^{21} & \alpha_{r \leftarrow n}^{22} \end{pmatrix}, \quad (7.21)$$

where $B_r = \text{diag}(\rho_{1,r}, \rho_{2,r})$ and $\circ$ denotes the Hadamard product.

The fourth construction uses the exact Lewis–Sendov Hessian of the smoothed spectral penalty [66]. If $H_{\mathcal{R}} = \nabla^2 \mathcal{R}$ denotes the resulting image-space Hessian, the

implemented approximation retains only its voxel-wise diagonal blocks,

$$\hat{H}_{\mathcal{R}}^{\text{LS-block}} = \text{blkdiag}([H_{\mathcal{R}}]_{11}, \dots, [H_{\mathcal{R}}]_{NN}), \quad [H_{\mathcal{R}}]_{nn} \in \mathbb{R}^{2 \times 2}. \quad (7.22)$$

It therefore retains PET/SPECT cross-curvature within each voxel but discards spatial off-diagonal blocks. Evaluating the smoothed spectral Hessian requires the voxel-wise singular-value decompositions used by the Lewis–Sendov construction. Recomputing this block model once per epoch consequently has a non-negligible cost.

7.4.3 Combining Data and Prior Curvature

Let $\mathcal{P}_{\text{D}} = \hat{H}_{\text{D}}^{-1}$ denote the data preconditioner and $\mathcal{P}_{\mathcal{R}} = \hat{H}_{\mathcal{R}}^{-1}$ the preconditioner obtained from one of the prior-curvature models above. For block methods, $\hat{H}_{\text{D}}$ is the 2×2 diagonal matrix containing the modality-specific data curvatures at each voxel. The principal comparison combines the operands through the inverse-sum, or parallel-sum, rule

$$\mathcal{P}_{\parallel} = (\mathcal{P}_{\text{D}}^{-1} + \mathcal{P}_{\mathcal{R}}^{-1})^{-1} = (\hat{H}_{\text{D}} + \hat{H}_{\mathcal{R}})^{-1}. \quad (7.23)$$

The same expression applies to diagonal and voxel-block operands, with numerical floors used before inversion. It is a combination of approximate curvature preconditioners, not a statement that either operand or their sum majorises the full objective.

The preconditioner is updated once per epoch and held fixed between updates. This limits curvature-computation cost, particularly for the Lewis–Sendov construction, but means the preconditioner becomes progressively stale within each epoch.

7.4.4 Preconditioner Comparison Methods

Five settings are compared: data-only BSREM, MM local diagonal, MM Gershgorin-inflated diagonal, MM local voxel-block, and Lewis–Sendov block-diagonal preconditioning. These settings are summarised in Table 7.1. The four prior-aware settings use the inverse-sum combination.

Table 7.1: Preconditioner models used in the principal comparison. The four prior-aware models use the inverse-sum combination in Equation (7.23).

Preconditioner	Prior-curvature construction	Combination
Data-only BSREM	None; EM data scaling only	Not applicable
MM local diagonal	Diagonal extraction of the local MM curvature surrogate	Inverse sum
MM Gershgorin-inflated diagonal	Local diagonal inflated by the magnitude of cross-modal coupling	Inverse sum
MM local voxel-block	Voxel-wise 2×2 local MM curvature surrogate	Inverse sum
Lewis–Sendov block-diagonal	Voxel-block diagonal of the exact smoothed spectral Hessian	Inverse sum

The comparison uses the one-bed anthropomorphic phantom, modality-separated subsets, and the folded prior schedule, i.e. the complete prior gradient at every update. All methods use the same objective, initialisation, long-run reference, initial step $\varsigma^{(0)} = 1$, and relaxation parameter $\eta_{\text{relax}} = 0.002$. Five stochastic realisations are evaluated for the first 25 matched data passes. Sensitivity-derived support masks are enabled with relative threshold 10^{-3} , and the data component is smoothed with a 5 mm Gaussian FWHM.

The reference is separate from this five-method comparison: it is a single 1000-epoch run with the same objective, a Gershgorin-diagonal MM preconditioner, and more conservative optimisation settings, including $\eta_{\text{relax}} = 10^{-4}$. It provides x^* for the NRMSE calculation rather than an additional candidate method.

7.4.5 Preconditioner Comparison

The four prior-aware constructions in Figure 7.3 behave broadly similarly. Their ordering varies between regions, and the differences are small compared with the distinction between prior-aware and data-only scaling; there is no universal winner. For count-limited PET, the whole-image data-only BSREM error grows by more than eight orders of magnitude within the first few passes and never recovers, indicating divergence under the tested schedule, whereas the prior-aware methods remain bounded and continue to reduce the hot- and cold-sphere errors. Prior curvature was therefore required for stable PET convergence under this common step-size

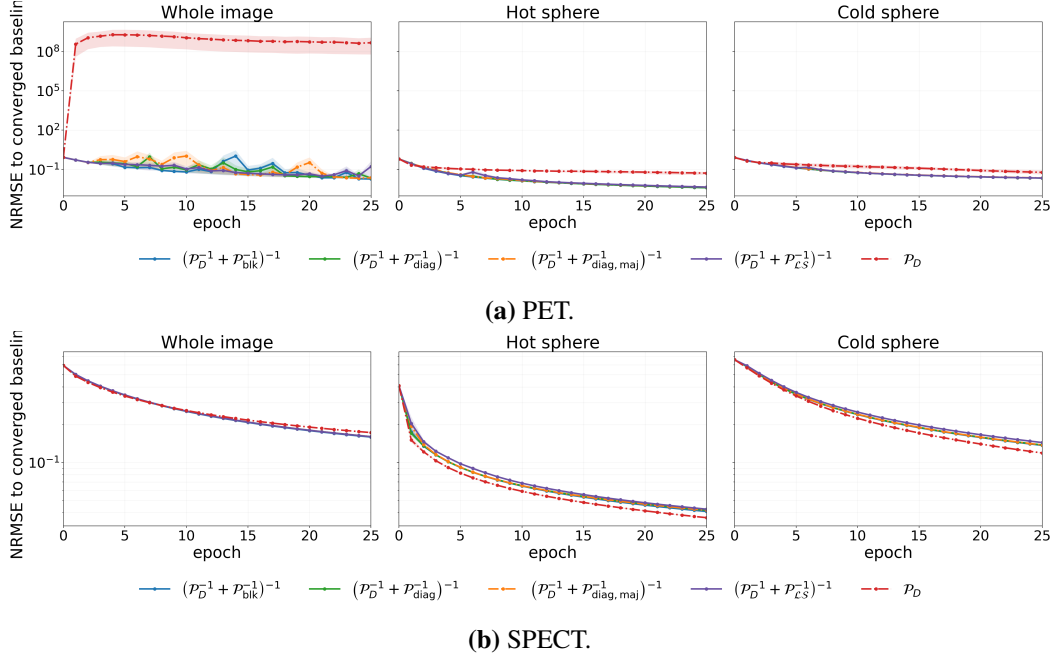


Figure 7.3: Convergence of data-only BSREM and the four prior-aware preconditioners in Table 7.1. All methods use the same objective and long-run reference within each modality. Lines show the mean per-run NRMSE over five stochastic realisations, and shading shows the pointwise minimum–maximum over the first 25 matched data passes. The PET vertical axis spans the full range reached by the data-only trajectory.

schedule.

SPECT is more robust. All five methods remain stable, and data-only BSREM is competitive with, or slightly faster than, the prior-aware alternatives in the sphere regions. This contrast suggests that the tested prior-aware scalings can be conservative for SPECT even though they are needed to stabilise the limited-count PET case.

7.4.6 Sensitivity to the Initial Step Size

Doubling the common initial step improves SPECT convergence across the three regions in Figure 7.4, but destabilises the limited-count PET reconstruction. A single shared relaxation schedule is therefore not equally well calibrated to both modalities. Modality-dependent initial steps or relaxation rates are a natural direction for future work, but were not tested here.

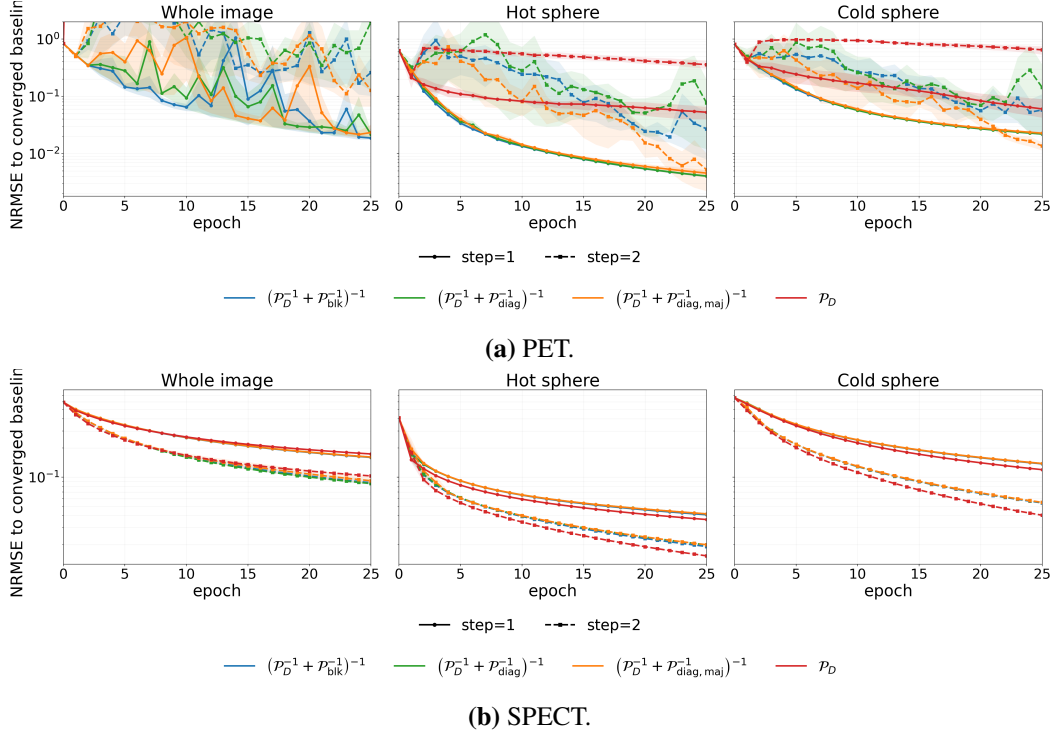


Figure 7.4: Sensitivity to the common initial step size for the displayed preconditioners. Solid curves use $\zeta^{(0)} = 1$ and dashed curves use $\zeta^{(0)} = 2$. Lines and envelopes again show the five-run mean and pointwise minimum–maximum over 25 matched data passes.

7.4.7 Lehmer Combination

The inverse sum is one member of a broader matrix Lehmer family. For the voxel-wise symmetric positive-definite preconditioners $\mathcal{P}_{\mathcal{R}}$ and $\mathcal{P}_{\mathcal{D}}$, define

$$C = \mathcal{P}_{\mathcal{D}}^{-1/2} \mathcal{P}_{\mathcal{R}} \mathcal{P}_{\mathcal{D}}^{-1/2}, \quad L_p(\mathcal{P}_{\mathcal{R}}, \mathcal{P}_{\mathcal{D}}) = \mathcal{P}_{\mathcal{D}}^{1/2} f_p(C) \mathcal{P}_{\mathcal{D}}^{1/2}, \quad f_p(t) = \frac{t^p + 1}{t^{p-1} + 1}, \quad (7.24)$$

where $f_p(C)$ is evaluated spectrally. The implemented preconditioner is $\frac{1}{2}L_p(\mathcal{P}_{\mathcal{R}}, \mathcal{P}_{\mathcal{D}})$. At $p = 0$ this reduces exactly to the parallel sum,

$$\frac{1}{2}L_0(\mathcal{P}_{\mathcal{R}}, \mathcal{P}_{\mathcal{D}}) = (\mathcal{P}_{\mathcal{D}}^{-1} + \mathcal{P}_{\mathcal{R}}^{-1})^{-1}. \quad (7.25)$$

For the Lewis–Sendov block model, $p = 0.01$ is marginally better than the inverse-sum result in these curves, but the difference is small. This comparison does not establish that the same order is preferable for the MM-derived preconditioners.

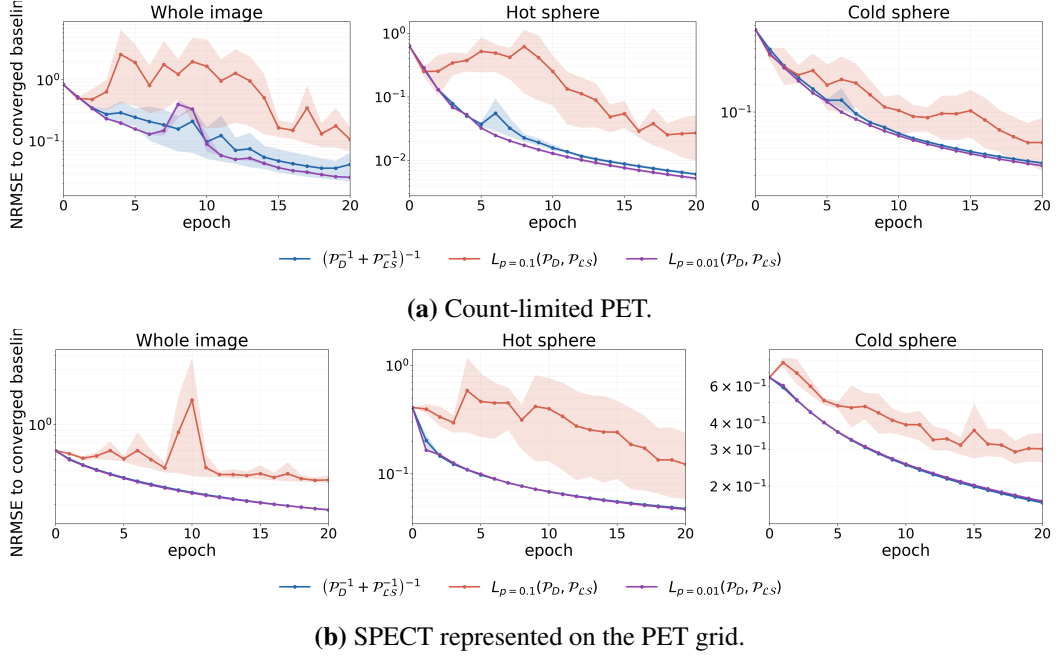


Figure 7.5: Parallel-sum and Lehmer combinations for the Lewis–Sendov block-diagonal prior model. This experiment changes only the combination rule; lines and shading show the five-run mean and pointwise minimum–maximum over 20 matched data passes.

Chapter 8 had already frozen the Lewis–Sendov block-diagonal model with $p = 0.01$ before the broader benchmark in this chapter. Retaining it therefore provides protocol continuity, not a post-hoc claim that Lewis–Sendov or $p = 0.01$ is universally optimal.

7.5 Conclusion

Joint reconstruction of the PET bed positions is required for the edge-preserving prior to act consistently across their overlap; the qualitative example showed the noise and discontinuities that can remain after separate reconstruction and fusion. For the one-bed convergence study, the prior-application schedule mattered more than the subset organisation: with the average prior weight per data pass matched, only the folded schedule, mathematically identical to evaluating the complete prior at every update, converged, while the sampled and once-per-epoch schedules stagnated under the common preconditioner and step-size protocol. Within the folded schedule, modality-separated subsets converged faster than paired subsets under matched data-

projection work, although they also performed twice as many parameter updates and prior-gradient evaluations per pass.

Prior-aware preconditioning was required for stable convergence of the count-limited PET reconstruction under the tested shared step schedule. SPECT was more robust, and data-only BSREM was sometimes competitive, suggesting that the prior-aware methods can be conservative for that modality. The four prior-curvature models behaved similarly overall, with no universal winner. Increasing the common initial step improved SPECT but destabilised PET, motivating modality-dependent relaxation as future work.

The Lewis–Sendov Lehmer experiment found only a small advantage for $p = 0.01$ over the parallel sum. Chapter 8 retains the previously frozen Lewis–Sendov block-diagonal/ $p = 0.01$ configuration for protocol continuity. Its use in that chapter should not be interpreted as selection of the best method after the wider comparison reported here.

Chapter 8

Weighted Directional Total Nuclear Variation for Joint ^{90}Y PET/SPECT Reconstruction with CTAC-derived Guidance

This chapter applies the weighted directional total nuclear variation (w -dTNV) prior developed in Chapter 6 to the clinically relevant problem of post-treatment ^{90}Y activity imaging after SIRT. We consider joint reconstruction of low-count ^{90}Y PET and bremsstrahlung SPECT, using computed tomography attenuation correction (CTAC) information directionally (via the CTAC-derived attenuation map μ) to stabilise edge geometry without enforcing intensity correlation.

Relative to Chapter 6, the focus here is validation and comparison with other methods: we specify modality-dependent forward models, co-registration operators, practical normalisation for heterogeneous PET/SPECT scales, and a reproducible evaluation protocol on a National Electrical Manufacturers Association (NEMA) International Electrotechnical Commission (IEC) phantom with PET bootstrapping and on a post-SIRT patient cohort.

8.1 Motivation and clinical context

Quantitative post-treatment imaging is required for personalised dosimetry following ^{90}Y SIRT [70]. Pre-treatment planning based on $^{99\text{m}}\text{Tc}$ -macroaggregated albumin (MAA) often mis-predicts microsphere deposition, motivating post-treatment verification of delivered activity distributions [71].

Two ^{90}Y modalities are clinically available:

- **^{90}Y PET**: detects the rare internal pair-production branch, providing comparatively high spatial resolution but extremely low count statistics [22].
- **Bremsstrahlung SPECT**: has substantially higher corrected counts (about $15\times$ for the data studied here) but suffers from severe blur, septal penetration, and scatter due to the continuous emission spectrum [61, 72].

Both modalities are typically acquired alongside CT for attenuation correction and anatomical localisation. Since PET and SPECT image the same underlying microsphere distribution while CTAC provides structural rather than activity information, independent reconstruction discards both a strong physical coupling and useful anatomical context.

The sequential hybrid kernelised expectation maximisation (SHKEM) method of Deidda *et al.* improves lesion contrast by reconstructing one modality to guide another [48, 73]. However, the sequential construction remains sensitive to iteration choice and noise amplification. This motivates a truly joint MAP formulation with a coupled regulariser that promotes shared geometry while allowing modality-specific contrast.

8.2 Joint MAP formulation for PET/SPECT

We use the joint MAP framework introduced in Section 3.2.3. For the present application we set $M = 2$ modalities, with $m = 1$ denoting PET and $m = 2$ denoting SPECT. Let $z_m \in \mathbb{R}_+^N$ denote the image for modality m represented on a shared grid and let $y_m \in \mathbb{N}^{I_m}$ denote the corresponding projection data.

The native image grid for modality m may differ from the shared grid. Once registration is fixed, let $W_m : \mathbb{R}_+^{J_m} \rightarrow \mathbb{R}_+^N$ denote the discrete resampling operator from

the native grid to the shared grid, and let $W_m^* : \mathbb{R}_+^N \rightarrow \mathbb{R}_+^{J_m}$ denote its discrete adjoint with respect to the chosen image-space inner products. The adjoint maps shared-grid variables back to the native grid before application of the system matrix. For each modality we therefore use the affine Poisson model

$$\bar{y}_m(z_m) = A_m W_m^* z_m + b_m, \quad (8.1)$$

where A_m is the system matrix and b_m collects additive corrections. For example, randoms and scatter for PET and simulated scatter/penetration terms for SPECT (see Sections 8.5.1 and 8.5.2).

The joint objective is

$$\hat{\mathbf{z}} = \arg \min_{\mathbf{z} \geq 0} \left\{ \sum_{m=1}^2 \mathcal{D}_m(y_m | A_m W_m^* z_m + b_m) + \mathcal{R}(\mathbf{z}) \right\}, \quad (8.2)$$

where $\mathcal{D}_m$ is the Poisson negative log-likelihood (up to an additive constant),

$$\mathcal{D}_m(y_m | \bar{y}_m) = \sum_{i=1}^{I_m} ((\bar{y}_m)_i - (y_m)_i) \log((\bar{y}_m)_i), \quad (8.3)$$

and $\mathcal{R}$ is the coupled prior. In this chapter, $\mathcal{R}$ is the smoothed *w-dTNV* prior $\tilde{\mathcal{R}}_{\text{w-dTNV}}$ defined in Section 6.2 (Chapter 6), specialised to PET/SPECT with CTAC guidance and practical global modality normalisation. The overall regularisation strength is carried by the modality weights introduced below, so no additional scalar λ is used in the implementation.

8.3 *w-dTNV instantiated for ^{90}Y PET/SPECT with CTAC guidance*

8.3.1 Directional guidance from the CTAC-derived μ -map

As in Section 6.2.1, we use CT information directionally rather than appending CT as a third coupled channel. Here the guidance image is the CTAC-derived attenuation map μ (resampled to the PET grid). We pre-smooth μ with a small blurring kernel

(FWHM = 2 mm) to reduce high-frequency CT noise while retaining high guidance edge resolution. We then compute a stabilised direction field in the discrete gradient space,

$$\xi_n = \frac{(D\mu)_n}{\sqrt{\|(D\mu)_n\|_2^2 + \eta^2}}, \quad \eta = 0.05 \cdot \max_n \|(D\mu)_n\|_2, \quad (8.4)$$

where D is the chosen directional finite-difference operator (specified in Section 8.3.3). The stabiliser η prevents amplification of small-magnitude μ differences in homogeneous regions.

We can then form the modality-wise directional operators $\Pi_{m,n} \in \mathbb{R}^{d \times d}$ using the construction already adopted in Chapter 6:

$$\Pi_{m,n} = \mathbb{I}_d - \gamma_m \xi_n \xi_n^\top, \quad \gamma_m \in [0, 1]. \quad (8.5)$$

Unless stated otherwise, we set $\gamma_1 = \gamma_2 = 1$. Since ξ_n is stabilised and generally has norm less than one, this should be interpreted as anisotropic shrinkage of gradient components parallel to the local μ -edge normal direction rather than as a hard projection.

8.3.2 Global modality normalisation as a constant diagonal weighting

Chapter 6 introduced a general voxel-wise diagonal modality weighting $B_n = \text{diag}(\rho_{1,n}, \dots, \rho_{M,n})$. In the present study we use a spatially uniform normalisation to mitigate the dominant scale disparity between ^{90}Y PET and bremsstrahlung SPECT:

$$B_n \equiv B := \text{diag}(\rho_1, \rho_2), \quad \rho_1, \rho_2 > 0. \quad (8.6)$$

Weights are computed from early-stopped unregularised reconstructions represented on the shared grid, z_m^{init} , via robust dynamic-range (percentile) scaling over a CTAC-derived foreground mask Ω' :

$$\rho_m = \frac{\beta_m}{q_{0.99}(z_m^{\text{init}}|\Omega') - q_{0.01}(z_m^{\text{init}}|\Omega')}, \quad m \in \{1, 2\}, \quad (8.7)$$

where $q_p(\cdot|_{\Omega'})$ denotes the p th percentile over voxels in Ω' , and β_m are user-controlled regularisation weights used for operating-point selection (Section 8.7.1). Percentile scaling provides a robust dynamic-range estimate and reduces sensitivity to outliers. This matches the intent of Section 6.2.2 while avoiding an additional spatially varying weight for this investigation [74].

8.3.3 Directional finite-difference operator for PET/SPECT coupling

We use the generic finite-difference definition from Chapter 6 (Section 6.1) but specify the set of offsets to better accommodate SPECT blur and modest registration error. Let $\Delta = (\Delta_x, \Delta_y, \Delta_z)$ denote the voxel sizes in SI units. We define a nine-direction half-stencil

$$\mathcal{E}_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \cup \{(\pm 1, 1, 0), (\pm 1, 0, 1), (0, 1, \pm 1)\}, \quad (8.8)$$

and use both one- and two-voxel steps, $\mathcal{E} = \mathcal{E}_1 \cup \{2e : e \in \mathcal{E}_1\}$, yielding $d = 18$ directions. For a set of offsets $\mathcal{E} = \{e_\ell\}_{\ell=1}^d$, define the d -directional forward difference operator D by

$$((Dz)_n)^{(\ell)} := \frac{z_{n+e_\ell} - z_n}{\|e_\ell\|_\Delta}, \quad \|e\|_\Delta := \sqrt{(e_x\Delta_x)^2 + (e_y\Delta_y)^2 + (e_z\Delta_z)^2}, \quad (8.9)$$

with Neumann boundary conditions. This follows the “extended Jacobian” motivation for TNV-type coupling [37] and empirically improved stability in the present PET/SPECT setting.

8.3.4 The w - d TNV Jacobian and regulariser (two modalities)

Using the notation of Chapter 6, we form the guided gradients $\Pi_{m,n}(Dz_m)_n \in \mathbb{R}^d$ and define the weighted directional Jacobian

$$\tilde{\mathcal{G}}_n := \tilde{\mathcal{G}}_n^{\text{w-dTNV}}(\mathbf{z}) = \left[\Pi_{1,n}(Dz_1)_n \mid \Pi_{2,n}(Dz_2)_n \right] B \in \mathbb{R}^{d \times 2}. \quad (8.10)$$

The regulariser is the smoothed spectral potential introduced in Chapter 6, weighted by the voxel volume V_{vox} ,

$$\tilde{\mathcal{R}}_{w\text{-dTNV}}(\mathbf{z}) = V_{\text{vox}} \sum_{n=1}^N \phi(\tilde{\mathcal{G}}_n), \quad \phi(Y) = \sum_{\ell=1}^2 g(\sigma_{\ell}(Y)), \quad (8.11)$$

where g is the Charbonnier smoothing (Equation 6.3) and $\sigma_{\ell}(Y)$ are singular values. We set the smoothing parameter ε relative to initial joint-gradient magnitudes:

$$\varepsilon = 0.05 \cdot \text{median}_{n \in \Omega'} \sigma_1(\tilde{\mathcal{G}}_n^{\text{init}}), \quad \tilde{\mathcal{G}}_n^{\text{init}} := \tilde{\mathcal{G}}_n^{w\text{-dTNV}}(\mathbf{z}^{\text{init}}), \quad (8.12)$$

with $\mathbf{z}^{\text{init}} = (W_1 x_1^{\text{init}}, W_2 x_2^{\text{init}})$.

If $\Pi_{2,n}(Dz_2)_n \approx 0$ at some voxel (e.g. SPECT locally uniform after guidance), then $\tilde{\mathcal{G}}_n$ has an effectively single nonzero column and $\phi(\tilde{\mathcal{G}}_n)$ reduces to a TV-type penalty on the remaining modality column. This is the same “no forced edges” behaviour discussed in Chapter 6.

8.3.5 Ablation comparators

To separate the contributions of PET/SPECT coupling and CTAC-derived directional guidance, we compare w -dTNV with two variational ablations. The first is weighted TNV (w -TNV), which keeps the two-modality Jacobian and the same modality normalisation but removes directional CTAC guidance by setting $\Pi_{m,n} = \mathbb{I}_d$ for both modalities:

$$\tilde{\mathcal{G}}_n^{w\text{-TNV}}(\mathbf{z}) = \left[(Dz_1)_n \mid (Dz_2)_n \right] B. \quad (8.13)$$

This tests PET/SPECT coupling without anatomical anisotropy.

The second is directional TV (dTV), which keeps CTAC-derived directional guidance but removes PET/SPECT coupling. It is obtained by applying the same directional operator only to the PET gradient,

$$\tilde{\mathcal{R}}_{\text{dTV}}(z_1) = V_{\text{vox}} \sum_{n=1}^N \phi(\rho_1 \Pi_{1,n}(Dz_1)_n), \quad (8.14)$$

which reduces to a smoothed single-channel directional-TV penalty. This tests

anatomical anisotropy without using SPECT as a coupled functional channel. All variational methods use the same forward models, registration, foreground mask, dynamic-range normalisation, and optimisation framework; only the regulariser structure is changed.

8.4 Optimisation and implementation

We minimise the joint objective (8.2) using a subset stochastic variance-reduced gradient (SVRG) method [51], consistent with the stochastic framework studied in Chapter 7. Let N_d be the total number of PET and SPECT data functions in the reconstruction being described, including separate functions for each PET bed position where applicable. The application implementation distributes the prior equally across these functions,

$$\mathcal{O}(\mathbf{z}) = \sum_{i=1}^{N_d} \mathcal{O}_i(\mathbf{z}), \quad \mathcal{O}_i(\mathbf{z}) = \mathcal{D}_i(\mathbf{z}) + \frac{1}{N_d} \tilde{\mathcal{R}}_{\text{w-dTNV}}(\mathbf{z}), \quad (8.15)$$

so that summing all stochastic functions recovers one complete prior term. This folded organisation is mathematically identical to holding the complete prior outside the data-function sampler and evaluating it at every update, as in the Chapter 7 estimator, and it matched the only prior-application schedule that converged reliably in the subset-selection comparison there (Section 7.3). Every stochastic function includes the scaled prior contribution; the prior is not sampled as a separate subset. Data functions are formed using staggered, view-interleaved partitioning and are split by modality, so each $\mathcal{D}_i$ contains either PET or SPECT data.

At iteration k , with anchor point $\tilde{\mathbf{z}}$ refreshed once per epoch, we use the SVRG estimator

$$\mathbf{g}^{(k)} = \nabla \mathcal{O}(\tilde{\mathbf{z}}) + N_d \left[\nabla \mathcal{O}_{i_k}(\mathbf{z}^{(k)}) - \nabla \mathcal{O}_{i_k}(\tilde{\mathbf{z}}) \right], \quad i_k \sim \text{Unif}\{1, \dots, N_d\}, \quad (8.16)$$

and apply a projected voxel-block-preconditioned update

$$\mathbf{z}^{(k+1)} = \left\{ \mathbf{z}^{(k)} - \varsigma^{(k)} \mathcal{P}^{(k)} \mathbf{g}^{(k)} \right\}_{\geq 0}, \quad (8.17)$$

with a decaying step size

$$\varsigma^{(k)} = \frac{\varsigma^{(0)}}{1 + \delta k}, \quad \varsigma^{(0)} = 1, \quad \delta = 0.005. \quad (8.18)$$

Reconstructions were run for 250 epochs, corresponding to approximately 500 full data passes because the full-gradient snapshot is recomputed once per epoch. This fixed stopping point was used for all methods rather than a case-specific convergence criterion.

8.4.1 Block-diagonal preconditioning

For modality m , the data component is the EM-type scaling [69, 40]

$$\mathcal{P}_{\text{data},m} = z_m \odot \left(W_m A_m^\top \mathbf{1} \right), \quad (8.19)$$

represented on the shared grid. At voxel n , these modality-specific values form the diagonal block

$$\mathcal{P}_{D,n} = \begin{pmatrix} \mathcal{P}_{\text{data},1,n} & 0 \\ 0 & \mathcal{P}_{\text{data},2,n} \end{pmatrix}. \quad (8.20)$$

The prior component $\mathcal{P}_{\mathcal{R},n}$ is obtained by inverting the Lewis–Sendov voxel-block diagonal of the exact smoothed spectral Hessian described in Section 7.4.2. It retains the PET/SPECT cross-curvature within each 2×2 voxel block while discarding spatial off-diagonal blocks. The prior blocks are recomputed once per epoch using the smoothed spectral Hessian and its voxel-wise singular-value decompositions, then held fixed within that epoch.

After applying an eigenvalue floor $\lambda_{\min} = 10^{-6}$ to both operands, the application combines them using the scaled matrix Lehmer construction from Equation (7.24). Specifically,

$$C_n = \mathcal{P}_{D,n}^{-1/2} \mathcal{P}_{\mathcal{R},n} \mathcal{P}_{D,n}^{-1/2}, \quad \mathcal{P}_n = \frac{1}{2} \mathcal{P}_{D,n}^{1/2} f_p(C_n) \mathcal{P}_{D,n}^{1/2}, \quad f_p(t) = \frac{t^p + 1}{t^{p-1} + 1}, \quad (8.21)$$

with $p = 0.01$. At $p = 0$ this expression is the parallel sum. The Lewis–Sendov/ $p = 0.01$ setting was frozen for this application before the broader optimisation bench-

mark in Chapter 7; its use here reflects protocol continuity rather than selection as the universally best preconditioner.

8.4.2 Comparator: sequential hybrid kernel EM

We compare against the sequential hybrid kernel expectation–maximisation (SHKEM) method of Deidda *et al.* [48], using the reconstruction order and hyperparameter settings reported as optimal in that work. The sequential order is CT-guided SPECT followed by CT/SPECT-guided PET: SPECT is reconstructed first using CT-derived anatomical features and hybrid kernel updates, and the resulting SPECT reconstruction is then used together with CT-derived features and the current PET iterate to construct the PET hybrid kernel [73]. To reduce noise amplification we use frozen kernels, halting kernel updates after 27 PET subiterations and 72 SPECT subiterations, matching the comparator configuration in [48]. All kernel parameters followed the original implementation from Deidda *et al.* For clinical PET data, the SHKEM implementation is applied in the same joint multi-bed setting as the variational methods.

8.4.3 Software implementation

All reconstructions were implemented in the Synergistic Image Reconstruction Framework (SIRF) [75] using Software for Tomographic Image Reconstruction (STIR) projector backends [76, 77, 78] and the Core Imaging Library [79], with custom code for the proposed regulariser. The SHKEM implementation follows [48], extended here to the same joint multi-bed PET reconstruction setting used by the variational methods.

8.5 Forward models, corrections, and co-registration

8.5.1 PET forward model

PET list-mode data were processed using the vendor toolchains to generate normalisation, attenuation, and additive corrections (randoms and scatter), which were held fixed during reconstruction [80, 81]. Reconstructions use a voxel size of $3.27 \times 2.13 \times 2.13 \text{ mm}^3$ (z,y,x) on a grid of $83 \times 255 \times 255$.

Patient PET data were acquired across two partially overlapping bed positions. For all methods, these bed positions were reconstructed jointly in a single combined-bed image space, with separate forward- and back-projections for each bed position; for the variational methods, the image-domain prior is evaluated over the combined volume. This is the multi-bed formulation derived in Chapter 7. It permits one edge-preserving prior to act over the combined field of view and was motivated by the qualitative overlap comparison in Figure 7.1. Bed positions overlapped by 11 voxels. Phantom PET data were reconstructed using a single bed position.

Projection data are partitioned into staggered, view-interleaved subsets. For SHKEM we use $S = 9$ PET subsets, following the comparator study. For SVRG-based reconstructions we use 18 PET subsets for phantom data and 18 subsets per PET bed position for clinical data, giving 36 PET subsets in total for two-bed patient reconstructions.

8.5.2 SPECT forward model

For Bremsstrahlung SPECT we use the fixed acquisition model developed and simulation-cross-checked in Chapter 4: a geometric projector with an analytic intrinsic-plus-geometric primary response, an image-space range-blur kernel and a simulated non-primary additive term. SPECT images are discretised on a $128 \times 128 \times 128$ grid with isotropic voxel size 4.18 mm.

The geometric collimator response is modelled as a depth-dependent Gaussian with linearly varying width, obtained by fitting an analytic collimator model [63]. Patient scatter, collimator scatter and septal penetration are excluded from this primary operator. Bremsstrahlung blur is represented by the fixed 6.9 mm-FWHM effective isotropic Gaussian calibrated and cross-checked against the ^{90}Y Monte Carlo response study in Section 4.2.3 [19].

Non-primary contamination is incorporated through an additive term prepared before the joint reconstruction. As described in Section 4.1.2, the term is estimated by an outer reconstruct–simulate loop using Simulation of Imaging Nuclear Detectors (SIMIND) [59, 82, 61] (Appendix A.1). We initialise it using a learned estimate

from a neural network [60] (Appendix A.2) and refine it for 10 outer iterations using

$$b^{(k+1)} = \alpha \hat{b}^{(k+1)} + (1 - \alpha)b^{(k)}, \quad \alpha = 0.2, \quad (8.22)$$

where $\hat{b}^{(k+1)}$ is the newly simulated additive estimate and $b^{(k)}$ is the previous estimate. The final additive estimate is fixed throughout the stochastic joint optimisation; no further Monte Carlo update is performed during that reconstruction. It is therefore a data- and activity-dependent plug-in estimate: fixing it defines a fixed conditional objective, but does not propagate uncertainty in the additive correction or guarantee consistency with the final jointly regularised SPECT image.

SPECT projection data are partitioned into staggered subsets, with $S = 12$ for both SVRG-based reconstructions and SHKEM.

8.5.3 Co-registration and warp operators

CTAC provides the anatomical reference for co-registration and for guidance via the μ -map. The shared reconstruction grid is the PET grid. The PET CTAC is therefore downsampled to the PET reconstruction grid and used both to define the attenuation map μ and as the anatomical reference space for the coupled prior, so W_1 is the identity on the shared grid.

The PET and SPECT CTAC volumes were retained in their native acquisition grids before registration. Since patient arm positioning differed between modalities (arms down in PET, arms up in SPECT), registration was restricted to the trunk. A trunk mask was obtained automatically from each CTAC using TotalSegmentator [TODO`totalsegmentator`2023] **TODO: Zenodo-managed bibliography entry pending for TotalSegmentator; replace TODO_totalsegmentator_2023 once imported.**, and the native-grid SPECT CTAC was registered to the PET CTAC using a rigid transformation followed by deformable registration initialised from the rigid solution. The resulting native-SPECT-to-PET deformation defines W_2 . The corresponding operator W_2^* in the data-fidelity term is the discrete adjoint of this resampling operator, not merely a matrix transpose; linear interpolation was used so that the forward and adjoint warp pair were implemented consistently.

8.6 Data and evaluation protocol

All quantitative image analysis is performed on PET reconstructions to preserve direct comparability with SHKEM, where PET is the target modality and SPECT provides guidance. SPECT reconstructions are shown for completeness but are not included in the quantitative endpoints. Reconstruction hyperparameters are tuned once using the clinical training patient (SIRT3) and then fixed for all phantom and clinical reconstructions. This single-patient tuning strategy avoids repeated adjustment on the evaluation cohort, but it does not constitute a robustness study of the selected weights. Throughout, dTV, w -TNV, and SHKEM are compared against the reference method w -dTNV.

8.6.1 Phantom data

A NEMA IEC Body Phantom with a sphere-to-background ratio of 8 was scanned at The Christie Hospital using a Siemens Biograph mCT PET/CT system and a GE Discovery 670 SPECT/CT system. PET was acquired over 15 h and SPECT over 15 min. From the 15 h PET acquisition, $N_{\text{bs}} = 20$ statistically independent PET datasets equivalent to 15 min were generated by sinogram bootstrapping [83]. The SPECT energy window was 39.84–128.16 keV.

We quantify PET reproducibility across bootstraps via voxel-wise coefficient of variation (CoV), and PET quantification via recovery coefficients (RCs) for spheres of diameters 10 mm, 13 mm, 22 mm, 28 mm, and 37 mm. Three volume of interests (VOIs) were defined (Figure 8.1): a high-count (HC) VOI comprising five filled spheres (one sphere was excluded due to filling issues), a warm-background (BG) VOI comprising the phantom volume excluding the spheres plus a 10 mm margin and excluding the lung insert plus a 5 mm margin, and the cold lung insert (LI).

Let $x_{\text{bs},j}^{(c)}$ be the reconstructed PET intensity at voxel j for method c in bootstrap realisation $\text{bs} \in \{1, \dots, N_{\text{bs}}\}$. Define the inter-bootstrap mean and standard deviation

$$\bar{x}_j^{(c)} := \frac{1}{N_{\text{bs}}} \sum_{\text{bs}=1}^{N_{\text{bs}}} x_{\text{bs},j}^{(c)}, \quad \text{SD}_j^{(c)} := \sqrt{\frac{\sum_{\text{bs}=1}^{N_{\text{bs}}} \left(x_{\text{bs},j}^{(c)} - \bar{x}_j^{(c)} \right)^2}{N_{\text{bs}} - 1}}, \quad (8.23)$$

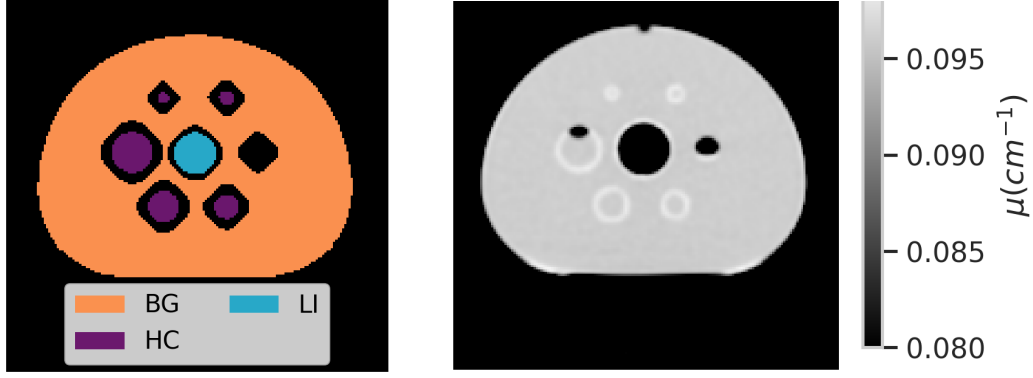


Figure 8.1: NEMA phantom analysis regions. Left: warm-background (BG), lung insert (LI), and high-count (HC) VOIs. Right: CTAC-derived attenuation map showing sphere edges, the lung insert, and the under-filled sphere excluded from the HC VOI.

and voxel-wise

$$\text{CoV}_j^{(c)} = \frac{\text{SD}_j^{(c)}}{\bar{x}_j^{(c)}}. \quad (8.24)$$

For a VOI $\mathcal{V}$ we report the VOI-averaged statistic

$$T_{\mathcal{V}}^{(c)} = \frac{1}{|\mathcal{V}|} \sum_{j \in \mathcal{V}} \text{CoV}_j^{(c)}. \quad (8.25)$$

For sphere v , let $\bar{C}_{\text{bs},v}^{(c)}$ denote the mean reconstructed intensity within the sphere and $\bar{C}_{\text{bs,bkg}}^{(c)}$ the mean within BG for the same (c, bs) . With true ratio $R_{\text{true}} = 8$, define

$$\text{RC}_{\text{bs},v}^{(c)} = \frac{\bar{C}_{\text{bs},v}^{(c)}}{\bar{C}_{\text{bs,bkg}}^{(c)} R_{\text{true}}}. \quad (8.26)$$

To assess statistical significance of inter-bootstrap CoV, we compare each method c with the reference $c_0 = w\text{-dTNV}$ using a paired row-swap permutation test on $\log(T_{\mathcal{V}}^{(c)})$. Let $X_{\mathcal{V}}^{(c)} \in \mathbb{R}^{N_{\text{bs}} \times |\mathcal{V}|}$ be the bootstrap-by-voxel stack restricted to VOI $\mathcal{V}$, and define the observed difference $d_{\text{obs}} = \log(T_{\mathcal{V}}^{(c_0)}) - \log(T_{\mathcal{V}}^{(c)})$. Under the null of exchangeability, each bootstrap index is independently swapped between the paired stacks with probability $1/2$ to form permuted differences d_{perm} . We use

$N_{\text{perm}} = 5000$ permutations and compute the two-sided p-value

$$p = \frac{1 + \sum_{i=1}^{N_{\text{perm}}} \mathbf{1}(|d_{\text{perm},i}| \geq |d_{\text{obs}}|)}{N_{\text{perm}} + 1}, \quad (8.27)$$

with Benjamini–Hochberg (BH) adjustment across methods within each VOI. No thresholding was applied to exclude near-zero voxels, so CoV estimates in the cold lung insert should be interpreted as a noise-realisation stability measure rather than a conventional percentage variability around a substantial mean signal.

Inference for RC uses a linear mixed model on

$$y_{\text{bs},c,v} := \log\left(\text{RC}_{\text{bs},v}^{(c)}\right), \quad \text{bs} = 1, \dots, N_{\text{bs}}, \quad (8.28)$$

with a random intercept for bootstrap replicate and fixed effects for method, sphere, and their interaction:

$$y_{\text{bs},c,v} = \beta_0 + \alpha_c + \gamma_v + (\alpha\gamma)_{c,v} + u_{\text{bs}} + \epsilon_{\text{bs},c,v},$$

$$u_{\text{bs}} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_{\text{bs}}^2), \quad \epsilon_{\text{bs},c,v} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2). \quad (8.29)$$

For each sphere size, many-to-one contrasts compare each method to the reference $c_0 = w\text{-dTNV}$, with Benjamini–Hochberg adjustment across planned comparisons within that sphere. Since the model is fitted on the log scale, $\exp(\Delta_{c,v}) < 1$ indicates lower recovery than $w\text{-dTNV}$ for sphere v .

8.6.2 Clinical data

Clinical data were acquired at Oxford University Hospitals from 10 patients undergoing ^{90}Y SIRT: one patient was used for hyperparameter selection and nine for testing. PET was acquired on a GE Discovery 710 PET/CT system and SPECT on a GE Discovery NM/CT 670 system. Informed consent is not necessary for retrospective image reviews of this nature at Oxford University Hospitals. PET was acquired with two bed positions (15 min each) and SPECT with 15 min. Energy windows were 50–150 keV.

Tumour VOIs were delineated once per patient on the vendor Q.Clear $\beta\text{-4000}$

PET reconstruction using an adapted automatic radial-gradient method based on Mikell *et al.* [84]. Candidate seed voxels were processed in descending order of intensity, with a minimum seed separation of 30 mm. From each seed, the lesion boundary was estimated along radial rays after Gaussian smoothing ($\sigma = 0.1$); each ray was limited to a maximum length of 25 mm and a background-adjusted intensity floor $I_{\text{bkg}} + 0.42(I_{\text{seed}} - I_{\text{bkg}})$, where I_{bkg} is the mean intensity in the manually defined homogeneous liver background VOI. The 42% fraction was motivated by the PET lesion-delineation thresholds reported by Erdi *et al.* [85]. Masks containing fewer than 10 voxels were discarded, and the five highest-ranked non-overlapping masks were retained; where candidate masks overlapped, the mask originating from the hotter seed took precedence. These VOIs were propagated unchanged across all reconstructions so that measured differences reflected reconstruction effects rather than VOI definition. Further implementation details are provided in Appendix C.

A homogeneous liver background VOI was drawn manually, avoiding motion and visible artefacts. The training-patient lesion and background VOIs are shown in Figure 8.2.

For patient p , lesion l , and method c , define tumour-to-background ratio (TBR)

$$\text{TBR}_{p,l,c} = \frac{\bar{C}_{p,l,c}}{\bar{C}_{p,\text{bkg},c}}, \quad (8.30)$$

and background noise (percentage CoV)

$$\text{CoV}_{p,c}(\%) = 100 \frac{\text{SD}_{p,c}}{\bar{C}_{p,\text{bkg},c}}, \quad (8.31)$$

where $\text{SD}_{p,c}$ is the within-background-VOI standard deviation.

We analyse $\log(\text{TBR})$ using a linear mixed-effects model with fixed method effects and random intercepts for patient and lesion-within-patient:

$$\begin{aligned} y_{p,l,c} &:= \log(\text{TBR}_{p,l,c}) = \beta_0 + \alpha_c + u_p + v_{p,l} + \varepsilon_{p,l,c}, \\ u_p &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_p^2), \quad v_{p,l} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_l^2), \quad \varepsilon_{p,l,c} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2). \end{aligned} \quad (8.32)$$

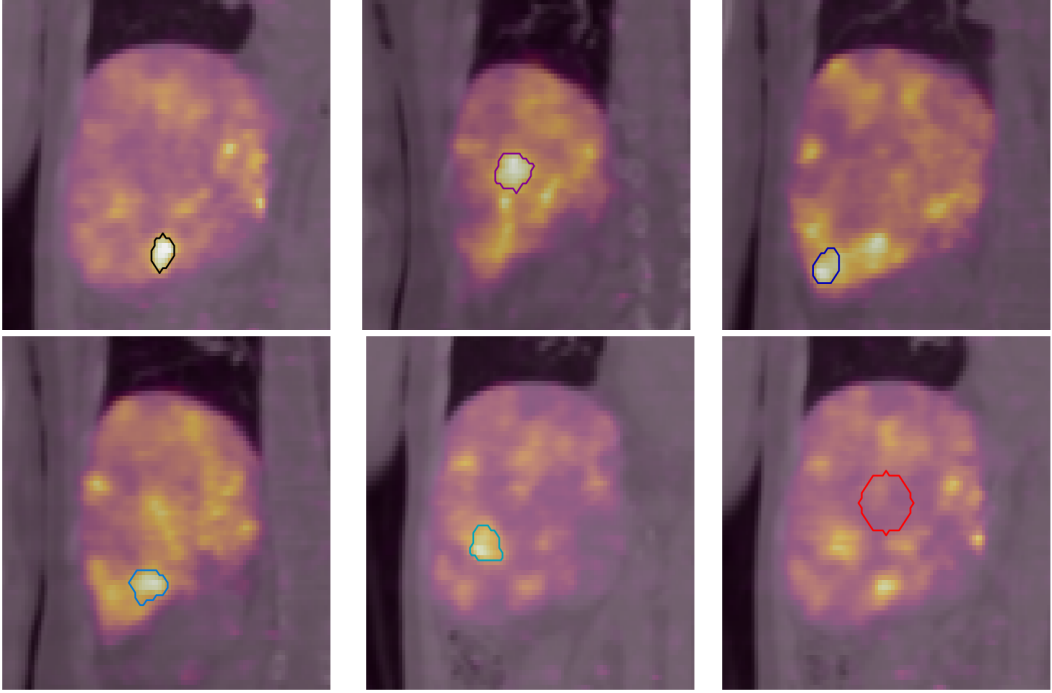


Figure 8.2: Training-patient VOIs. The five identified lesions and the background reference region are shown on the vendor PET reconstruction with registered attenuation maps. Top row: lesions 1–3. Bottom row: lesions 4–5 and background.

Background noise is analysed analogously at the patient level using

$$\begin{aligned} y_{p,c}^{\text{CoV}} &:= \log(\text{CoV}_{p,c}(\%)) = \beta_0 + \alpha_c + u_p + \varepsilon_{p,c}, \\ u_p &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_{p,\text{CoV}}^2), \quad \varepsilon_{p,c} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \psi^2). \end{aligned} \quad (8.33)$$

Planned many-to-one contrasts compare dTV, w -TNV, and SHKEM to w -dTNV, with BH adjustment across the three comparisons within each endpoint. Results are reported as ratios $\exp(\Delta)$ on the original scale, where values below one indicate lower TBR or lower background CoV than w -dTNV.

8.7 Results

8.7.1 Hyperparameter selection

Regularisation weights β_{PET} and β_{SPECT} in (8.7) were tuned on a single training patient by sweeping the TBR–CoV_{bkg} trade-off and selecting an operating point on the Pareto frontier [86] with background CoV approximately matched to the vendor

reconstruction. The selected weights were then fixed for all phantom and clinical experiments:

Table 8.1: Selected regularisation weights (tuned on a single training patient) used for all phantom and clinical experiments.

Method	β_{PET}	β_{SPECT}
w -dTNV	0.01	0.02
w -TNV	0.0075	0.02
dTV	0.02	–

For SHKEM, hyperparameters followed Deidda *et al.* [48]; the PET subiteration displayed and analysed here (27) was chosen to approximately match the background CoV of the vendor reconstruction.

Figure 8.3 shows the training-patient trade-off curves for two representative lesions. Lesion 1 shows only a modest initial benefit from synergistic information and is largely insensitive to further increases in SPECT weighting until over-regularisation introduces negative bias. Lesion 2 shows a clearer synergistic benefit: increasing SPECT weighting shifts the TBR–CoV_{bkg} curve towards higher TBR at comparable background CoV up to a distinct elbow region, after which excessive coupling reduces TBR and/or increases background CoV. For lesion 2, the PET image obtained with w -dTNV also exhibited features consistent with a CTAC artefact; this is discussed in Section 8.8.

8.7.2 Phantom results

A representative reconstruction from a single bootstrapped PET realisation is shown in Figure 8.4 (PET and corresponding SPECT; for SHKEM, SPECT is the separately reconstructed guidance image; dTV does not reconstruct SPECT).

Figure 8.5 and Table 8.2 summarise sphere recovery. Quantitative analysis demonstrated that w -dTNV achieved significantly higher recovery than dTV and w -TNV for all sphere sizes ≤ 22 mm. w -dTNV showed similar recovery to SHKEM in the larger spheres but significantly improved recovery for the two smallest spheres at the matched operating point.

Figure 8.6 summarises voxel-wise reproducibility across bootstraps. At

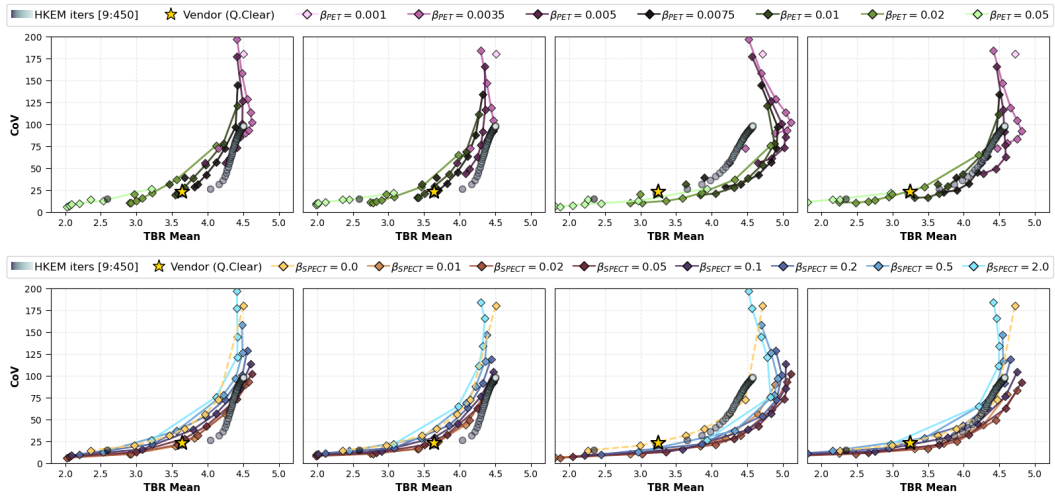


Figure 8.3: Training-patient hyperparameter sweep. Impact of varying PET weighting while holding SPECT regularisation constant (top row) and varying SPECT weighting while holding PET regularisation constant (bottom row) on TBR and CoV_{bkg} for lesions 1 and 2. The first and third columns show w -dTNV; the second and fourth show w -TNV.

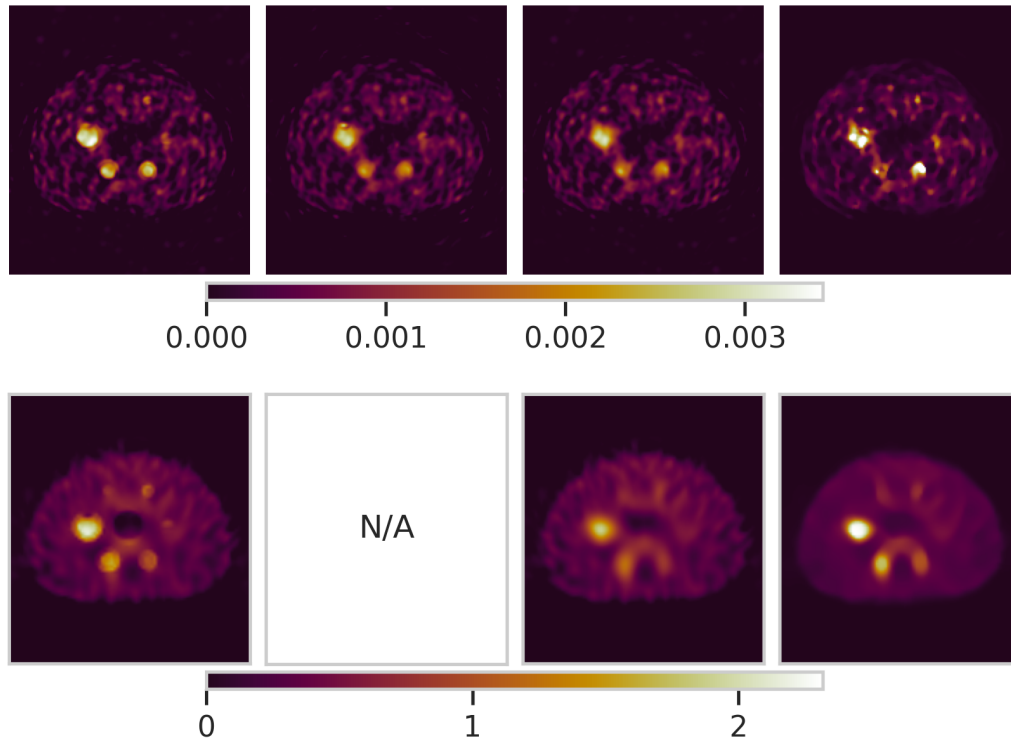


Figure 8.4: Visual comparison of NEMA phantom reconstructions for a randomly chosen bootstrap. From left to right: w -dTNV, dTV, w -TNV, SHKEM (subiteration 27). PET reconstructions are shown on the top row and SPECT on the bottom row (SPECT guidance for SHKEM). Images are displayed in reconstruction intensity units, proportional to activity concentration.

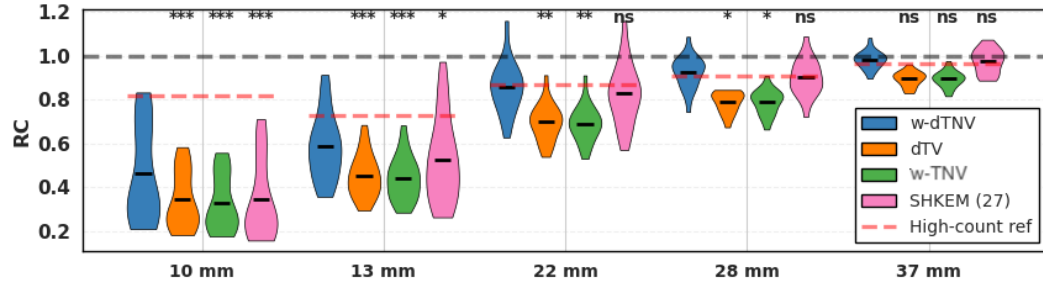


Figure 8.5: Recovery coefficient (RC) for NEMA phantom spheres for w -dTNV, dTV, w -TNV and SHKEM (subiteration 27). Also shown are the equivalent RCs for a high-count, long-OSEM reconstruction. The grey line indicates perfect recovery. Statistical significance is reported for paired comparisons versus w -dTNV (Benjamini–Hochberg adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Method	v	$\exp(\Delta_{c,v})$	95% CI	p_{BH}
dTV	37 mm	0.9113	[0.8001, 1.0380]	0.2472
	28 mm	0.8543	[0.7500, 0.9730]	0.0267
	22 mm	0.8165	[0.7168, 0.9300]	0.0034
	13 mm	0.7819	[0.6865, 0.8906]	0.00043
	10 mm	0.7706	[0.6766, 0.8777]	$< 10^{-4}$
w -TNV	37 mm	0.9108	[0.7997, 1.0374]	0.2472
	28 mm	0.8544	[0.7502, 0.9732]	0.0267
	22 mm	0.8090	[0.7103, 0.9215]	0.0034
	13 mm	0.7598	[0.6671, 0.8654]	0.0001
	10 mm	0.7384	[0.6483, 0.8410]	$< 10^{-4}$
SHKEM (27)	37 mm	0.9931	[0.8720, 1.1312]	0.9175
	28 mm	0.9739	[0.8551, 1.1093]	0.6906
	22 mm	0.9616	[0.8443, 1.0953]	0.5559
	13 mm	0.8723	[0.7659, 0.9936]	0.0397
	10 mm	0.7381	[0.6480, 0.8407]	$< 10^{-4}$

Table 8.2: Effect ratios $\exp(\Delta_{c,v})$ (comparator vs. baseline w -dTNV) with 95% confidence intervals and BH-adjusted p -values, reported by sphere diameter v . Values below one indicate lower recovery coefficient than w -dTNV.

the matched operating point, w -dTNV and w -TNV yield lower voxel-wise CoV than w -dTNV in the evaluated VOIs, consistent with stronger effective smoothing. SHKEM exhibits higher variability in background and high-count regions but improved stability in the cold lung insert. Sphere-level CoV trends were consistent with the voxel-wise analysis, with w -TNV and dTV generally providing the lowest variability across the evaluated sphere diameters and SHKEM the highest.

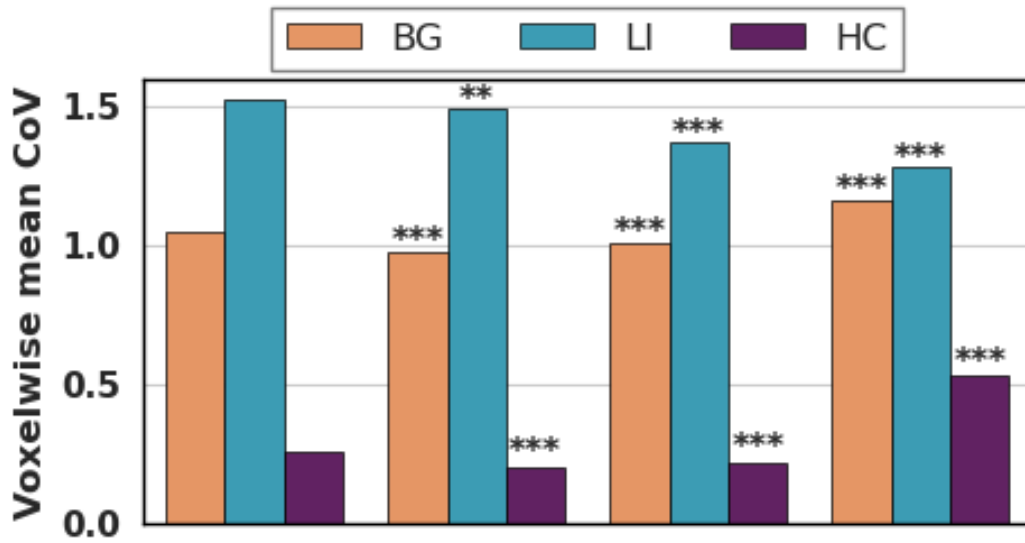


Figure 8.6: Mean voxel-wise coefficient of variation (CoV) across 20 bootstrapped PET realisations within each VOI. Statistical significance is reported for paired comparisons versus w -dTNV (BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

8.7.3 Clinical results

Figure 8.7 shows representative PET and SPECT reconstructions for the training patient. These images illustrate the operating-point behaviour used to select the fixed regularisation weights: w -dTNV improves lesion contrast where the SPECT and CTAC-derived guidance contain useful structural information, while the response is weaker where there is no clear anatomical correlate.

Across the test cohort, w -dTNV yields higher lesion TBR than dTV, w -TNV, and SHKEM at the matched operating point, while background CoV differences are smaller and method-dependent. SHKEM reconstructions appeared visually smoother and, as in the phantom, appeared more effective at suppressing cold-background

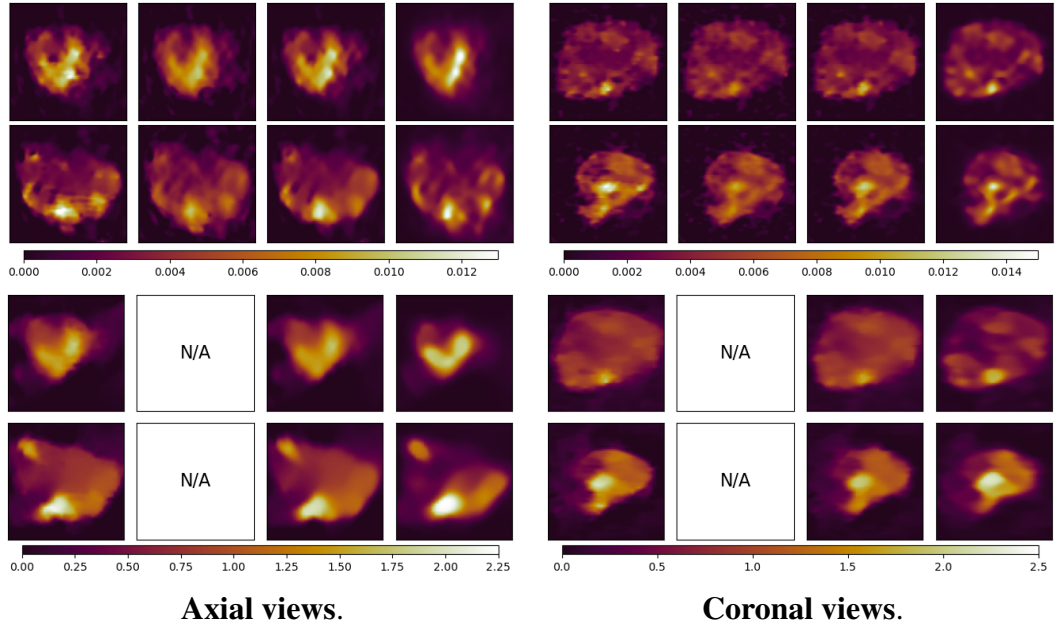


Figure 8.7: Visual comparison of reconstruction methods for training patient SIRT3. Columns show w -dTNV, dTV, w -TNV and SHKEM (subiteration 27). The upper two quadrants show PET reconstructions; the lower two show SPECT reconstructions or guidance images. Images are displayed in reconstruction intensity units, proportional to activity concentration.

noise:

Method	$\exp(\Delta)$	95% CI	p_{BH}
dTV	0.802	[0.758, 0.849]	$< 10^{-4}$
w -TNV	0.915	[0.865, 0.968]	0.004
SHKEM (27)	0.935	[0.884, 0.990]	0.020

Table 8.3: Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p -values for mean tumour-to-background ratio (TBR), comparing each method to w -dTNV. Values below one indicate lower TBR than w -dTNV.

8.8 Discussion

The phantom results indicate that functional coupling alone (w -TNV) is insufficient to recover small-sphere boundaries at the matched operating point, and that anatomical guidance (explicit in w -dTNV, implicit in SHKEM through CT-informed kernel features) is beneficial for boundary localisation. The recovery gain for w -dTNV relative to w -TNV and dTV is most pronounced for the smallest spheres, consistent with guidance stabilising edge geometry when PET statistics are poor and SPECT resolution

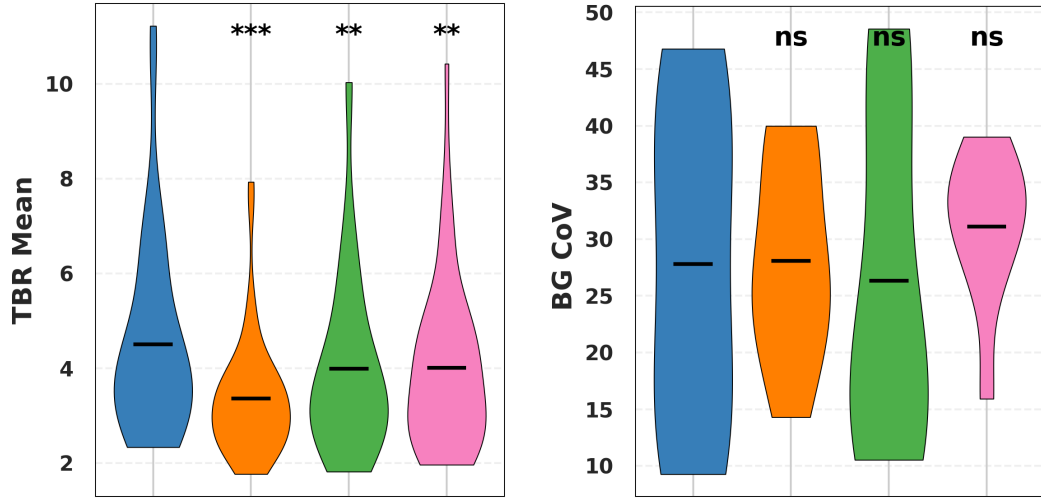


Figure 8.8: Clinical cohort summary: lesion-wise mean TBR (left) and patient-wise background CoV (right). Statistical significance is reported for comparisons versus w -dTNV(BH-adjusted): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Method	$\exp(\Delta)$	95% CI	p_{BH}
dTV	1.183	[0.903, 1.549]	0.35
w -TNV	1.007	[0.833, 1.167]	0.984
SHKEM (27)	1.339	[0.933, 1.913]	0.093

Table 8.4: Effect ratios $\exp(\Delta)$ with 95% confidence intervals and BH-adjusted p -values for background CoV, comparing each method to w -dTNV. Values above one indicate higher background CoV than w -dTNV.

is limited.

The reproducibility analysis highlights a trade-off: dTV and w -TNV provide lower voxel-wise CoV than w -dTNV at the selected operating point, consistent with stronger effective smoothing. All three variational methods outperform SHKEM in background and high-count repeatability, whereas SHKEM is most stable in the cold lung insert. A plausible mechanism is that SHKEM kernel weights in near-zero regions favour local averaging among consistently low voxels, so “zeros reinforce zeros” under positivity constraints, improving cold-region stability while not necessarily improving reproducibility elsewhere.

Notably, dTV and w -TNV produced broadly similar PET appearances in the phantom, despite dTV including μ -directional guidance and w -TNV including PET/SPECT coupling. This suggests that, in this experiment, neither CTAC-derived

directional guidance alone nor isotropic PET/SPECT coupling alone was sufficient to recover the smallest structures. The improved recovery observed with w -dTNN is therefore consistent with a complementary mechanism: the higher-count SPECT image provides additional functional support for the location of activity transitions, while the μ -derived directional field reduces penalisation of gradients aligned with anatomical boundaries.

Clinically, lesion responses to synergy and directional guidance are heterogeneous. Lesions with clear anatomical correlates (e.g. calcification or liver capsule boundaries visible in the μ -map) tend to benefit more from μ -guided anisotropy and PET/SPECT coupling, while lesions lacking such correlates may show limited benefit. This heterogeneity is consistent with the design goal of w -dTNN: CTAC guidance steers orientation preferences, but cannot invent edges unsupported by the functional data.

For example, lesion 2 in the training patient shows substantial improvement in the TBR–CoV trade-off as SPECT weighting is increased (Figure 8.3). Post-hoc examination indicates that this lesion lies adjacent to a calcified region visible in the μ -map, providing a sharp anatomical edge that coincides with the functional boundary. Lesion 3 also improves under both w -dTNN and w -TNN; this lesion abuts the liver on three sides, where the soft-tissue interface generates reasonably strong anatomical contrast. For w -dTNN, the μ -map liver boundary provides explicit directional guidance aligned with the functional uptake margin, while for w -TNN the SPECT boundary may be sufficiently well defined to support PET/SPECT coupling even without directional CTAC guidance.

Conversely, lesion 1 shows little benefit from increasing SPECT weighting until excessive weighting introduces negative bias. Visual inspection shows no obvious anatomical correlate at the tumour margin, and the PET and SPECT images appear already well aligned. The reason for this lesion’s limited response to synergistic reconstruction remains unclear.

A practical limitation is sensitivity to CTAC artefacts. Since ξ_n in (8.4) depends on $D\mu$, structured CT artefacts (e.g. arms-down streaking) can imprint into $\|\xi\|$

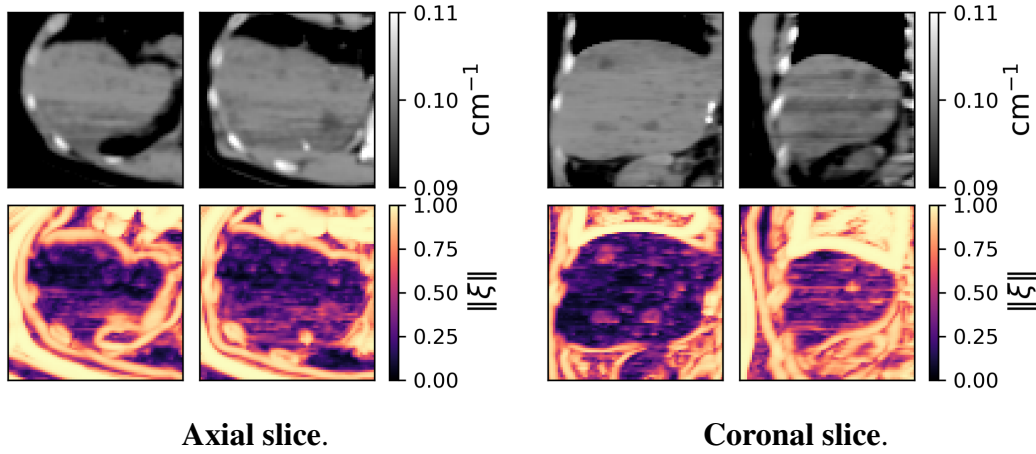


Figure 8.9: Training-patient CTAC-derived guidance. The CTAC-derived attenuation map μ (top row; cm^{-1}) and magnitude of the w -dTNV directional weighting field $\|\xi\|$ (bottom row) are shown for lesions 1 and 2. CTAC streak artefacts are visible in μ and are reproduced in $\|\xi\|$, providing a pathway for artefact transfer into the emission reconstruction.

and therefore into the directional weighting field, providing a pathway for artefact transfer into the emission reconstructions. In the training patient, the low-dose PET CTAC was acquired with arms alongside the torso, introducing streaking and beam-hardening artefacts in the μ -map. The stabiliser η mitigates amplification of small gradients but cannot reliably suppress coherent high-contrast artefacts. SHKEM anatomical guidance was derived from the SPECT attenuation map, acquired with arms raised, rather than from the low-dose PET CTAC used to construct the w -dTNV directional weights; this difference in guidance input may contribute to the absence of streak-related artefacts in the SHKEM reconstructions.

The contrast with the NEMA phantom is instructive (Figure 8.10). In the phantom, the CTAC is largely free of coherent streak artefacts and the resulting $\|\xi\|$ follows the true boundaries and insert edges. In this setting, directional smoothing attenuates small high-frequency fluctuations without introducing structured bias.

These observations motivate future work on robust direction-field estimation and artefact-aware guidance strategies. From a practical perspective, if a low-dose CTAC is used to provide directional information, acquisition protocols that minimise artefacts, such as arms raised where feasible, are preferable.

To preserve direct comparability with SHKEM, regularisation weights were

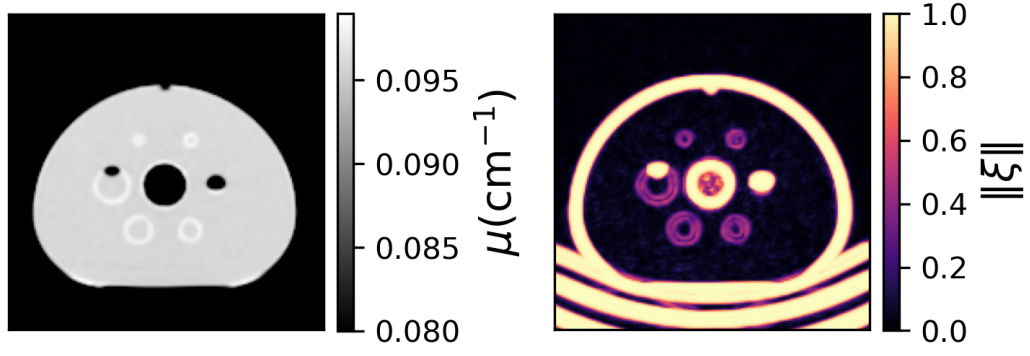


Figure 8.10: NEMA phantom, central axial slice. CTAC-derived attenuation map μ (left; cm^{-1}) and magnitude of the w -dTNV directional weighting field $\|\xi\|$ (right). With a high-quality CTAC free of prominent streak artefacts, $\|\xi\|$ concentrates around true object boundaries and insert edges, enabling stable μ -map-guided directional smoothing.

selected once on the clinical training case and quantitative evaluation focused on PET. This means the robustness of the chosen weights across acquisition protocols and lesion presentations was not fully characterised. The jointly reconstructed SPECT images were assessed qualitatively only. Although they appeared visually less noisy while maintaining lesion conspicuity, a rigorous assessment of SPECT quantification would require modality-specific lesion delineation and evaluation protocols. The sensitivity of w -dTNV to residual PET/SPECT/CT registration error also remains an important topic for future work.

More broadly, this chapter focuses on the favourable case in which PET and SPECT image the same underlying ^{90}Y microsphere distribution. The same framework may be extendable to settings with non-identical functional distributions but shared anatomy, such as planning–treatment image pairs ($^{99\text{m}}\text{Tc}$ -MAA versus post-treatment ^{90}Y), multi-nuclide imaging, or multi-timepoint studies. In such cases, the directional/anatomical component remains well motivated, but the inter-modality coupling strength may need to be relaxed or made spatially adaptive to avoid transferring modality-specific or temporally evolving uptake features.

8.9 Conclusion

This chapter validated the w - d TNV prior of Chapter 6 in joint ^{90}Y PET/SPECT reconstruction with CTAC-derived directional guidance. The key modelling choice is to use CTAC information directionally (via $\Pi_{m,n}$) while preserving a two-modality Jacobian, retaining the computational advantages of the 2×2 Gram structure (Chapter 6) and avoiding unstable intensity coupling to CT.

On a bootstrapped NEMA IEC phantom, w - d TNV improves small-object recovery relative to w -TNV, d TV, and a sequential hybrid kernel baseline at a matched noise operating point, with significant gains over the isotropic and non-synergistic variational baselines. Repeatability improves relative to SHKEM in the background and high-count regions, although it is reduced relative to the more strongly smoothing w -TNV and d TV baselines and SHKEM remains most stable in the cold lung insert. In a post-SIRT patient cohort of 45 lesions in nine patients, w - d TNV yields significantly higher lesion TBR than the comparators at approximately matched background CoV. Lesion responses are heterogeneous: lesions with clear anatomical correlates, such as calcification or liver boundaries, benefit most from synergy and μ -map-informed edge alignment, whereas lesions without corresponding anatomical structure show more limited improvement. These findings support CTAC-guided synergistic variational coupling as a practical route towards more reliable post-SIRT ^{90}Y activity imaging for personalised dosimetry, while highlighting the need for careful handling of CT artefacts and PET/SPECT/CT registration in clinical translation.

Chapter 9

Discussion, Conclusion & Future Work

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Appendix A

Software Contributions

This appendix documents the software developed or substantially adapted for the work in this thesis. The largest contribution, `simind-python-connector`, is described first; two smaller packages — a SIRF-adapted deep-learning scatter estimator and a kernelised Richardson–Lucy deconvolution package built on the Core Imaging Library (CIL) — are described briefly thereafter.

A.1 `simind-python-connector`: Python Workflows for SIMIND Simulations

This section documents the `simind-python-connector` package used to support reproducible SIMIND-based simulation workflows in this thesis. The inspected package version was 1.0.1, released on 16 April 2026 and archived on Zenodo with DOI 10.5281/zenodo.19605413. The public repository is available at GitHub.

TODO: Replace the DOI URL above with the Zenodo-managed citation key for DOI 10.5281/zenodo.19605413 once the automatic bibliography import has added it.

A.1.1 Purpose and Scope

SIMIND is a Monte Carlo code for simulating clinical SPECT scintillation-camera imaging and related system effects [87, 82], and is used throughout this thesis because ^{90}Y bremsstrahlung SPECT requires careful modelling of scatter, attenuation, collimator response, and septal penetration (Chapters 4 and 8). The role of

`simind-python-connector` is not to replace SIMIND, but to make it easier to drive from Python and easier to connect to reconstruction software. The package is independent of, and not affiliated with, the SIMIND project or Lund University: SIMIND is not distributed with the Python package, and a separately licensed installation must be available on the local machine or inside the configured container environment. This separation aids reproducibility, since the open Python workflow can be version controlled while the licensed simulation executable remains an external dependency.

The package provides two main capabilities. First, `SimindPythonConnector` offers a direct Python interface for configuring voxel phantoms, writing SIMIND input files, setting runtime switches, running SIMIND, and loading projection outputs as NumPy arrays. Second, adaptor classes translate the resulting data into reconstruction-library-native representations for STIR, Synergistic Image Reconstruction Framework (SIRF), and PyTomography workflows [76, 88]. The package therefore sits between simulation and reconstruction: SIMIND produces projection data, while reconstruction is performed by the target framework.

TODO: Add Zenodo-managed BibTeX entry for PyTomography if the PyTomography adaptor is retained in the final cited text.

TODO: Add Zenodo-managed BibTeX entry for the Schneider CT-density conversion paper if this utility is discussed beyond software documentation.

A.1.2 Workflow and Interfaces

The core workflow is deliberately explicit: a user constructs a connector from a base SIMIND configuration, output directory, and output prefix; supplies a source image and attenuation or density information; specifies energy windows; applies runtime switches such as radionuclide, collimator, photon multiplier, and random seed; and calls `run()`. The connector writes SIMIND-compatible source and density files, saves the generated configuration, executes SIMIND, and loads the resulting Interfile projection data. The main interfaces are summarised in Table A.1.

The package can be installed from PyPI with `pip install simind-python-connector` and depends only on standard Python scientific and medical-image utilities such as NumPy, pydicom, and PyYAML; STIR, SIRF, PyTomography, and SIMIND itself

Table A.1: Main *simind-python-connector* interfaces relevant to thesis workflows.

Interface	Role
<code>SimindPythonConnector</code>	Backend-agnostic connector that runs SIMIND and returns projection outputs as NumPy arrays with header and data-file paths.
<code>RuntimeOperator</code>	Container for per-run SIMIND runtime switches and optional orbit-file selection.
<code>StirSimindAdaptor</code>	Converts connector inputs and outputs for STIR-based reconstruction workflows.
<code>SirfSimindAdaptor</code>	Provides the analogous adaptor path for SIRF, which builds on STIR for PET/SPECT reconstruction.
<code>PyTomographySimindAdaptor</code>	Accepts torch tensors, executes the SIMIND connector path, and returns PyTomography-compatible projection tensors and headers.
<code>hu_to_density_schneider</code>	Converts CT Hounsfield units to material density using the packaged Schneider-style tissue lookup.

are optional or external dependencies needed only for the corresponding adaptor or simulation paths.

A.1.3 Conventions and Validation

A central difficulty in connecting simulation and reconstruction software is that array-order, projection-axis, and unit conventions differ between packages, and an apparently small axis flip or millimetre-to-centimetre error would change the simulated collimator response, attenuation path lengths, or reconstructed orientation. The connector documentation therefore makes these conventions explicit — STIR/SIRF image arrays are handled as (z, y, x), PyTomography object-space tensors as (x, y, z), and raw Interfile projections as (views, bins, axial) — and the connector derives the centimetre-based SIMIND geometry switches automatically from the millimetre voxel sizes used by reconstruction libraries.

Validation separates lightweight pure-Python unit tests (configuration handling, connector behaviour, Interfile parsing, density conversion, and optional-backend guards) from backend-dependent tests, which are marked per framework and skipped automatically when the corresponding dependency is unavailable — important for continuous integration, since public CI runners cannot include a licensed SI-

MIND executable. Dedicated Docker environments are supplied for STIR, SIRC, and PyTomography, and the included examples cover basic NumPy simulations, runtime-switch comparisons, multi-window simulations, SCATTWIN/PENETRATE scoring comparisons, Schneider-style density conversion, adaptor-based OSEM reconstructions, and DICOM-driven setup paths. Together these define the intended reproducibility contract: SIMIND runs that are scriptable, recordable, and portable across reconstruction frameworks.

A.2 Deep-Learned Scatter Initialisation for SIRC

The bremsstrahlung reconstructions of Chapters 4 and 8 initialise the preparatory additive-term estimation using a convolutional neural network that predicts scatter projections directly from the measured emission projections and CT-derived attenuation information, following Xiang et al. [60]. The code used in this thesis is a fork of the original authors' implementation, adapted to run inference within SIRC workflows [88]: it adds SIRC-facing configuration and inference scripts driven by a YAML configuration file, alongside the pre-trained network weights and example notebooks. As discussed in Chapter 4, the pre-trained network was developed under a different scanner and protocol configuration than the datasets used in this thesis; the scatter-derived output is therefore used only as a heuristic warm start for the broader non-primary term. It is refined by the damped reconstruct–simulate loop driven by `simind-python-connector` (Section A.1), and the final estimate is frozen before the target reconstruction.

The adapted repository is available at GitHub and retains the structure of the upstream implementation by Xiang et al., to which all credit for the network architecture and training belongs; only the SIRC integration and inference path are contributions of this thesis. **TODO: Archive a tagged release of spect-scatter-deep-learning on Zenodo and replace this GitHub link with the DOI-backed citation.**

A.3 Kernel_RL: Kernelised Richardson–Lucy

Deconvolution with the Core Imaging Library

The anatomy-guided deconvolution experiments of Chapter 5 were performed with `Kernel_RL`, a Python package implementing RL deconvolution and its anatomy-aware variants — KRL, HKRL, and RL–DTV — within the optimisation framework of the Core Imaging Library (CIL) [79]. The kernel operator realising the image model $x = K\alpha$ is provided with interchangeable backends — PyTorch/CUDA on GPU and Numba on CPU — selected automatically at runtime; sparse neighbour masking and an optional single-precision mode together allow 256^3 volumes to be processed on consumer GPUs.

The package is released under the MIT licence and is available at GitHub. Installation is supported via conda (with CIL from the conda-forge and ccpi channels) or via the provided Docker environment, which reproduces the software configuration used for the experiments of Chapter 5. **TODO:** Archive a tagged release of `Kernel_RL` on Zenodo and replace this GitHub link with the DOI-backed citation.

Appendix B

Convergence Diagnostics for the Long-Run Reference Reconstruction

The NRMSE comparisons in Chapter 7 use the final iterate of one long-run reconstruction as their reference image. This one-bed phantom reconstruction used the same objective as the 20-pass comparisons, but was continued for 1000 epochs with 36 stochastic updates per epoch, a Gershgorin-diagonal MM preconditioner, and a more conservative relaxation rate of $\eta_{\text{relax}} = 10^{-4}$. Only one such reference run was performed.

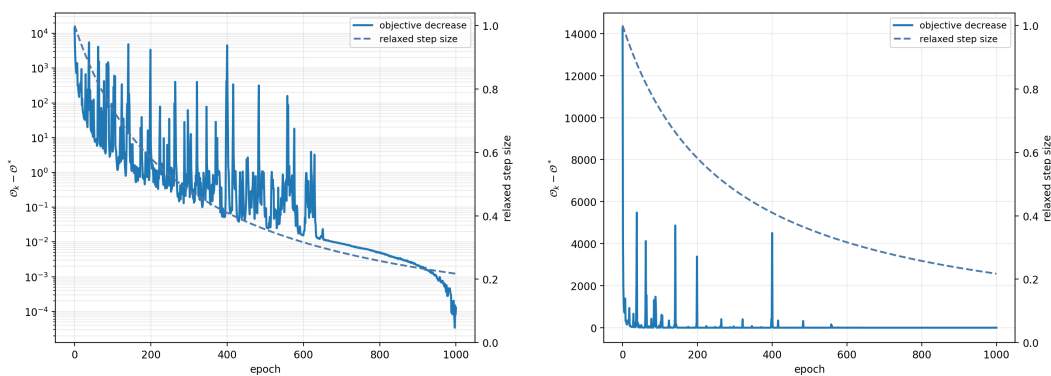
Let $\mathcal{O}_k = \mathcal{O}(\mathbf{z}^{(k)})$ be the recorded objective after epoch k , and define

$$\mathcal{O}^* = \min_{0 \leq k \leq 1000} \mathcal{O}_k. \quad (\text{B.1})$$

Here $\mathcal{O}^*$ is the best objective observed during this run. It is not a proven global minimum, and the reference image should likewise be understood as a long-run numerical reference rather than ground truth.

The objective contains stochastic excursions early in the run, but its gap relative to the best observed value becomes negligible. Table B.1 reports the final objective and iterate diagnostics. The run had achieved 99.9% of its total observed objective decrease by epoch 45, when the relaxed step remained 0.8606.

Together, the vanishing objective gap, relative objective change, and iterate velocity indicate that the reference was effectively stationary relative to the best



(a) Objective gap on a logarithmic scale.

(b) Objective gap on a linear scale.

Figure B.1: Long-run objective diagnostics. The solid curve is $\mathcal{O}_k - \mathcal{O}^*$ and the dashed curve is the relaxed step size. The linear view shows that most of the observed objective decrease occurs early, while the logarithmic view resolves the residual decrease over the complete 1000-epoch run.

Table B.1: Stationarity diagnostics for the 1000-epoch reference reconstruction.

Quantity	Value
Final objective gap / observed objective decrease	8.819×10^{-9}
Final relative objective change per epoch	1.717×10^{-11}
Final PET iterate velocity per epoch	7.285×10^{-8}
Final relaxed step size	0.2174
Epoch reaching 99.9% of observed decrease	45
Step size at that epoch	0.8606

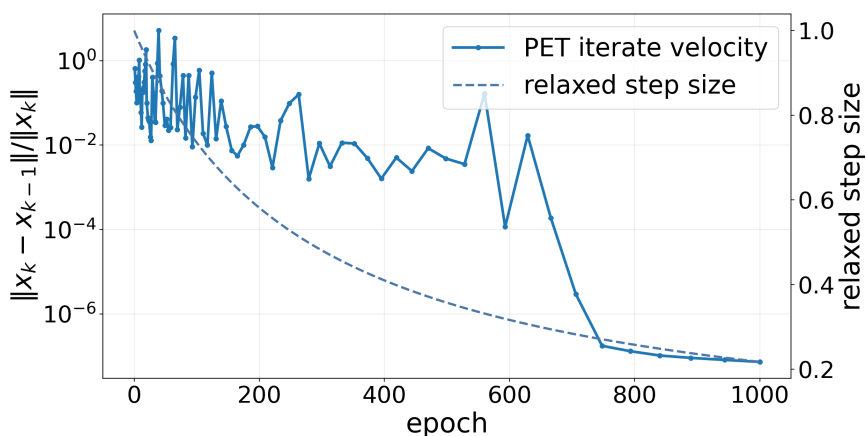


Figure B.2: Relative PET iterate velocity, $\|x^{(k)} - x^{(k-1)}\| / \|x^{(k)}\|$, and relaxed step size during the long-run reference reconstruction. The final velocity is 7.285×10^{-8} per epoch while the step size remains 0.2174.

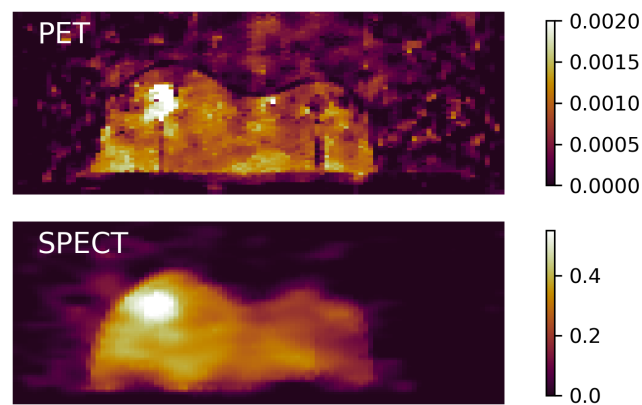


Figure B.3: Final PET and SPECT iterates of the long-run reference reconstruction, shown on a matched slice with independent colour bars. This pair is the reference x^* against which the NRMSE curves of Chapter 7 are measured; it is the long-run baseline rather than one of the candidate preconditioner settings compared there.

observed iterate. The appreciable final step size makes step-size starvation an unlikely explanation for the plateau. These diagnostics support its use as a numerical reference for the short-run NRMSE comparisons, without asserting convergence to a global minimiser.

Appendix C

Automatic Lesion VOI Delineation

This appendix documents the automatic lesion-mask construction used to define fixed inputs for the patient analysis in Chapter 8. The procedure was adapted from the gradient-based PET segmentation method of Mikell *et al.* [84], with pragmatic bounds added for the present post-SIRT data. It was not evaluated as a separate methodological contribution, and no delineation-accuracy or parameter-sensitivity study was performed.

C.1 Candidate selection

For each patient, candidate seed voxels were identified on the vendor Q.Clear β -4000 PET reconstruction and ordered from highest to lowest intensity. Seeds were required to be separated by at least 30 mm. Candidate masks were grown in this order, and masks containing fewer than 10 voxels were discarded. The five highest-ranked non-overlapping masks were retained. If two candidates overlapped, the mask grown from the hotter seed was retained and the lower-ranked candidate was discarded.

C.2 Radial-gradient growth

The vendor image was smoothed with a Gaussian kernel ($\sigma = 0.1$). Rays were cast outwards from each seed, and the candidate boundary in each direction was located from the maximum image-gradient magnitude within the admissible search interval. Figure C.1 illustrates the seed-centred radial search and the corresponding gradient-magnitude image.

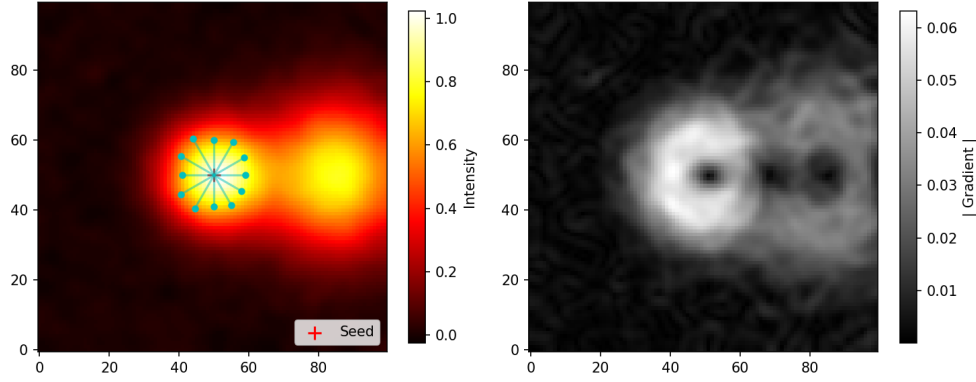
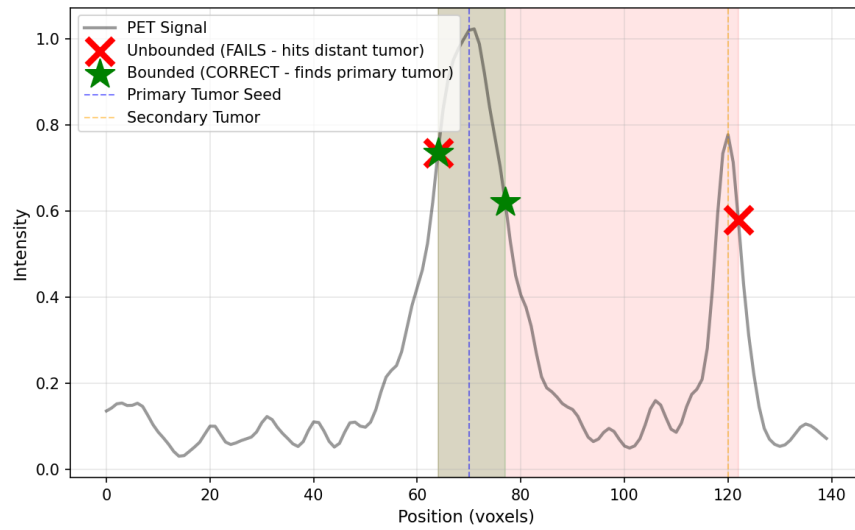


Figure C.1: Two-dimensional illustration of radial-gradient lesion delineation. Left: radial searches extend from the seed voxel to candidate boundary locations. Right: the corresponding image-gradient magnitude used to identify the lesion edge. This schematic illustrates the search principle; patient masks were generated in the image volume.

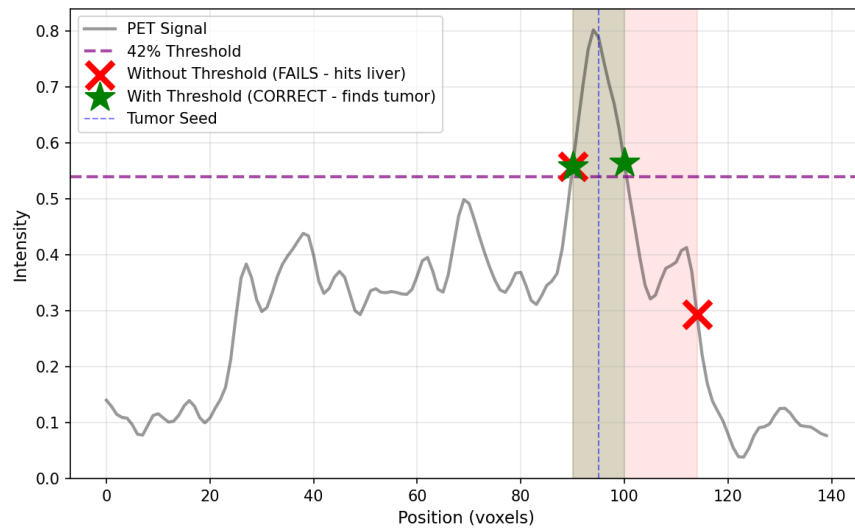
Two safeguards bounded each radial search. First, the maximum ray length was 25 mm, preventing a ray from reaching a spatially separate hot lesion. Second, samples were restricted to intensities above the background-adjusted floor

$$I_{\text{floor}} = I_{\text{bkg}} + 0.42 (I_{\text{seed}} - I_{\text{bkg}}), \quad (\text{C.1})$$

where I_{bkg} is the mean intensity in the patient-specific liver background VOI. The floor therefore retains 42% of the contrast between the seed and the background. This fraction lies within the 36–44% range reported by Erdi *et al.* for PET lung-lesion delineation [85]. For those data the background was comparatively low; here, the non-zero liver background was retained explicitly. Because the published thresholds depended on imaging conditions and were not derived for post-SIRT ^{90}Y liver lesions, Equation (C.1) was used as a pragmatic stopping bound rather than as a separately validated threshold.



(a) Maximum-distance bound. Without a bounded search, a ray can encounter the gradient associated with a separate hot lesion.



(b) Background-adjusted intensity bound. Restricting the admissible search interval prevents a gradient in the surrounding liver background from being selected as the lesion boundary.

Figure C.2: Illustrative safeguards used to bound the radial-gradient search. The curves are schematic one-dimensional profiles rather than patient measurements.

Appendix D

Colophon

This is a description of the tools you used to make your thesis. It helps people make future documents, reminds you, and looks good.

(example) This document was set in the Times Roman typeface using L^AT_EX and BibT_EX, composed with a text editor.

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